

Smart Technology Rollator for Independence, Detection, and Empowerment (STRIDE)

Final Project Report

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Abstract

The Lafayette College Department of Electrical and Computer Engineering has requested that the class of 2026 design a [STRIDE](#) system. Mobility devices are designed to meet the needs of their users, and there is a recognized difficulty in navigating stairs with most mobility devices; additionally, there is a recognized potential advantage in incorporating obstacle detection systems to aid in navigation tasks. Thus, the senior design team developed a device that encompasses three systems: a rollator, a stair-climbing mechanism, and a sensor detection system.

The completed system is a motor-driven rollator with the ability to navigate both flat ground and stairs. This system is comprised of: an aluminum tubing frame with aluminum and acrylic paneling; four tri-wheel assemblies, two of which are powered and interconnected by gear-chain systems, the other two composed of omnidirectional wheels; two [DC](#) brushless motors, powering the aforementioned tri-wheels; a 24 V [DC](#) battery and a custom power-routing system; an obstacle-detection system; a physical interface; a remote control interface; and a [MCU](#) hub. The rollator can operate under direct user control through a set of physical buttons and braking mechanisms, as well as autonomously through a remote control terminal on a password-protected website. It supports continuous operation on flat ground for up to 8 hours before recharge. The system is to be used indoors only on ADA-compliant staircases and surfaces. This document provides an introductory problem statement; the design requirements and considerations; the evolution of the design; the final details of the design; an overview of the testing and evaluation process; conclusions and future work; and an overview of team management.

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List of Acronyms

AC Alternating Current

ADA Americans with Disabilities Act

ADC Analog-to-Digital Converter

AEC Acopian Engineering Center

AI Artificial Intelligence

ASME American Society of Mechanical Engineering

ATP Acceptance Test Plan

BLDC Brushless DC

BMS Battery Management System

BPS Bits Per Second

CAD Computer-Aided Design

CC Constant Current

CUDA Compute Unified Device Architecture

CV Constant Voltage

DAC Digital-to-Analog Converter

DC Direct Current

DMM Digital Multimeter

E-stop Emergency Stop

ECE Electrical and Computer Engineering

EMI Electromagnetic Interference

ESP32 Expressif 32-bit

EZ DRIVE Electrocraft Motorcontroller

FOV Field of View

GPIB General Purpose Interface Bus
GPIO General-Purpose Input/Output
GPR General Project Requirements
I²C Inter-Integrated Circuit
I²S Inter-Integrated Sound
IEEE Institute of Electrical and Electronics Engineers
IMU Inertial Measurement Unit
IPC Institute for Printed Circuits
ISO International Organization for Standardization
LED Light-Emitting Diode
LIDAR Light Detection And Ranging
LiFePO₄ Lithium Iron Phosphate
MCU Microcontroller
MSPS Mega Samples per second
NPLC Number of Power Line Cycles
NXP Next eXPerience Semiconductor
PCB Printed Circuit Board
PDB Power Distribution Board
ROI Region of Interest
SCL Serial Clock
SDA Serial Data
SDK Software Development Kit
SoftAP Soft Access Point
SPI Serial Peripheral Interface
SPR Specific Project Requirements
STM32 ST Microelectronics 32 Bits

STRIDE Smart Technology Rollator for Independence, Detection, and Empowerment

TI Texas Instruments

TOF Time of Flight

TSAL Tractive System Active Lights

TSMS Tractive System Master Switch

UART Universal Asynchronous Receiver-Transmitter

UI User Interface

USB Universal Serial Bus

WBS Work Breakdown Structure

Standards and Governing Documents

ADA Accessibility Requirements Americans with Disabilities Act Chapter 4: accessibility requirements accessible routes, including surfaces, pathways, and doorways.

ADA Accessibility Requirements Americans with Disabilities Act Chapter 5: accessibility requirements for general site and building elements, including stairways.

ASTM A240/A240M-24 Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications

ASTM B221-21 Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes

IEC 61000-4-2:2025 Electromagnetic compatibility (EMC) - Part 4-2: Testing and measurement techniques — Electrostatic discharge immunity test

IPC-2152 Standard for Determining Current-Carrying Capacity in Printed Board Design

IPC-2221C Generic Standard on Printed Board Design

ISO 11199-2:2021 Walking aids manipulated by both arms — Requirements and test methods — Part 2: Rollators

ISO 12100:2010 Safety of machinery - General principles for design - Risk assessment and risk reduction

ISO 14001:2015 Environmental management systems - Requirements with guidance for use

ISO 7176-1:2014 Wheelchairs - Part 1: Determination of static stability

ISO 9001:2015 Quality management systems - Requirements

Lafayette College Electrical Safety Program College electrical safety rules and procedures for work on or near electrical equipment

Lafayette College General Project Requirements General engineering, documentation, safety, and reliability requirements for Lafayette ECE senior design projects

NFPA 70 (NEC) National Electrical Code

OSHA 29 CFR Part 1910 Subpart O OSHA machinery and hazard requirements referenced by the project requirements

RoHS Directive 2011/65/EU Restriction of the use of certain hazardous substances in electrical and electronic equipment

UN 38.3 UN Manual of Tests and Criteria, Part III, subsection 38.3, transportation testing requirements for lithium cells and batteries

1 Introduction

Enhancing an individual's mobility is crucial for supporting independence and empowerment among seniors and individuals who have lost mobility due to injury or illness. In recent years, designers and companies have recognized the need expressed by these people for mobility support devices that are specifically tailored to an individual's level of mobility; canes and wheelchairs are both useful, potentially life-changing devices, but are only appropriate for certain users. The emphasis on designing devices aligned with users' variable needs has resulted in the popularization of the rollator, which provides a level of support between a cane and a wheelchair.

A typical rollator is a four-wheeled device with handles, used to support a user's balance when navigating flat or moderately sloped terrain. It is more mobile on flat ground than a walker, but can become more burdensome on sloped terrain, where it either resists the motion of the user when going uphill or threatens to run away from the user when going downhill. It is bulky and hard to carry up or down stairs, especially for target users. Sloped surfaces and elevation changes, especially with an added load, are some of the first obstacles that people begin to have trouble with as they age or lose mobility. Additionally, typical rollators do not provide any assistance with navigation for users, who might have impaired vision or be experiencing cognitive decline.

Thus, a niche can be identified within the existing scope of mobility devices for a rollator that can better support users on sloped terrain; that can traverse stairs, removing the need to lift it up and down; that can carry light loads, such as groceries, up and down slopes and stairs; and has an obstacle-detection system to assist users in navigation. The goal of the [STRIDE](#) system is to provide a finished prototype that addresses all of these needs. The three main components of [STRIDE](#) are the rollator, the stair climbing mechanism, and the safety sensors. The development of these components is outlined in [Section 3](#), the final design details of the finished device are listed in [Section 4](#), and the testing and evaluation of these components is summarized in [Section 5](#).

The team was given a list of quantitative metrics as constraints and goals. [STRIDE](#) should support users of up to 200 lbs, and adjust its height to accommodate users from 5'0" to 6'3". The maximum dimensions must not exceed 35 in long by 25 in long by 45 in. The rollator should allow walking speeds of up to 3 mph, and should be able to operate on flat ground at a 25% duty cycle for at least 8 hours. Power should be supplied by a single 24 V battery. [STRIDE](#) should abide by [ISO 11199-2:2021](#), the standard governing assistive products for walking. A full list of design requirements is provided in [Section 2](#). The conclusion of how well the team met these requirements is given in [Section 6](#). Finally, an overview of team management practices is given in [Section 7](#).

2 Design Requirements

The initial design constraints as presented to the [STRIDE](#) senior design team are available as Appendix 1. Below are listed the major constraints the team considered when developing the rollator. Some constraints have been adjusted from the initial constraints, as the team developed a better sense of the design and what was achievable with the resources available. All metrics presented here are within the boundaries presented by [ISO 11199-2:2021](#), which governs rollators and similar walking-assist devices.

2.0.1 Weight

The weight of the rollator must not exceed 75 lbs. The total weight of the device, occupant, and carried load should not exceed 300 lbs in total at any given time. The front and rear wheels of the rollator must be capable of supporting up to the weight of the rollator + 200 lbs to ensure the user and the rest of the rollator are properly supported. The rollator's storage compartment will be capable of supporting up to 25 lbs in load without deformation or other risks.

2.0.2 Dimensions

The maximum physical dimension of the rollator shall not exceed 35 in long by 25 in wide by 40 in high. These are considered standard dimensions for commercial rollators. Current dimensions of the rollator are 27 in long by 22 in wide by 40 in high (at max handle height).

2.0.3 Material

The frame will be composed of 6061 square aluminium tubing, which balances strength and weight. This aluminium frame will be able to support the system without deforming. The frame shall include bright, high-contrast visual elements to ensure visibility at all times. The rollator's storage compartment will be capable of supporting up to 25 lbs in load without deformation or other risks. The tri-wheel structure will be machined from steel, which will provide strength and easy machining ability. The wheel tires should be composed of rubber to maintain sufficient friction between the wheels and the ground. The seat should be made of a soft, comfortable material capable of holding up to 200 lbs.

2.0.4 User Adaptability

The rollator shall accommodate individuals ranging in height from 5'0" to 6'3" by providing an adjustable handlebar within this range. The rollator shall support a seated user weighing up to 200 lbs without structural or functional failure. Users outside of this specified height/weight range, including children, are not considered part of the intended user population and the rollator shall not be configured for their use. The controls of the rollator will be tactile interfaces, such as textured buttons, toggle switches, or embossed directional indicators, to accommodate users with impaired vision, in addition to audio cues.

2.0.5 Mobility and Performance

The rollator shall reach speeds of up to 3 mph to accommodate the average adult walking speed. It shall remain steady when traveling up or down ramps of grades up to and including the 6 degrees incline set out by ADA standards. The rollator should be able to ascend and descend a staircase with riser heights ranging from 4 in to 7 in and a minimum tread width of 11 in, as to align with ADA standards [1].

- **Speed Control:** The rollator shall maintain adaptive motor speed for stability across varying terrain. The rollator shall not move backwards automatically to minimize the risk of injuries.
- **Safety System and Collision Avoidance:** The rollator shall integrate a hand brake for manual speed control. The rollator shall also include an automatic braking system, controlling tractive motor power through a normally open main contactor. The rollator shall automatically park if an obstacle is detected within 6 in of the minimum stopping distance, which varies based on speed, or the user releases the handlebars.

2.0.6 Power Capacity

Power will be distributed from a single 24 V battery to all the components of the system as required. The battery shall have sufficient battery capacity to operate continuously for 8 hours without recharging, assuming a 25% duty cycle of active mobile operation. The rollator shall constantly monitor its battery level and alert the caregiver when the level drops to 20% of the total charge. It would provide a second critical warning if the battery level drops to 10%. The rollator shall not be in operation if the battery level falls to 5% and should initiate shutdown procedures if that occurs.

2.0.7 Navigation

The proposed rollator is intended for indoor use. It is not designed to operate on complex outdoor terrain, e.g, gravel or grass. Under no circumstances should the rollator be used to carry a user up or down stairs; stair traversal shall always be purely autonomous.

2.0.8 Sounds

The operating noise emitted by the system shall not exceed 53 dBA measured 3 feet away on a hard floor, excepting intentional communication functions. While in motion, the system shall continuously emit a calm warning sound. Sound options must be user-selectable. When operating, the system shall make audible announcements mirroring status updates made by lights and visual cues, so as to facilitate use by users with impaired vision.

2.0.9 Engineering Standards

Whenever relevant, the system shall follow appropriate engineering standards in terms of part quality and assembly procedure.

2.0.9.1 Raw material:

All raw materials purchased for the project shall meet ASTM quality standards. All purchased aluminum meets [ASTM B221-21](#): Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes [2] All purchased stainless steel meets [ASTM A240/A240M-24](#): Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications [3]

2.0.9.2 Electrical & Electronic Components:

The motor manufactory is compliant under ISO standard [ISO 9001:2015](#), which governs the design, manufacture, distribution and sales of motors and motion control devices [4]. The battery is in [UN 38.3](#) compliance, which certifies the safety of lithium batteries for transport [5]. The battery manufactory is compliant under ISO standard 9001:2015 and ISO14001 for environmental management [4][6].

Where possible, electronics components were selected to be compliant with [RoHS Directive 2011/65/EU](#) to avoid the use of hazardous materials wherever possible.

3 Design Evolution

3.1 Prior Work

The beginning stages of the project involved background research and literature reviews. The team researched existing mobility devices with stair-climbing capabilities and obstacle detection capabilities. Through this literature review, several pertinent prior works were identified:

- The FreeMobility Z-Walker, a rollator-type device that uses tri-wheels and telescoping supports to traverse stairs[7]
- The Scewo Bro wheelchair, a wheelchair that uses a track system to climb stairs [8]
- Bhaskar et. al’s remote-controlled wheelchair, which uses a sensor system to dynamically detect and avoid obstacles [9]

These prior works allowed the team to gain a sense of feasibility for the future plans of the project, and provided a reference point in the proceeding stages of design.

3.2 Climbing Mechanism Selection

Through the background research stage and several team-professor discussions, the team determined that the most important early decision to make would be the selection of the stair-climbing mechanism. This decision would affect the direction of design for all future components, and would be extremely hard to pivot from, so sufficient research and consideration had to be put into the decision. Firstly, the team considered whether the climbing mechanism should be fully integrated into the design, or whether it should be a separate module that could be connected as needed. A decision matrix, shown in Table 1, was developed.

Climbing Mechanism	Integrated	Modular
Power Management	Difficult	Less demanding
Sensor Communication	Easy	Difficult
Testing	Easy	Difficult

Table 1: Decision matrix for integrating or breaking out the climbing mechanism

At the end of this phase, the team leaned towards an integrated climbing mechanism, as there was doubt as to how easy it would be for an end user to safely and securely connect the climbing mechanism properly to the rest of the rollator. In addition, sensor communication in order to control the climbing mechanism was also reasoned to be easier in a fully integrated system. However, this consideration was ultimately only a contributing factor to the overall question of what climbing mechanism should be selected.

The previously-mentioned prior works were considered, as well as additional works specifically detailing different climbing mechanisms. The most applicable of these works

included Y. Cho, B. Balan, and L. M. Smith’s papers on the tri-wheel (alternatively called delta-wheel and star wheel) structure [10] [11] [12], as well as a tri-spoke configuration that used leg-like structures to perform stair climbing [13] and a track system by A. Pappalettera et al [14]. Through this research, the team developed the decision matrix in Table 2.

Mechanism	Tri-Wheel	Tri-Spoke	Conveyor Belt
Connection Style	Integrated	Integrated	Modular
Speed	Medium	Medium	Slow
Stability	Medium	Low	High
Weight	Medium	Low	High
Difficulty of Design	Low	Low	High

Table 2: Decision matrix for the climbing mechanism

Using the metrics in the decision matrix, the team decided to pursue the integrated tri-wheel as the method of stair climbing. This method harnesses the advantages of an integrated climbing system detailed above, as well as balancing speed, stability, and weight. This system also avoids the additional moving parts that would be required of an adjustable conveyor belt, which would need to both roll as well as pivot up and down to align with stairs; in contrast, the tri-wheel structure only requires direct power to be transmitted to the wheels.

3.3 Motor Specification

The motor specification process was driven by a key number of factors, including system power constraints, required torque output, and ease of integration with the overall **STRIDE** platform. The team began by analyzing different types of **DC** motors, guided primarily by the project requirement that all subsystems be powered from a single onboard 24 V battery. This immediately constrained the design space to **DC** motor solutions, as they are inherently compatible with battery-powered systems. Following this, a comparison was made between brushed and brushless **DC** motors. Although brushed motors offered simplicity and lower cost, they were ultimately rejected due to their mechanical wear, lower efficiency, and maintenance requirements. For a system intended to operate reliably in assistive mobility applications, these drawbacks posed significant risks. **Brushless DC (BLDC)** motors were therefore selected due to their higher efficiency, longer lifespan and reduced maintenance.

The critical factor that ultimately guided motor choice was the system torque requirements. The team made initial torque estimates using principles from mechanics:

The power needed to support the rollator ascending is

$$P_g = Mgz,$$

where M is the mass of the rollator and z is the vertical distance that the rollator’s center of mass moves up.

Kinetic energy of rolling motion:

$$KE = \frac{1}{2}(I\phi^2 + Mv^2),$$

where I is the moment of inertia and ϕ is a subwheel's angular position.

The power needed to support the forward motion of the rollator is:

$$P_h = \frac{dE_h}{dt} = \frac{d}{dt}\left(\frac{1}{2}Mv^2\right) = M \cdot \dot{x} \cdot \ddot{x},$$

where x is the horizontal displacement of the center of mass.

The power needed to rotate the wheel is:

$$P_{roll} = \frac{dE_{roll}}{dt} = \frac{d}{dt}\left(\frac{1}{2}I\dot{\phi}^2\right) = I \cdot \dot{\phi} \cdot \ddot{\phi}$$

Through combining these values, the total power needed for the system is:

$$\sum P = P_{roll} + P_g + P_h + P_{loss}$$

Rotational work-energy theorem:

$$W = \Delta KE = \int \tau d\phi,$$

where τ is torque.

$$\tau = \frac{P}{\dot{\phi}}$$

$$\sum \tau = Mg \frac{dz}{d\phi} + I\ddot{\phi} + M \frac{dx}{d\phi} \cdot \ddot{x} + \frac{P_{loss}}{\dot{\phi}}$$

finds the sum of torques required to lift the rollator up one step of a specified length and height.

The theoretical calculations, taking into account the expected total load of the system of 125 lb (the initial 100 lb system weight limit, plus 25 lb for additional carrying load), showed torque requirements of up to 45 Nm per meter motorized tri-wheel. This system coupled with a 3:1 gear ratio meant the chosen motor would provide up to 15 Nm of torque at its shaft. At the same time, the motors had to operate at speeds compatible with the 3 mph walking requirement, meaning the final solution required an appropriate combination of motor speed and torque. These constraints eliminated many off-the-shelf and lower-cost motor options, as they could not deliver sufficient torque without excessive gearing.

Finally, the team evaluated commercially available [BLDC](#) motor systems. While several candidate motors and controllers were considered, as shown in the decision matrix in Table [3](#), many required additional custom development for control and feedback integration. The team reached out to Engineers and sales persons of different companies for quotes and queries about their motor systems and pricing. The ElectroCraft MP24 series solution was ultimately selected because it provides an integrated motor–drive system with built-in [BLDC](#) control, Hall sensor and tach feedback, and simple analog/digital interfacing. This not only reduced system complexity but also ensured that the motors could reliably meet both the torque requirement and the speed requirement.

Table 3: Motor comparison matrix.

Motor Option	Rated Torque (N·m)	Rated Speed (rpm)	Rated Power (W)	Rated Current (A at 24 V)	Controller Requirement	Cost (\$)
ElectroCraft MP24 (21:33)	12.8 continuous; 26.2 for 15 minutes with gearbox	190	229–416	12–24	Requires external drive, ElectroCraft EZ Drive	740
BLMR6200SKM-GFV-F	16 with gearbox	130	200	10.5	Requires dedicated BLVD series driver	614
ElectroCraft MPW86	20 continuous	150	300	20–24	Complete wheeled solution, not suitable for STRIDE; requires external ElectroCraft EZ Drive-compatible drive	1262
DB90-24-120-25S/5LGN150K	20	17	120	6.2	Requires separate BLDC controller, not integrated	—
DB86-24(300-20SBZPLF90-12)	15.47	166	300	15.6	Requires external BLDC controller	700

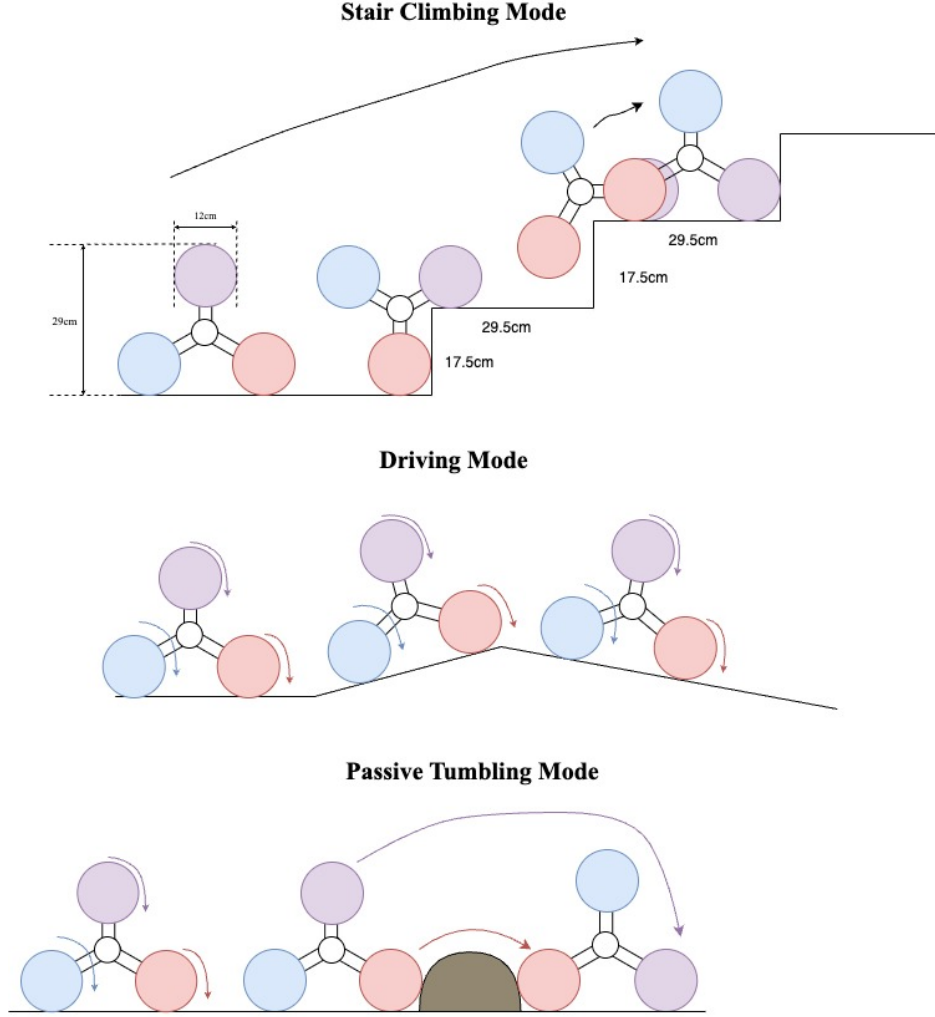
While several options met the minimum torque requirement of 15 Nm, the ElectroCraft MP24 (21.33:1) was chosen due to its strong overall performance and reliability. It has up to 12.8 Nm of current continuously which is sufficient for flat ground movement and goes up to 26.2 Nm for extended operation (15 minutes) offering a critical safety margin during demanding conditions such as incline traversal or transient loading. In addition, it supports output speeds of up to 190 rpm, aligning well with the system’s 3 mph target when paired with appropriate wheel sizing. This motor was the most affordable option that provided high torque at the relatively low speeds STRIDE requires.

3.4 Tri-Wheel Specification

The team selected a tri-wheel mechanism as the primary solution for enabling both stair climbing and efficient ground locomotion, based on the low-cost stair-climbing wheelchair design proposed in Riascos et al [15]. A tri-wheel consists of 3 sub-wheels arranged in an

equilateral triangle, mounted on a frame that rotates about a central drive axis. This structure allows the system to operate in two distinct modes: driving mode and stair-climbing mode. Figure 1 depicts the tri wheel’s driving mode and tumbling mode.

Figure 1: Tri-Wheel Driving Mode & Tumbling Mode



In driving mode, two of the three sub-wheels remain in contact with the ground; the system behaves like a conventional rollator. The triangular frame is not locked in orientation relative to the chassis, so it can pivot about the central axis to adapt to slopes and minor surface irregularities. In stair-climbing mode, when the front wheel encounters an obstacle, such as a stair riser, its motion is restricted. As a result, the motor torque increases within the transmission system. Once the torque exceeds a certain threshold, it induces rotation of the entire tri-wheel structure about its central axis. This rotation allows the system to “flip” over the step and continue climbing. The mechanism then repeats this process to ascend subsequent steps.

Two tri-wheel mechanisms were evaluated for stair-climbing capability: the delta-wheel from Riascos [15] and the tri-wheel from Smith et al [12]. To avoid confusion, Smith et al’s

device will be referred to here as a tri-spoke system. Both concepts use three orbital wheels arranged in a triangular configuration to enable rolling on flat surfaces and tumbling over obstacles, but they differ in power transmission, mode switching, and mechanical complexity.

The delta-wheel mechanism uses timing belts to transmit power from a central drive shaft to the three orbital wheels. When the wheels encounter an obstacle too large to roll over, the increased resistance causes the entire triangular frame to rotate around the drive shaft, lifting the rollator over the obstacle. The mode transition is completely passive; it occurs based on the reaction forces and friction at the wheel-obstacle contact point.

The tri-spoke mechanism uses a multi-stage gear train with separate driving and tumbling gear sets connected through a common shaft. In driving mode, two wheels contact the ground and roll normally with a 1:1 gear ratio. To initiate climbing, a cam-operated drum brake locks a central tumbling gear, redirecting power through the tumbling gear set at a 3:1 torque ratio, causing the assembly to walk over obstacles. This mode transition requires an operator command to engage the brake. The trade-off matrix in Table 3.4 summarizes how each mechanism performs.

Table 4: Decision matrix for the type of tri-wheel mechanism.

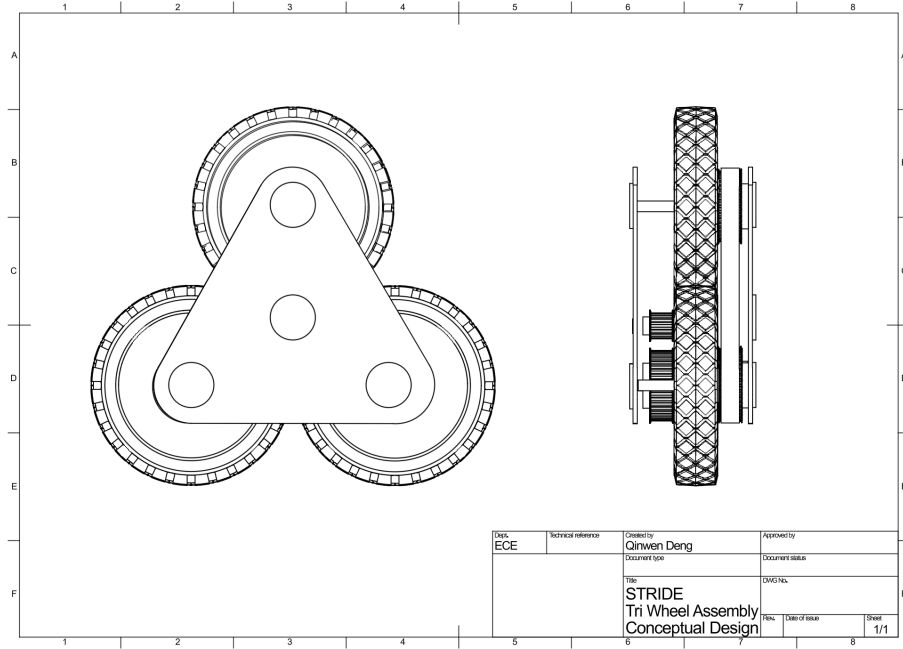
Mechanism /Criterion	Delta-Wheel Mechanism (Riascos)	Tri-Spoke Mechanism (Smith et al.)
Manufacturability	Easy: Belt-and-sprocket transmission uses off-the-shelf components.	Difficult: Precision gear train with six gears per branch.
Mode Switching	Simple: Passive transition driven by obstacle reaction forces.	Complex: Requires a cam-operated drum brake, dedicated actuator, and operator input.
Climbing Stability	Less controlled: Tumbling depends on sufficient friction between the sub-wheel and obstacle.	More controlled: Active brake engagement forces tumbling regardless of surface-contact conditions.
Cost	Lower: Additional components estimated at approximately 20% above a conventional wheelchair.	Higher: Custom steel gears, machined housing, and brake mechanism significantly increase cost.
Weight	Lighter: Fewer components; the full wheelchair prototype weighed 78 kg in total.	Heavier: 6.4 kg per tri-wheel unit.

The tri-spoke mechanism offers better climbing stability due to its controlled mode switching. However, this advantage comes at the cost of greater mechanical complexity and weight. Each tri-spoke branch requires a precision gear train with custom-machined steel gears enclosed in a machined aluminum housing, and each braking mechanism adds an additional actuator and control system.

The Delta-Wheel mechanism was selected because of the priorities of manufacturability, cost, and mechanical simplicity. This design uses commercially available components and does not require a separate mode-switching control system.

Each tri-wheel assembly is constructed in a sandwich configuration, with two triangular side plates forming the outer frame and the sub-wheels, sprockets, and chains housed between

Figure 2: The conceptual design of the tri-wheel assembly structure.



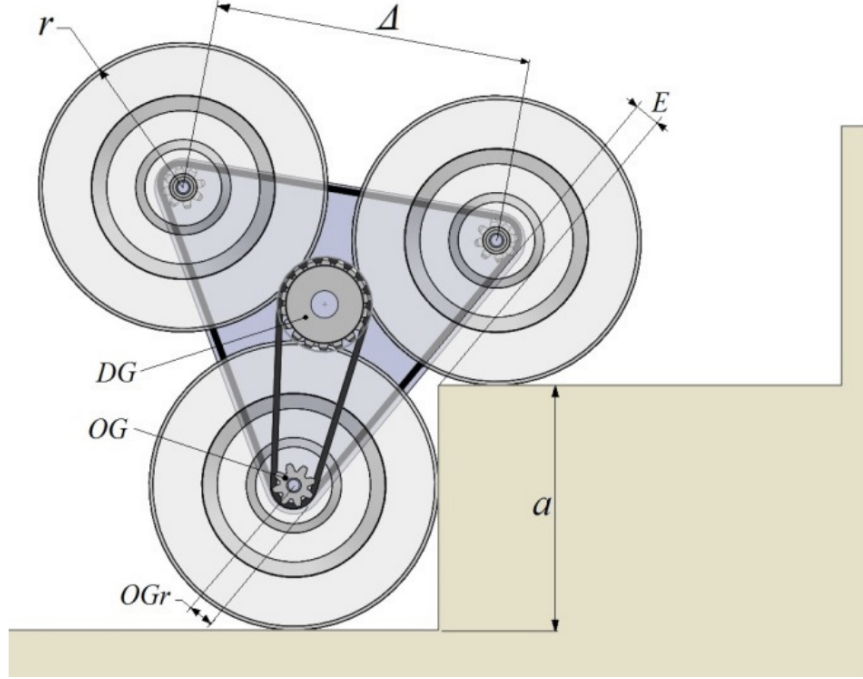
them. A central drive shaft passes through both side plates, attaching the assembly to the motor shaft. Power is transmitted from the motor via a drive sprocket and chain to a master wheel, which drives the other two sub-wheels through a second chain loop on the opposite side. Figure 2 shows the conceptual design of the tri-wheel assembly structure.

The tri-wheel assembly's climbing capability is determined by three interdependent geometric parameters: the delta-side length (Δ), the sub-wheel diameter R , and the orbital sprocket diameter OGr . The delta-side length (Δ) defines the maximum stair height the assembly can clear. The sub-wheel diameter and sprocket diameter together determine the clearance E between the triangular side plate and the stair edge during tumbling. E must remain positive to successfully tumble over a step without collision. If E reaches zero or becomes negative, the triangular plate contacts the stair edge and the assembly jams during rotation. Figure 3 shows tri-wheel assembly geometric parameters and stair-edge clearance during climbing.

A larger delta-side length (Δ) increases E and also allows the assembly to reach taller stair heights. A larger sub-wheel radius increases E by pushing the plate geometry further from the stair edge. Conversely, a larger orbital sprocket radius decreases E because the sprocket extends the effective profile of the frame closer to the stair edge. These parameters are not independent in the design.

The sprocket ratio was set at 2:1 to meet the torque and rpm requirement for climbing, which requires a relatively large orbital sprocket. This large OGr significantly reduces the available clearance. To compensate, the team increased the delta-side length to 8.4 in and selected a sub-wheel diameter of 8 in. This combination yields a clearance of approximately 1 inch for the target stair height of 6.75 in, measured from the front stair of the Acopian Engineering Building. This combination was verified through a Fusion simulation of the

Figure 3: Tri-wheel assembly geometric parameters and stair-edge clearance during climbing, showing the delta-side length (Δ), sub-wheel radius (r), stair height (a), orbital sprocket radius (OGr), drive sprocket radius (DGr), and plate-to-stair-edge clearance (E).



tri-wheel geometry against the stair profile, as shown in Figure 3, and further validated with a physical prototype using cardboard side plates and 3D-printed 8-inch wheels to confirm that the approximately 1-inch clearance is sufficient to avoid plate-to-stair collision during climbing.

The required torque for stair climbing was estimated using the work-energy theorem applied to the tri-wheel assembly as it tumbles over a stair edge.

$$W = \Delta KE = \int \tau d\phi,$$

The total power required consists of gravitational, rotational, translational, and loss components. This total power demand is then converted into the required driving torque using the relation

$$P = \tau \dot{\phi},$$

The required torque consists of a gravitational term, a rotational inertia term, a horizontal inertia term, and a loss term:

$$\sum \tau = Mg \frac{dz}{d\phi} + I \ddot{\phi} + M \frac{dx}{d\phi} \ddot{x} + \frac{P_{\text{loss}}}{\dot{\phi}}.$$

During stair climbing, the tri-wheel assembly operates at a relatively low and constant angular velocity, so the angular acceleration ($\ddot{\phi}$) and the linear acceleration ($\ddot{x}$) are close to 0. Under

this assumption, the torque can be simplified to

$$\sum \tau \approx Mg \frac{dz}{d\phi} + \frac{P_{\text{loss}}}{\dot{\phi}}.$$

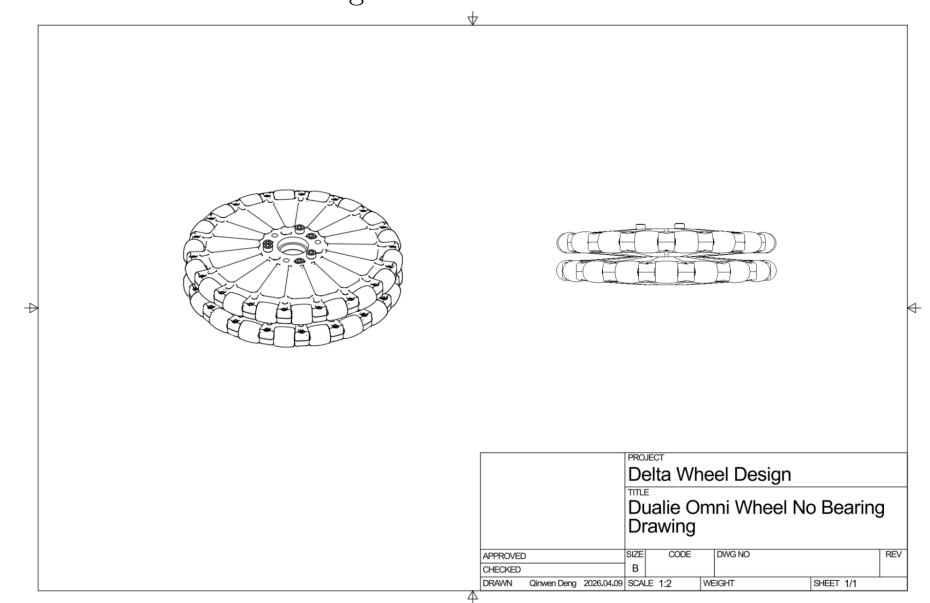
The dominant contribution is from the gravitational term. The center of mass rises as the tri-wheel assembly tumbles over the stair edge. $\frac{dz}{d\phi}$ describes how much height the center of mass gains over one unit of rotation. The center of mass traces a circular arc with radius r (the distance from the center of mass to the pivot point). The worst case is when the pivot point is horizontal from the center of mass, which is the radius of the wheels. Also, it is hard to precisely calculate the power loss (P_{loss}) due to friction and system inefficiencies, but some margin of torque is added. A detailed torque calculation can be found in Appendix 4.

Assuming a climbing load of 200 lbs, the maximum required torque at each powered tri-wheel unit is approximately 45 Nm. The selected motor provides 12.8 Nm of output torque, so a 2:1 sprocket ratio was adopted to meet this requirement. The sprocket ratio was initially targeted at 3:1. However, a 3:1 ratio reduces the motor output speed by a factor of three, which drops the rollator’s ground speed below the 3 mph design target. A 2:1 ratio maintains sufficient speed while relying on the motor’s peak torque output of 26.2 Nm to meet the climbing torque requirement.

The 3:1 ratio was also rejected for mechanical reliability reasons. All sprockets use #25 chain ($\frac{1}{4}$ in pitch), which is the smallest standard roller chain size. A larger chain pitch would increase the sprocket outer diameter for the same tooth count, enlarging the orbital sprocket profile and reducing the stair-edge clearance. With #25 chain, the largest orbital sprocket that fits within the tri-wheel assembly while maintaining positive clearance is 30T. However, first, at typical center distances within the assembly, only 4–5 teeth on the 10T sprocket are engaged with the chain at any given time, increasing the risk of chain skipping under the high torque loading. By comparison, the 15T sprocket in the 2:1 configuration engages 7–8 teeth, doubling the load distribution. Second, commercially available #25 10T sprockets are limited to a 0.25in bore. A shear stress analysis of the drive shaft shows that a 0.25in steel shaft would experience around 521 MPa of shear stress at 26.2 Nm, exceeding the allowable shear stress of 72.5 MPa. The minimum shaft diameter required to safely transmit this torque is approximately 0.58 inches. A 15T sprocket accommodates a larger bore, allowing a shaft diameter that meets this structural requirement. Based on these speeds, torques, teeth engagement, and structural constraints, the team selected a 30T:15T #25 sprocket combination.

The team also evaluated timing belts as an alternative to roller chains for power transmission, but ended up using roller chains. This is because, during evaluation and physical testing, timing belts require a closed loop of a specific length to match the center distances within the assembly. Off-the-shelf closed-loop timing belts are only available in fixed lengths, which may not match the geometry of the tri-wheel layout. Joining an open-loop timing belt into a closed loop requires specialized tooling and produces a splice that is weaker than the rest of the belt. By contrast, roller chains can be easily cut to any length and joined with a master link, allowing precise adjustment to the required center distances. Second, under the high torque loading experienced during stair climbing, timing belt teeth are more susceptible to skipping than roller chain, which engages with the sprocket teeth through mechanical interlocking.

Figure 4: Omni-wheel.



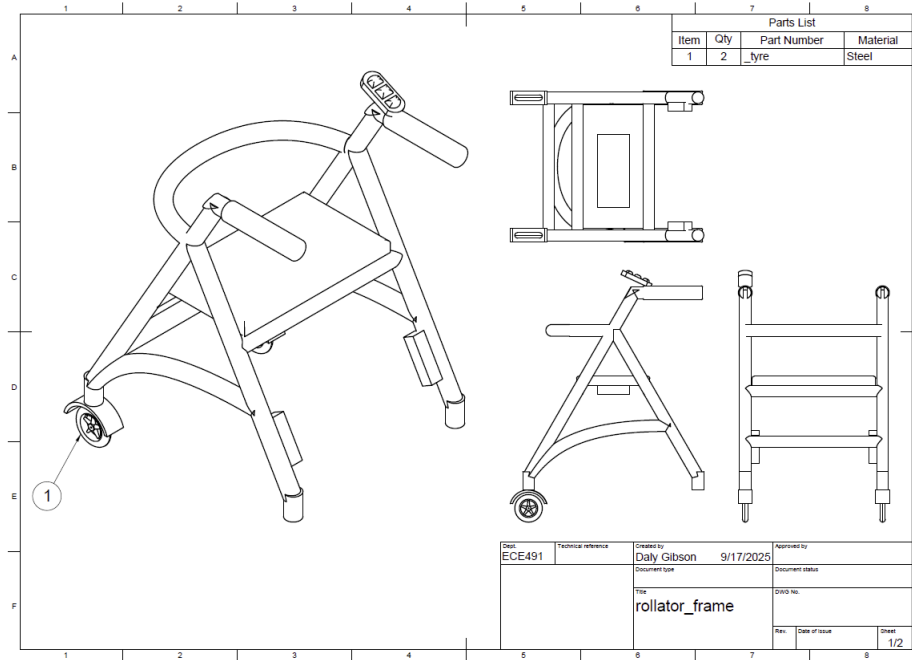
The rear wheels were initially designed as caster wheels to allow steering and directional changes. However, this configuration places the heavy climbing burden on the front powered tri-wheels. With caster wheels at the rear, the front tri-wheels must generate enough torque to lift not only themselves over each step, but also drag the entire rear structure up and over the stair edge. The caster wheels provide no mechanical assistance during climbing; they simply act as dead weight that the front must overcome. To mitigate this issue, the team replaced the rear caster wheels with unpowered tri-wheel assemblies. When the front tri-wheels pull the chassis forward, and the rear tri-wheels reach the stair edge, the triangular geometry of the rear assemblies passively guides them over the step without requiring any motor power. Rather than needing to be lifted vertically over the stair riser, the rear tri-wheels tumble over the edge as they are pulled forward. This significantly reduces the torque requirement on the front motors. To maintain the steering capability that caster wheels originally provided, the team used omnidirectional wheels (Figure 4) as the sub-wheels in the rear tri-wheel assemblies. Omnidirectional wheels have rollers along their circumference that allow lateral motion perpendicular to the wheel’s rolling direction. This enables the rollator to turn and navigate on flat surfaces while retaining the passive climbing assistance of the tri-wheel geometry.

3.5 Frame Design

The frame design largely depended on the team’s choice of climbing mechanism and motor; thus, it began with largely conceptual steps. Through referencing existing rollator designs, the frame design team developed an initial sketch to accompany the project proposal, seen in Figure 5.

This design includes several elements that will be carried over for future iterations, such as the triangular strut shape and the central seat. However, many of the features would

Figure 5: First design of the rollator frame. This design was never expected to be final, but was used for illustrative purposes.



eventually be changed as the team developed a better understanding of all of the requirements of the rollator and the components necessary to make it functional. The tri-wheel was placed in the back, as this was expected to give better maneuverability on flat ground; however, this was assuming that the motors would be much smaller than they were eventually spec'd as. Additionally, the members of the frame were cylindrical and curved, as this was observed on many commercial rollators; however, for the purposes of manufacturing this prototype, a different form factor would eventually be observed as more reasonable.

Until the motors, as the largest and most location-specific components, were selected, only preliminary work could be done on the frame. After consultation with the machine shop and mechanical engineering professors, the team selected aluminum tubing as the main material for manufacturing. Aluminum is much lighter than steel (2.7 g/cm^3 vs. 7.83 g/cm^3), which reduces the needed motor torque to lift the frame up stairs, while still having a comparative level of strength and ease of manufacturing. The team was additionally notified that the machine shop would be unable to create the curved members seen in the original drawing, so a design composed of straight beams would need to be created.

Once the tri-wheel was selected as the primary climbing mechanism, the team developed specifications for the overall size of the frame and the size of the tubing. The size of the frame is controlled by two major factors: the height and width of the operator, and the width of standard ADA-compliant doorways. The rollator must be a height that aligns with the intended userbase, which is defined as being from 5'2" to 6'3", and be a width that accommodates users weighing up to 200 lbs. Standard ADA-compliant doorways have a minimum width of 32" [1]. Based on these metrics, and accounting for the additional height and width added by the tri-wheel structure, the next iteration of the rollator design was

designated with a width of 21.5", allowing for a maximum tri-wheel width of 4" when allowing 0.75" of clearance on either side. The height of the rollator was designated at around 30", allowing room for the tri-wheel to lift the rollator. The width of the tubing was preliminarily selected at 1" circumference; the machine shop later advised the team to use square tubing, for ease of manufacturing, and to use larger tubing for the main triangular struts for strength. Thus, the following strength calculation was made using the specifications for square tubing and techniques from statics.

Maximum force on a member was estimated at 200 lbs of force, from a user seated on a singular seat member, which converts to 893 N of force. This force is distributed to the ends of the member, where the top of the member is under maximum tension and the bottom of the member is under maximum compression. Estimating this as 900 N of force, the following equation is applied:

$$\sigma = F_n \times A$$

where σ normal stress (Pa (N/m²)), F_n = normal force acting perpendicular to the area (N), and A = cross-sectional area (m²).

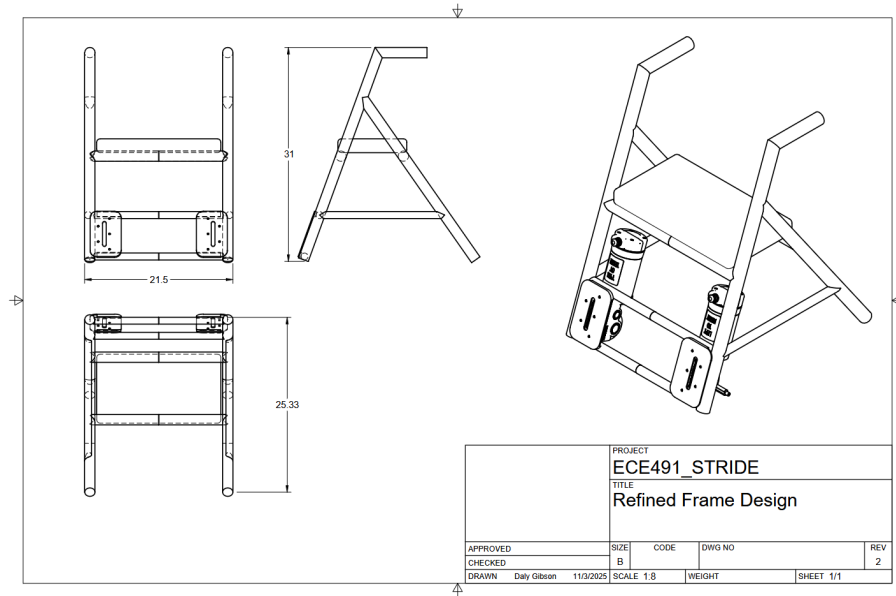
The result of this equation is compared to the Young's modulus of the material being evaluated, which is the quantity of stress the material can be put under before it undergoes a plastic (permanent) response, i.e. buckling or bending. Aluminum-6061 has a Young's modulus of 70 GPa, so the maximum stress on any member must never exceed 70 GPa.

From examining available stock aluminum, the team decided it would be most cost-effective to evaluate aluminum tubing with a wall thickness of 1/8". For a member with outer dimensions 1.5" x 1.5", inner dimensions 1.25" x 1.25", the total cross-sectional area A is 0.6875 in², or 0.000444 m²; for a member with outer dimensions 1" x 1", inner dimensions 0.75" x 0.75", A is 0.4375 in², or 0.000282 m². Thus, when taking the smallest possible cross-sectional area, the maximum tensile stress on a member is 3.191 MPa. As this is 5 orders of magnitude below the Young's modulus of aluminum-6061, this gives a very large margin before there is risk of plastic deformation. The team elected to use this thickness of aluminum because of the large margin of safety in its strength, its cost-effectiveness, and the machine shop's familiarity with this thickness of aluminum.

Additionally, the team designated where the powered tri-wheel would be placed. As the motors were specified by this point, the team recognized that placing the tri-wheel at the back of the rollator—and thus, the motors—would result in too much potential interference between the motors and the user's legs. Additionally, the support structure needed to support the large motors needed would cause further interference with the user. Thus, the team elected to move the tri-wheel to the front of the frame. This allowed for a strong attachment point to be designed, where the motors are directly screwed into custom motor plates, which are welded on three sides to other members of the frame. This distributes the force applied by the motors, especially when climbing.

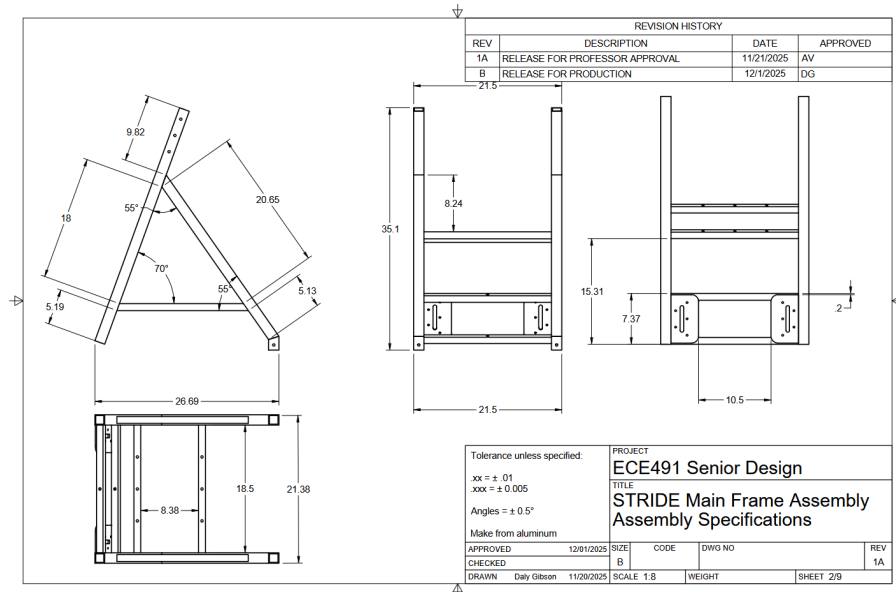
At this point, the team brought the design documents seen in Figure 6 down to the machine shop for review. This is the point where the machine shop advised the team to use square tubing instead of round, to reduce manufacturing time. This change also resulted in changes to the motor plate to account for the differences in position of the members it would be welded to. During the ensuing revision process, the team also decided to remove

Figure 6: Second major design of the rollator frame.



the handles and instead leave a place for telescoping handles to be attached later, to allow for adjustability in height. This resulted in the final design of the rollator sent to the machine shop for manufacture, seen in Figure 7.

Figure 7: Assembly view of the rollator design sent to the machine shop. Component drawings are available in Appendix 2.



From the 3D model of the final design, the team calculated that the rollator frame would require approximately 10.8' of 1.5" diameter square tubing and 9.5' of 1" diameter square tubing. This allowed the team to buy two 12' lengths of aluminum tubing, which allowed for

a reasonable amount of extra material in case of mistakes or revisions. The weight of this aluminum was calculated as

$$(24 \text{ ft} * 12 \text{ in/ft} * (0.4375 \text{ in}^2 + 0.6875 \frac{\text{in}^2}{2}) * 0.0975 \frac{\text{lb}^2}{\text{in}^3} = 15.795 \text{ lbs},$$

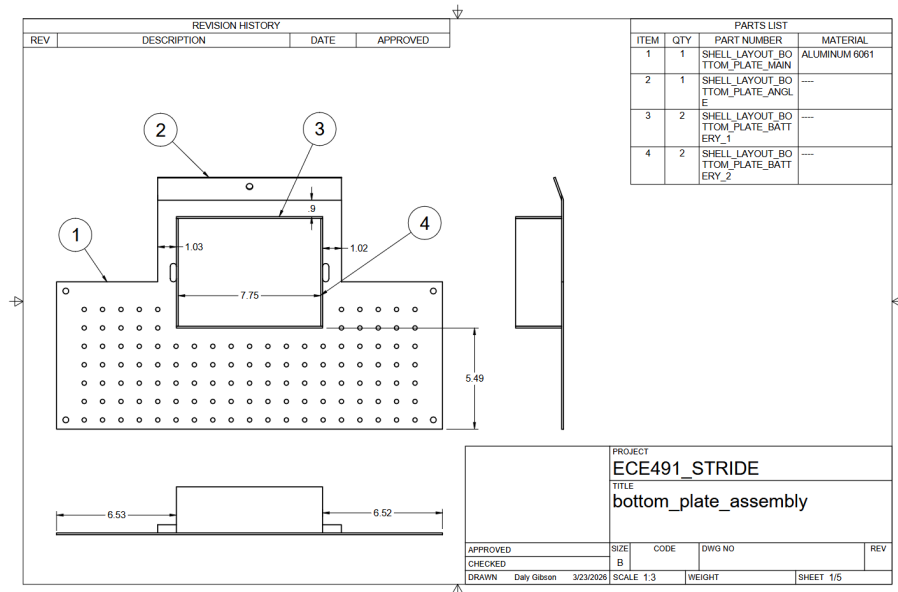
which was lighter than early estimates used to specify the motors. The final estimate for the frame weight was made at 13 lbs.

At this point, the frame was manufactured, with initial assembly completed on December 15th. One necessary revision was noted at this time; the CAD of the motor that was used to develop the motor plate did not include a small extrusion on the face of the motor. This required a revision to be sent out to cut a hole in each motor plate. Once this was done, the motors fit onto the frame. A final diagram of the main frame is available in Section 4.1.

In the spring, the frame team moved to designing plating to encase the rollator. Two main types of plates needed to be designed: support plates, machined from aluminum with regular thru-holes to allow for component attachment and support, and cladding plates, laser-cut from acrylic to enclose the electrical components of the rollator. The bottom and top plates are support plates, while the side plates are cladding plates. One side of the cladding plates includes a door to allow access to the interior without full dismantlement of the plating.

Initially, the bottom plate was designed with upward extrusions at the front to partially encase the battery, as shown in Figure 8. The battery would then be fully attached by a ratchet strap passing over the top to provide security during stair climbing. These extrusions were eventually removed for ease of manufacturing and weight, and instead, a second ratchet strap attachment point was added. The battery remained in its original position to bias the mass of the rollator towards the front in alignment with the powered tri-wheel.

Figure 8: Initial bottom plate design, with extrusions to hold battery. The final design is pictured in Section 4.1.



The original handle design was a telescoping pole that reached to the top of the rollator

and included handle bars on the top. This design would extend the height of the handles from 0 in above the top of the rollator up to 8 in above the top of the rollator.

Through the design process, the rollator gradually grew taller and taller; additionally, the tri-wheels increased in size. While the larger rollator size was useful for component arrangement, it required the team to reevaluate the handle style. Instead, the handles were designed to sit lower on the frame. The brake design component of the handles will be discussed in Section 4.8.1.

Final assembly was conducted by the team in the lab. Mounting holes were drilled under supervision in the machine shop, adhering to the machine shop safety standards [appendix]. The rollator was then brought up to the lab and parts were assembled with vibration-resistant fasteners. Bolt-nut fastenings were all tightened by hand evenly before being fully tightened by wrench.

3.6 Sensor Specification

The first step in the sensor system design was identifying the specific tasks that sensors needed to support. This design document is available as Appendix 18. It was determined that sensors must enable functions such as determining the operating speed of the rollator, measuring the force applied by the user through an accelerometer/[Inertial Measurement Unit \(IMU\)](#), and detecting turning behavior. Additionally, sensors are required to detect human actions, including catching or stopping motion when a user falls. Some of these functions can be inferred from motion detection of wheels, which is detected by integrated sensors on the selected motors. For stair traversal, the system must detect staircases, including individual stair height and individual stair length from both the top and bottom of the stairs; detect that the rollator is correctly oriented to begin stair traversal; and determine when the rollator has cleared a stair both vertically and horizontally. This orientation detection uses the user interface system to provide feedback so the user can reposition the rollator properly. Simple obstacle detection is also required when traveling on flat ground to detect collision risks. Based on these requirements, the team chose several sensors to research further: an [IMU\(Accelerometer\)](#), distance sensors, and a camera. The [IMU](#) was chosen to be used during the stair climbing stage and detecting the slope. With the detection of the current orientation of the rollator, the system can detect if it is at the stage of stair traversal or finished with traversal.

Next, the team researched and tested multiple distance sensing technologies for obstacle detection, including [Time of Flight \(TOF\)](#), depth cameras, [Light Detection And Ranging \(LIDAR\)](#), and ultrasonic sensors. Braking distance was used as the primary specification to evaluate these options. In order to be viable, a sensor needs to accurately detect any given obstacle from a far enough distance that the rollator can safely stop before collision. The team used the maximum specified speed of 3 mph (5 kph) and a range of acceleration values, up to the maximum specified deceleration of 3 m/s². Using the equation

$$v_f^2 = v_i^2 t + 2ad$$

the team could determine the stopping distance for each acceleration value.

Table 5: Stopping distance calculation.

Acceleration (m/s ²)	Distance (m)
-0.5	1.8
-1	0.9
-2 (E-stop)	0.45
-3 (E-stop)	0.3

Ideally, the rollator should not have to brake abruptly, and should operate in the normal ranges of no more than -1 m/s² acceleration. Thus, the team specified a minimum reliable obstacle detection range of 6 ft (2 m).

Based on this assumption, the team first considered [TOF](#) sensors due to their low cost, robustness in dark environments, relatively high accuracy at short range, computational simplicity, and typically wider [Field of View \(FOV\)](#). However, experimental testing revealed that the uncertainty/error of the sensor increases dramatically beyond the range of 1.5 ft [Appendix 10](#). This prompted a change in sensor selection and thus reevaluation of the braking distance model.

To do this, the team developed a model using user reaction time and typical walking speed. Using a typical walking speed of 2.2 mph (0.8 m/s) and an estimated reaction time of around 2 second, the team recalculated that the minimum emergency braking distance should be approximately 4 m.

Minimum braking time:

$$t_1 = \frac{4 \text{ (m)}}{3 \text{ (mph)} \times 0.447 \left(\frac{\text{m}}{\text{s} \cdot \text{mph}} \right)} = 2.98 \text{ (s)}$$

Maximum time duration at typical walking speed:

$$t_2 = \frac{4 \text{ (m)}}{0.8 \left(\frac{\text{m}}{\text{s}} \right)} = 5.00 \text{ (s)}$$

Reaction time:

$$t_{\text{reaction}} = t_2 - t_1 = 5.00 \text{ (s)} - 2.98 \text{ (s)} = 2.02 \text{ (s)}$$

Using these numbers, the team developed a schema to handle obstacle avoidance. The rollator will warn users at a distance of 4 m of a detected obstacle. If there is no decrease in speed after the designated reaction time, the rollator will automatically stop.

This updated requirement significantly impacted sensor selection. Although [TOF](#) sensors claim a maximum range of up to 7 m, their optimal and reliable range in practical applications is often closer to 2 m due to increasing error beyond this distance. As a result, ToF sensors were deemed insufficient, and the team began evaluating depth cameras and [LIDAR](#) systems, which can reliably operate at longer ranges, considering both technical performance and system-level constraints.

A key design challenge arose in caregiver mode, where the rollator functions as a compact motorized wheelchair. In this mode, the rollator moves backwards, meaning the user faces

the rear, which limits a forward-facing depth camera to detecting only obstacles behind the rollator. To address this, the team considered two solutions: (1) implementing a 360-degree [LIDAR](#) system for full environmental awareness, or (2) modifying the user manual to include a safety warning requiring caregiver vigilance when the user is seated. Literature such as “Walk-IT: An Open-Source Modular Low-Cost Smart Rollator” demonstrates the use of a [LIDAR](#) (RPLidar A1M8) with a 360-degree field of view and approximately 6-meter range, though concerns remain regarding detection range and placement. [16] Another study, “An Intelligent Rollator for The Elderly,” combines both depth cameras and [LIDAR](#), using [LIDAR](#) for step detection and adaptive pacing[17]—an approach that introduced a new possibility for satisfying the requirement of adapting to the user’s pace.

The team explored multiple [LIDAR](#) placement strategies. Mounting the [LIDAR](#) under the seat or on the rollator frame introduces occlusion issues, as the user’s body blocks critical detection regions, especially when seated. Alternatively, mounting the [LIDAR](#) above the user’s head (similar to prior wheelchair projects) provides optimal coverage, as supported by placement studies [18]. However, this approach introduces significant mechanical and ergonomic challenges, and would require a large revision to the project size constraints. Considering these trade-offs, the team ultimately favored Solution 2: modifying the safety manual to require caregiver presence during seated operation, and using a forward-facing depth camera to perform obstacle detection tasks while operating as a standard rollator.

Based on this analysis, the team selected a Depth Camera as the primary obstacle detection sensor and evaluated several candidate devices, detailed in Appendix 18. Options included Intel RealSense D Series, Orbbec devices, and a Yahboom stereo camera. The RealSense option was limited to about 4 m detection range (and will vary from lighting conditions), which was insufficient for the team needs. Among Orbbec models, one supported up to 10 m but lacked clear optimal specifications, while another had more defined performance but only reached about 3 m. The Yahboom camera was cheaper and suitable for embedded use, but lacked software support and would require careful calibration. The final winner is [19][ZED 2i](#) from StereoLabs not only because of its maximum detection range of 20 m, but also its full open-source [Software Development Kit \(SDK\)](#) which can be easily implemented with Jetson Orin Nano. The depth camera has to be paired with a mini-computer (Jetson Nano) due to its high computational requirements and existing [SDK](#) from the manufacturer. Additionally, the ZED 2i has an onboard [IMU](#), which removes the need to buy and integrate a separate sensor.

However, depth cameras have two major limitations: inability to detect glass objects and poor performance in dark environments. To address these gaps, the team incorporated ultrasonic sensors as complementary sensors. The selecting criteria includes: maximum and minimum distance range, [FOV](#), whether resistance, and input voltage. The maximum reliable range should be more than 2 m to ensure the system has enough room to brake. The minimum range requirement is not specified, but compared with the rest of the ultrasonic sensor, A02YYWU stands out with 3 mm minimum range and 4.5 m maximum detection range. The main concern with an ultrasonic sensor is the time delay; in order to detect objects 4 meters away, the propagation of sonic waves results in a 200 ms delay between readings. Based on the maximum speed of 1.3 m/s, a worst-case scenario of a 200 ms delay before detecting an obstacle would result in traveling 0.26 m before obstacle detection and avoidance procedures would begin. Since this is shorter than the minimum braking distance

calculated above, this result is acceptable.

Finally, the team refined the sensor configuration and system architecture. The depth camera serves as the primary forward-facing obstacle detection sensor, while ultrasonic sensors provide redundancy and compensate for edge cases such as transparent obstacles and low-light conditions. The depth camera’s on-board [IMU](#) is used for motion and orientation detection, particularly during stair climbing. On the software side, several key components were developed in the unit testing stage: the team studied depth-processing and stereo-matching principles to understand disparity, calibration, and noise behavior; implemented [Region of Interest \(ROI\)](#) masking (to the minimum width of doorway specified in [1]) to reduce false positives and computational load; and developed the obstacle detection algorithm as an integration-ready C++ module. The team initially explored a top-down occupancy grid aligned with the system reference frame but determined it was unnecessary; as [STRIDE](#) does not perform navigation, it is less intuitive to debug, adds computational stress to the Jetson NANO, and provides limited improvement in obstacle detection.

The team defined and validated detection constraints with the chosen sensors, including a range of ~ 4 m (validated at 3.9–4.1 m) and a width of 32 inches (doorway-compliant filtering). Testing demonstrated $\geq 95\%$ classification accuracy in static scenarios and $< 1\%$ depth error in dynamic conditions, including diagonal motion tracking. Power analysis showed an average consumption of 4.7 W with peaks around ~ 6 W. The team also identified performance risks, particularly FPS limitations under dynamic conditions. To further improve robustness, wall detection was added to filter ultrasonic sensor data. The ultrasonic sensors do not differentiate between obstacles detected to the side and toward the front, so this additional function allows the system to ignore misleading ultrasonic readings. This functionality has been integrated into the main program.

Overall, the sensor system evolved from a broad exploration of multiple sensing modalities into a carefully justified hybrid configuration. Each design decision—from task identification, to sensor selection, to final integration—was driven by experimentally validated constraints and real-world usage considerations, resulting in a system that balances performance, cost, and reliability.

3.7 Motor Control Design

The design scheme for controlling the motors was considered to be largely dependent on the specific motor chosen. This design was therefore deferred until the team settled on a specific motor. Following the selection of the MP24 ElectroCraft motor, its electrical and control requirements were analyzed. Unlike brushed DC motors, the MP24 requires three-phase power excitation (phases A, B, and C) and cannot be driven directly from a DC supply. Instead, it requires electronic commutation, in which the phase currents are switched in a precise sequence based on feedback from the integrated Hall sensors on the motor. Beyond phase driving and feedback handling, the motor must be operated within a controlled voltage and current range, with protection against overcurrent, undervoltage, and thermal faults. To satisfy these requirements, the ElectroCraft [Electrocrafft Motorcontroller \(EZ DRIVE\) BLDC](#) controller was selected as the accompanying drive.

It provides dedicated three-phase outputs for the motor windings, integrated Hall sensor input circuits, and optional tachometer/encoder input support, enabling proper trapezoidal

commutation of the BLDC motor. Additionally, the drive includes internal current limiting, overvoltage protection, and fault detection, removing the need for complex external circuitry. The drive accepts a 12–48 V DC supply and internally generates the switching signals required to energize the motor phases, while exposing a simplified control interface consisting of an analog speed input (0–10 V) and digital control signals such as enable, direction, and braking.

With the low-level motor requirements handled by the drive, the system Microcontroller (MCU) could then be selected to interface at a higher level. The MCU does not directly drive motor phases; instead, it generates command signals and reads status outputs such as tachometer pulses and fault flags.

The STM32H7A3ZI-Q was selected due to its integrated Digital-to-Analog Converter (DAC), high processing capability, and sufficient General-Purpose Input/Output (GPIO)s for interfacing with the motor driver’s control signals. The DAC facilitates direct analog speed command generation; however, its 0–3.3 V output range does not meet the motor controller’s 0–10 V input specification. To address this, a custom signal conditioning board was developed using an op-amp in a non-inverting configuration with a gain of 3, effectively scaling the DAC output to the required voltage range. The non-inverting configuration was selected as it enables the signal to be fed directly to the op-amp input terminal, avoiding the loading effects that may arise from the input resistor in an inverting configuration. The conditioning board is interfaced directly with the EZ DRIVE controller, consolidating all analog and digital control signal connections.

3.8 Microcontroller Specification

3.8.1 STM32

The STRIDE system required a MCU with robust on-board analog interfaces, specifically integrated glsadc and glsdac modules, to directly interface with signals such as user brake inputs, motor control references, and sensor outputs. This MCU would handle all low level functions of STRIDE. By selecting a MCU with built-in analog capabilities, the design avoids the need for external Analog-to-Digital Converter (ADC) or DAC chips, which would otherwise increase circuit complexity, wiring overhead, and potential sources of noise and failure. Additionally, fewer external components translate to easier debugging, lower power consumption, and more compact hardware design.

Beyond analog requirements, the MCU needed to support a wide range of peripherals, including GPIO, Inter-Integrated Circuit (I²C), Universal Asynchronous Receiver-Transmitter (UART), and hardware timers, to interface seamlessly with subsystems such as the battery monitoring circuit, motor drivers, sensors, and communication modules. This requirement led to the selection of ST Microelectronics 32 Bits (STM32)-based boards, which offer a high density of peripherals alongside strong real-time performance. The STM32 platform is specifically designed for embedded control applications, providing deterministic timing, precise interrupt handling, and tight hardware-software integration features that are essential for motor control and safety systems.

In comparison, platforms such as the ESP32 and Raspberry Pi were considered but found less suitable for this application. The ESP32 offers good wireless capabilities and ease of use shown in table 6, but it has more limited and less precise analog performance compared to the

STM32, particularly in ADC accuracy and DAC availability. From a development perspective, the STM32 ecosystem provides strong support through tools such as STM32CubeIDE and HAL libraries, enabling structured and scalable firmware development. The STM32 platform provides greater control over hardware, better scalability for complex embedded systems.

Criteria	STM32H7A3ZI-Q	STM32G4 / L4 Series	ESP32-S3	Raspberry Pi Pico
ADC Performance	16-bit (3.6 MSPS) High accuracy 2 × 12-bit (true DAC)	12–16 bit Adequate	12-bit Adequate	12-bit (roughly 500 kSPS) Noisy
DAC Capability	Stable motor control output 4+ I2C, 6 SPI, 8 UART, 100+ GPIO	12-bit, variant dependent	8-bit, low resolution	Requires filtering workaround
Peripheral Availability	Plenty 280 MHz (Cortex-M7)	2–3 I2C, fewer UART/SPI Moderate	2 I2C, 3 SPI, 3 UART (45 GPIO) Moderate	2 I2C, 2 SPI, 2 UART (26 GPIO) Limited
Processing Power	Single core Very high	80–170 MHz (Cortex-M4) Sufficient	240 MHz dual-core Very high	133 MHz dual-core Sufficient
Cost	12~18 Higher upfront, lower total system cost	5~10 Moderate	5~13 Low cost	3~5 Lowest cost but highest system complexity

Table 6: Comparison between Microcontrollers

The STM32H7A3ZI-Q provides the best overall performance, offering high-resolution DACs, true DAC outputs, extensive peripherals, and strong processing capability, making it ideal for precise control and full system integration.

3.8.2 Jetson Orin Nano

The inclusion of the Jetson Orin Nano in the system was primarily driven by the computational requirements of the depth camera. The ZED 2i relies heavily on GPU acceleration for real-time depth sensing. This immediately imposed a hardware requirement that could support Compute Unified Device Architecture (CUDA)-based parallel processing, which typical MCUs or low-power CPUs cannot provide. The team selected the Jetson Orin Nano not only because Stereolabs provides optimized SDK support specifically for the Jetson Nano platform, ensuring seamless integration and reliable performance, but also because of its relatively high computational capability within an embedded form factor. Importantly, this choice leaves significant headroom for future expansion without being constrained by hardware limitations, which are typically harder to revise later in the design process.

Despite the presence of such a powerful embedded system, the system still required STM32 MCUs. The Jetson Orin Nano, while strong in high-level computation, lacks low-level hardware interfaces such as ADCs. The motor drivers require analog input signals and deterministic timing behavior that cannot be reliably achieved through a Linux-based system like the Jetson Nano. Therefore, STM32 MCUs were introduced as dedicated low-level controllers to handle motor actuation, sensor interfacing, and real-time signal processing. This separation ensures that time-critical operations remain stable and predictable, while the Jetson Nano focuses on computationally intensive tasks.

Finally, the Jetson Orin Nano was designated as the main controller in the system architecture due to its processing power and flexibility. With substantial computational slack, it is capable of executing high-level decision-making algorithms, integrating sensor data, and coordinating overall system behavior. In this role, the Jetson Nano effectively functions as a “watchdog” or supervisory controller, issuing commands to the STM32 units while monitoring system performance. This hierarchical control structure leverages the strengths of both

platforms: the Jetson Nano for intelligence and adaptability, and the [STM32](#) for reliability and precise control. As a result, the system achieves both robustness in low-level operations and scalability in high-level functionality.

3.8.3 ESP32

From a caregiver’s perspective, easy accessibility to phones, laptops, and other devices with wireless connectivity is very convenient. Hence, the use of a device with Wi-Fi capabilities was ideal for the project. To accomplish this task, the ESP32 was selected as an ideal option for its price and [Soft Access Point \(SoftAP\)](#) connectivity. The following table details the 3 options of ESP32 devkits discussed; finally, the team settled on the ESP32-S3-DevKitC-1N8R8 for its cheaper price and the need of at least 4 MB of memory to allocate sounds for device operations.

Table 7: ESP32 model configurations for comparison.

Distributor	Part Number	Module	Integrated Flash	PSRAM	SPI	Voltage
	ESP32-S3-DevKitC-1N8R8	ESP32-S3-WROOM-1N8R8	8 MB	8 MB	QD	3.3 V
	ESP32-S3-DevKitC-1N32R16V	ESP32-S3-WROOM-2N32R16V	32 MB	16 MB	OT	1.8 V
	ESP32-S3-DevKitC-1UN8R8	ESP32-S3-WROOM-1UN8R8	8 MB	8 MB	QD	3.3 V

From a system integration perspective, this setup allows the rollator to interface with external devices, providing caregivers with valuable real-time metrics and remote controls through a mobile UI. The ESP32’s low power consumption and wireless data transmission make it a perfect candidate for this application. The embedded WROOM module hosts a 2.4 GHz Wi-Fi access point, enabling the caregivers with real-time remote control through the user interface seamlessly. For high-reliability data transfer between ESP32 and Jetson Nano, the SPI protocol can be used to communicate via a carrier board. As previously detailed in our comparison with the ESP32 design, the ESP32 offers superior cost-efficiency while meeting all the functional requirements of the team.

3.9 Power System Development

3.9.1 Battery

In the project requirements, the team was given several specifications for the battery. The battery was required to be 24 V, as to align with the [Lafayette College Electrical Safety Program](#), and to be a [Lithium Iron Phosphate \(LiFePO₄\)](#)-composed battery. These guidelines were used by the group in motor specification, and later to specify the battery. The battery was specified once the motors, sensors, and [MCUs](#) were generally specified, as the total required power draw needed to be known. The team listed out all anticipated power requirements:

Item	Voltage (V)	Current Consumption (A)	Power Consumption (W)	Quantity
L Motor	24	N/A	275*	1
R Motor	24	N/A	275*	1
Ultrasonic	5	0.0144	0.072	6
STM32	5	0.5	2.5	5
Jetson Nano	12	2.5	30	1
ESP32	3.3	0.24	0.792	1
PDB Losses	24	0.4	9.6	1
Depth Camera (USB)	5	0.5	2.5	1
Class D Amp	5			1
Speaker	5	Low	4	1
Control Cluster LED's			0.2	5
Total Power Consumption: 610.8 W				

Table 8: Power Budget.

As illustrated above, the team needed to find a battery that could provide at least 610 W out continuously. An important caveat to this is that the motors will only reach their maximum wattage pull of 275 W very briefly, if at all, while stair climbing. Thus, the selected battery can afford to be quite close to the power ceiling, since the motors will be operating at a much lower average wattage (204 W).

When searching for a battery that met the above specifications, the team encountered two main types of battery: trolling batteries, for use on boats, and e-mobility batteries, for use on e-scooters or e-bikes. The team compared four potential batteries, with the main considerations being capacity (Ah), max power out (W), and weight:

Table 9: Battery comparison matrix.

Parameter	Timeusb Trolling (Amazon)	Timeusb Trolling (2)	Bioenno Emobility (1)	Bioenno Emobility (2)
Voltage (V)	25.6	25.6	24	24
Capacity (Ah)	50	25	40	30
Maximum current output (A)	60	50	80, 5 s	60, 5 s
Continuous current output (A)	60, 30 min	—	40	30
Maximum power output (W)	1536, calculated	1280*	960	720
Weight (lb)	21.2	13.3	20.9	15.8
Battery cost (\$)	229	150	580	550
Charger cost (\$)	99	99	10	40
Shipping (\$)	Free	Free	30	30
Total cost (\$)	328	249	620	620
Terminal type	Screw	Screw	Wired	Wired

*Value retained from vendor comparison notes.

* This wattage was reported at a 2C discharge rate.

Based on these values, the team elected to buy the 24 V 30 Ah battery from Bioenno. This battery provides sufficient capacity and wattage, as well as a more usable Anderson connector.

3.9.2 High Power Architecture

The [STRIDE](#) power delivery system is designed as a split high-power and low-voltage architecture built around a single 24 V battery bus. At the system level, battery power first enters the protected upstream power path, then passes through the master shutdown and safety hardware, and is finally divided into separate traction and electronics branches. This arrangement was chosen so that high-current motor power and sensitive low voltage electronics would not share the same unprotected distribution path. It also improved serviceability by allowing faults to be isolated by branch rather than forcing the entire robot into one undifferentiated power domain.

The high-power path was treated primarily as a safety and power-handling problem rather than simply as a wiring exercise. The traction system can demand much higher instantaneous current than the rest of the robot, and it must also be shut down in a predictable and visible way under fault or emergency conditions. For that reason, tractive motor power is controlled through the emergency-stop safety chain and main contactor rather than through

the low voltage electronics distribution path. In practice, this means that the motors are powered from a supervised 24 V traction branch whose energized state depends on the status of the [Emergency Stop \(E-stop\)](#) circuit, the [Tractive System Master Switch \(TSMS\)](#) and the required reset logic. This arrangement separates overcurrent protection from functional safety: fuses and breakers protect wiring and hardware, while the contactor and [E-stop](#) chain create a controlled safe state for the machine. A second design priority was staged protection. Instead of feeding every subsystem directly from the battery, the architecture was organized so that major functions could be protected and serviced independently. The upstream battery path provides primary fault protection and shutdown capability, after which the system divides into a traction branch and an electronics branch. The traction branch carries the 24 V motor bus, while the electronics branch feeds the Power Distribution Board, which generates the regulated rails needed by controllers, sensors, and interface hardware. This separation reduces the risk that traction transients or motor-side faults propagate directly into the control electronics, and it also makes debugging more manageable because the low-voltage system can be evaluated independently of full tractive power.

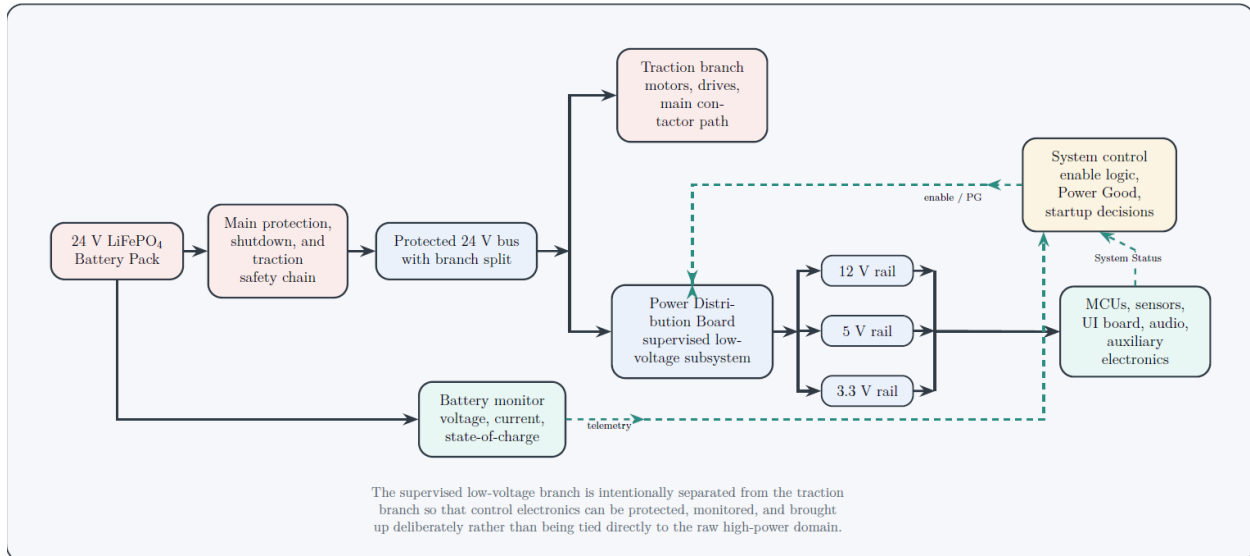


Figure 9: System Level STRIDE Power Architecture

Battery monitoring was also treated as part of the high-power architecture. Because the selected [LiFePO₄](#) battery maintains a relatively flat discharge voltage over much of its operating range, voltage alone is not a reliable indicator of remaining charge. As a result, the battery subsystem was treated as a managed energy source whose condition needed to be measured explicitly and interpreted at the system level. This was especially important for a device like STRIDE, where low battery state can affect whether the system should continue normal operation, enter a reduced-power state, or stop before attempting high demand functions such as stair climbing. Finally, the high power architecture was developed with serviceability in mind. [STRIDE](#) is a complex prototype that must be inspected, debugged, repaired, and demonstrated repeatedly. For that reason, the upstream power design favored clearly separated branches, visible protection devices, and deliberate fault recovery rather

than hidden or ambiguous behavior. A serious fault should create an obvious condition that can be investigated before the robot is returned to service.

3.9.3 Power Distribution Board Design Evolution

All low-voltage electronics, including MCUs, sensors, user-interface elements, and auxiliary circuits, are powered from the Power Distribution Board (PDB). With the exception of the dedicated traction branch, no low-voltage subsystem is supplied directly from the raw 24 V battery bus. This architecture centralizes the generation of the required 12 V, 5 V, and 3.3 V rails and eliminates the need to distribute unregulated battery voltage throughout the system.

This partitioning provides several engineering advantages. It reduces the number of distributed DC–DC conversion stages, limits exposure of sensitive electronics to battery-bus transients, and reduces the direct coupling of high-current motor disturbances into the control and sensing domain. It also improves fault isolation, staged bring-up, serviceability, wiring discipline, connector standardization, and maintenance access by establishing a clear boundary between the upstream high-power distribution network and the supervised low-voltage electronics domain.

Because the battery supply is nominally 24 V, while the principal electronic loads require lower regulated rails, centralized conversion on the PDB provides a more controlled and maintainable implementation than distributed point-of-load regulation or widespread raw-bus distribution.

From the outset, the PDB was intended to do more than generate three voltages. It serves as the electrical interface between the battery system and the low-voltage electronics domain. In addition to regulated rail generation, the board was required to provide rail supervision, controlled startup and shutdown, and status indication for each rail. In practice, this made enable and Power Good functions important design features. A regulated rail is only useful if the rest of the system can determine whether it is present, in regulation, and suitable for use. Because the low-voltage subsystem supports MCUs, communication hardware, sensors, and user-interface electronics, the power board must operate as a supervised subsystem rather than as a passive supply.

The original PDB concept was based on discrete TI LMR33630 buck converters to generate regulated 12 V, 5 V, and 3.3 V rails [20]. This approach was attractive for several reasons. First, it allowed the power board to remain fully custom and aligned with the team’s early switching-power-supply design work. Second, it followed a familiar buck-converter design process: select the output voltages, size the supporting passive components, implement decoupling, and distribute the resulting rails to the rest of the robot. Third, the expected low-voltage power demand appeared modest relative to the available battery power, making the discrete approach seem practical during the conceptual design stage.

At the conceptual stage, three 3 A regulated rails appeared straightforward. Table 10 summarizes the theoretical full-load contribution of each rail and the resulting total output power if all three rails are loaded simultaneously.

Table 10: Conceptual full-load power by regulated rail

Rail	Voltage	Current	Power
12 V rail	12 V	3 A	$12 \times 3 = 36 \text{ W}$
5 V rail	5 V	3 A	$5 \times 3 = 15 \text{ W}$
3.3 V rail	3.3 V	3 A	$3.3 \times 3 = 9.9 \text{ W}$
Total theoretical output:			60.9 W

Even with conversion losses included, this output remained well below the battery’s continuous power capability. The limiting factor was therefore not battery capacity, but whether the power board could be implemented in a manner that was electrically quiet, stable, manufacturable, testable, and maintainable within a mixed-power mobility system.

This marked the beginning of the design evolution. As the design matured, it became clear that the primary challenge of the [PDB](#) was not the nominal converter calculations, but the physical implementation of multiple switching regulators on a single board. Once the project progressed beyond schematic capture and into board layout, factors such as high- di/dt loop area, return-current control, switch-node containment, grounding strategy, and noise coupling became dominant concerns. A custom multi-rail switching supply would place substantial performance and schedule risk on [Printed Circuit Board \(PCB\)](#) layout quality, particularly in the switching-current loops, return paths, switch-node regions, [EMI](#) behavior, and overall bring-up process. The design question was no longer whether the converters operated correctly in schematic form, but whether the complete board could be laid out, fabricated, debugged, and trusted in a mixed-power system.

This work also made clear that power [PCB](#) design is fundamentally a physical problem, not only a schematic one. Current return under [DC](#) conditions is not equivalent to current return at switching frequencies, and a board can be correct at the netlist level while still performing poorly if switching loops are large, return paths are poorly controlled, or noisy copper couples into sensitive reference or logic regions [21], [22], [23]. This shifted the design focus away from component selection alone and toward current paths, geometry, and noise containment.

A second major design consideration was the need for rail supervision in addition to rail generation. The low-voltage system was required not only to produce nominal voltages, but also to enable or disable individual rails, isolate faults, and indicate whether each rail had reached a valid operating state. That is the practical role of Power Good in this design. Although a Power Good signal does not create power, it provides the rest of the robot with a reliable indication that a rail is present and in regulation. In a system containing controllers, sensors, and communication hardware, that indication is important because downstream subsystems should not be treated as available until their supply is valid. Per-rail enable control is similarly important because it supports staged startup, shutdown of nonessential loads, and localized fault diagnosis. Once these requirements were incorporated, the [PDB](#) was no longer simply a three-output regulator board, but a supervised low-voltage power distribution subsystem.

The design also evolved toward a more deliberate fault philosophy. As the role of the board became clearer, it became evident that low-voltage faults needed to be visible, localizable,

and recoverable during bring-up and later reuse. This led to the inclusion of front-end protection, branch protection, deliberate enable behavior, and serviceable fault isolation, rather than relying on a design that attempted to recover silently from all abnormal conditions. In practice, the [PDB](#) was therefore designed not only for nominal operation, but also for debugging, inspection, and human intervention when faults occurred.

Realistic engineering constraints also shaped the final direction of the design. The low-voltage power system had to be manufacturable within the senior design schedule, testable with the team’s available equipment and experience, and serviceable enough that faults could be diagnosed efficiently. It also had to satisfy broader project requirements related to safety and maintainability. These constraints pushed the design away from the most layout-sensitive fully discrete implementation and toward a more robust architecture in which the highest-risk conversion functions were assigned to integrated modules, while the custom board retained the distribution, supervision, protection, adjustment, and status functions specific to [STRIDE](#).

This design evolution was also informed by standard engineering practice. From a [PCB](#) perspective, the board had to be more than electrically plausible. It had to follow accepted printed-board design practice, particularly with respect to conductor sizing, layout discipline, and electrical spacing. In that context, [IPC-2221C](#) and [IPC-2152](#) provided an appropriate framework for the design problem [24]. The issue was not merely generating 12 V, 5 V, and 3.3 V outputs in schematic form, but doing so with copper widths, thermal performance, spacing, and routing discipline consistent with reliable fabrication and predictable operation. Likewise, the broader [STRIDE](#) requirements related to safety, documentation, accessibility, serviceability, and validated performance were not secondary considerations introduced after the fact. They were active design constraints that directly influenced the final architecture.

The transition away from a fully discrete multi-rail buck-converter implementation should therefore be understood as an engineering decision, not a reduction in engineering scope. It reflected the conclusion that the [PDB](#) would be more robust if the highest-risk conversion functions were implemented with well-characterized power modules, while the custom board design remained focused on the system-specific functions that still required original engineering effort: protected distribution, filtering, supervision, trim adjustment, enable control, Power Good reporting, connectorization, and serviceable fault isolation. Documenting this design evolution is important because it explains not only the final form of the board, but also the reasoning behind the architectural change and why the final implementation is more defensible than the original concept.

3.10 User Interface and Interaction

The original design for the user interface was a three-pronged system, with a physical control, physical feedback, and remote component. The physical component consists of two separate mechanisms; mode control and speed control. The remote component was planned to be an HTML-based web interface, with data transmission handled by the ESP32 (discussed in Section 3.8.3.)

Mode control will only be activated when the rollator is stopped; thus, the team relegated this control to a button panel. A large-button panel was chosen because the tactile nature of buttons makes the system more accessible for people with reduced vision or hand dexterity.

The team decided to use arcade-style buttons because they are large, do not require much pressure to press, provide tactile feedback when pressed, and allow for RGB lighting to indicate system status. The team debated whether to put this button panel at the front of the rollator, where the wiring would be more convenient, or on one of the handles, for easier access. The handle position was decided upon to facilitate easy access to mode control without having to circle around the rollator.

The button panel also provided a convenient place to locate the [E-stop](#), so that the [E-stop](#) control would be visible and accessible from anywhere around the system.

Speed control is composed of the user detection system and the braking system. As the system automatically moves forward when active, there must be a mechanism in place to detect whether the user is in position to begin walking, as well as a mechanism to control the speed and direction of the rollator. The team examined several different potential sensors for this task. Strain gages and force-sensitive resistors were considered as one-device solutions, where the user would be detected via light pressure and be able to control the system through heavier pressure. However, these sensors have a high amount of drift, which could lead to misread commands or drift outside of software detection ranges. Thus, the team decided to use a two-part system for detection and speed control. Detection would be handled with a touch sensor, and speed control would be handled by e-bike brakes, which use a Hall-effect sensor to send a digital-low signal when the brake is pressed.

Physical feedback is an important component of the system for safety and accessibility reasons. To ensure that the system can be operated and is safe for users and bystanders with visual or auditory disabilities, the team designed a system which used both light-based and audio-based cues. To this end, the team specced safety lights to place on both sides of the rollator, the aforementioned arcade buttons, and a speaker to play warning sounds and audio cues. These components would be controlled by the ESP32 as the main user-control system.

The remote terminal is able to perform all of the functions of the physical control system, including mode control and speed control. The team also decided that this terminal would be able to change system metrics; for example, the base speed that the rollator would operate at. Finally, the page would display metrics about the system, like battery percentage. This is facilitated by the ESP32.

3.10.1 User Interface Board Design Evolution

The [User Interface \(UI\)](#) board, also referred to here as the control cluster [PCB](#), was never intended to be a temporary breakout or a convenience adapter. From the beginning of the [STRIDE](#) project, the user-facing control system was expected to rely on large arcade-style buttons because they provided a practical physical interface for users with limited mobility. That decision immediately implied the need for a dedicated control board. The design problem was therefore never simply to connect a few switches and indicator lights to an [MCU](#); instead, it was to create a defined, accessible, and maintainable electrical subsystem that could sit between the user and the rest of the system.

From the earliest concept, the board had to support three major user inputs—Mode, Stair, and Reset—while also providing illuminated feedback that carried actual operating meaning. Because the visible indicators were intended to communicate system state rather than simply confirm that power existed, the design could not remain a loose collection of direct controller

connections. The control-cluster color map, in which green indicates Walk Assist Mode, blue indicates Caregiver Remote Mode, yellow indicates Stair Climbing Mode, and amber indicates Charging or Low Battery, pushed the design toward a more structured output architecture. Once the LEDs were expected to serve as part of the user interface rather than as generic lamps, a dedicated interface board became the more defensible engineering choice.

This evolution was also shaped by the formal expectations placed on the [STRIDE](#) interface. The project requirements explicitly state that the user interface must be accessible to individuals with disabilities, including users with visual or hearing impairments. The same system context also places [STRIDE](#) within the framework of [ISO 11199-2:2021](#) and use on [ADA accessibility requirements Chapter 4](#)-compliant surfaces and environments. Those requirements helped justify the use of large illuminated buttons and reinforced the idea that the control cluster should be treated as a serious subsystem rather than as a minimal control panel. In that sense, the board’s evolution was not merely about cleaning up wiring. It was about translating accessibility expectations into hardware that could provide repeatable inputs, legible visual feedback, and a predictable subsystem boundary [25], [26].

The move to a PCA9685-based architecture addressed several of these needs at once. The PCA9685 was attractive because it was a mature and well-understood LED-driver solution, it provided enough available channels for the control cluster with margin for future growth, and it offered a convenient [I²C](#)-controlled architecture for managing multiple LED outputs from a single device [27]. Just as importantly, it allowed the illuminated-button behavior to be handled through a standard [I²C](#) interface rather than through a large set of direct [MCU](#) outputs. This reduced controller side complexity and allowed the [UI](#) board to behave more like a defined subsystem and less like an assembly of special-case wiring [27].

As the design matured, the board became more intentionally partitioned. Rather than treating the control cluster as one undifferentiated collection of switches and LED wiring, the final architecture separated the logic domain from the LED-power domain. This produced a 3.3 V logic side for controller interaction and a separate 5 V LED side for the illuminated hardware. That separation simplified bench testing, improved integration clarity, and supported the larger architectural idea that the board should form a clean interface boundary between low-voltage logic and the user-facing electromechanical hardware.

The board also evolved in response to maintainability and system-integration concerns. A loose harness approach could have been made to work, but it would have created a less reusable and less supportable assembly. By contrast, a dedicated board made it possible to centralize labeling, define connector roles clearly, and support a more disciplined harnessing scheme. This mattered not only for initial assembly, but also for future troubleshooting and reuse. In keeping with the broader [STRIDE](#) electrical architecture, connector commonality was also treated as a design objective. Output-side interconnects were standardized around JST connectors, while board power input and output connections were standardized around the Molex Micro-Lock Plus family [28]. That commonality reduced unnecessary variation across subsystems and made the [UI](#) board fit more naturally into the overall robot wiring philosophy. It also reinforced the idea that the board should be maintainable and understandable as part of a larger integrated electrical system rather than as a one-off convenience design.

As the team refined the board, another distinction became clearer: the control cluster had to be robust not only electrically, but also procedurally. The board needed to be easy to wire correctly, easy to test on the bench, and resilient to normal handling and vibration

once installed on the rollator. That requirement influenced the move toward dedicated connectors, cleaner silkscreen labeling, partitioned logic and LED power domains, and later, solder-jumper-based configuration rather than loose header shunts. In other words, the board evolved from the idea of “buttons and lights on a PCB” into a deliberately engineered user-interface subsystem.

3.11 Overall System Architecture

The following system diagram, designed in Lucidchart, shows the original configuration planned in the general system:

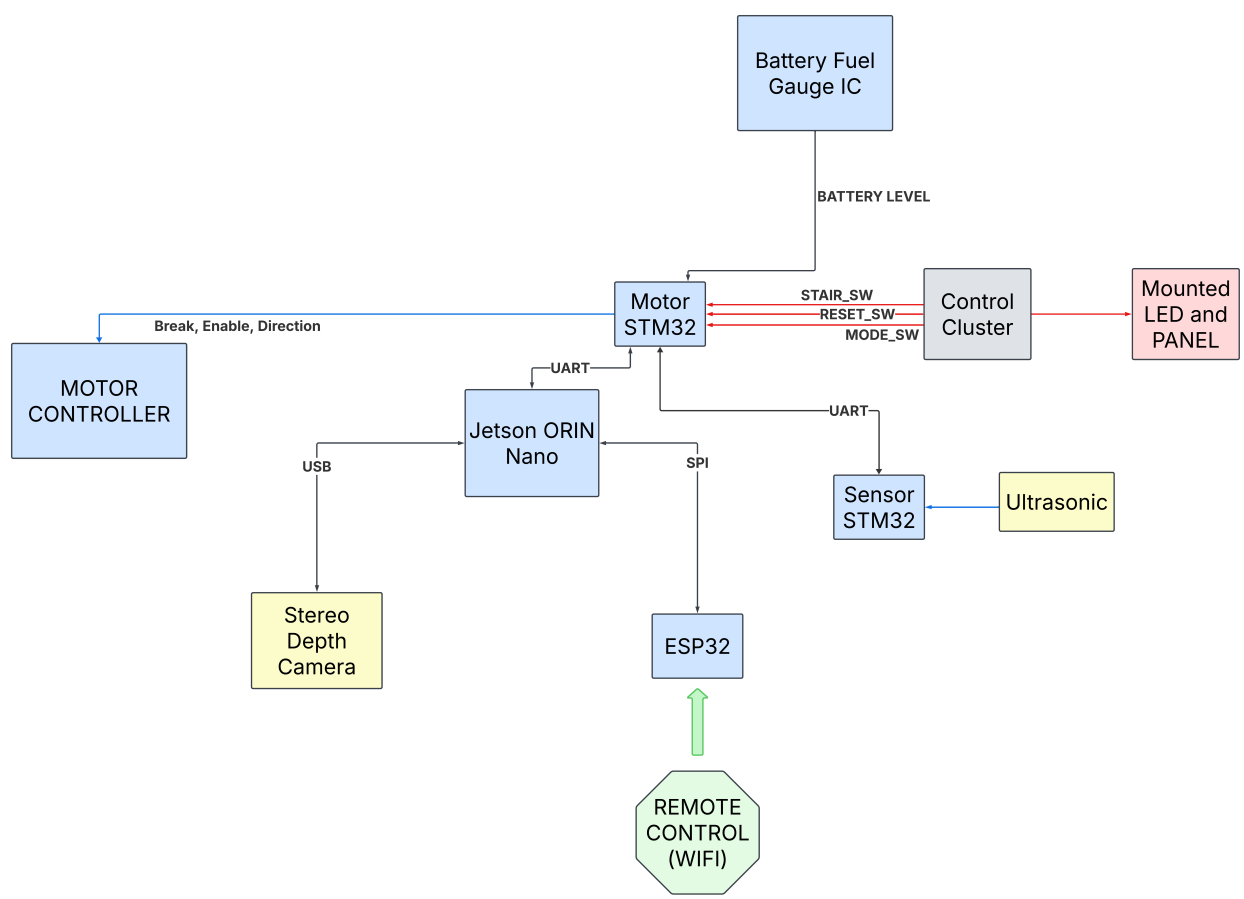


Figure 10: MCUs Communication System v2.

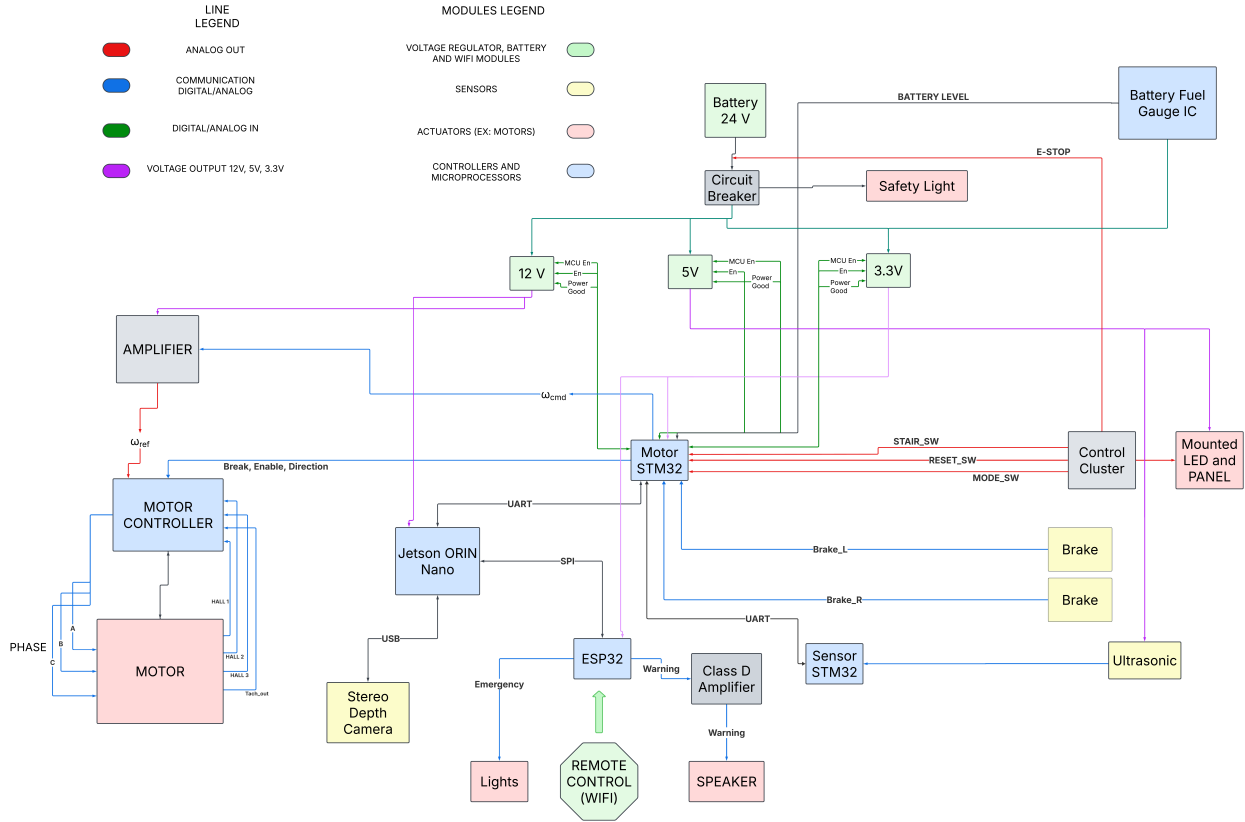


Figure 11: General System Design.

3.11.1 Power Architecture

The output voltages of the [PDB](#) were selected from the actual input requirements of the downstream loads rather than from a desire to maximize the number of available rails. Most of the [STRIDE](#) electronics naturally clustered around standard low-voltage domains, namely 3.3 V for controller and logic functions, 5 V for many sensors and peripheral circuits, and 12 V for higher-power electronics. The main outlier was the Jetson Orin Nano developer-kit platform, which is commonly supplied from a 19 V adapter. However, NVIDIA’s official carrier-board specification states that the carrier board’s main DC input accepts 9 V to 20 V at the DC jack, and the alternate back-power input is likewise specified for 9 V to 20 V. That meant the Jetson Nano could be supported from a regulated 12 V rail without introducing a dedicated 19 V rail solely for one subsystem. Using 12 V for the Jetson Nano therefore reduced conversion redundancy, simplified distribution, and kept the low-voltage power architecture aligned with the rest of the robot while still remaining within the documented input range of the NVIDIA carrier board. Also, given that the load requirements are much more stringent on the 3.3 V and 5 V rails therefore the Jetson Orin Nano has a bigger power margin if needed.

From a safety standpoint, Lafayette’s Electrical Safety Program requires that energized parts be de-energized, locked out, tagged out, and verified as de-energized before work proceeds. The Electrical Safety Plan also states that students are considered unqualified

persons and may not work on or near exposed energized electrical equipment of 50 V or more. It also requires review with a qualified person even for lower-voltage energized work and emphasizes that control switches and interlocks are not substitutes for lockout/tagout. The General Project Requirements likewise frame safety and good practice as formal project obligations and require machinery hazards to be addressed in a plan consistent with OSHA 29 CFR Part 1910 Subpart O. Finally, from a board-design standpoint, the [PDB](#) was developed with explicit attention to accepted printed-board practice, including conductor sizing, layout discipline, and safe spacing, which is exactly why IPC-2221C and IPC-2152 belong in the governing-standards discussion for this subsystem.

3.11.2 Microcontroller Architecture

Based on the design considerations in the previous section, the team decided to set Jetson Orin Nano as the central decision-making [MCU](#). The Jetson Nano is connected to the ESP32 through a physical SPI communication port in addition to an alternative WiFi access point in case the physical port malfunctions.

The ESP32 behaves as a WiFi host, providing the rollator's data directly from the Jetson Nano, such as battery, speed, and driving mode, through a website interface. The [STM32](#) connects both the motor controller and Jetson Nano through a carrier board behaving as a bridge between the [UART](#) communication protocols of both boards.

One of the biggest challenges the team faced was the motor vibration across the rollators, raising concerns on pin connection stability under rollator use. Hence, a modification in the connector's structure was needed across all [MCU](#). To achieve this task, two carrier boards were designed as critical components in the communication systems. The team named this board the ESP32 Wireless Board and the [STM32](#) Carrier Board. In addition to the [PCBs](#), the team also updated the LucidCharts for the specific modules. The final drawing configurations of the ESP32 and [STM32](#) can be seen in the following figures:

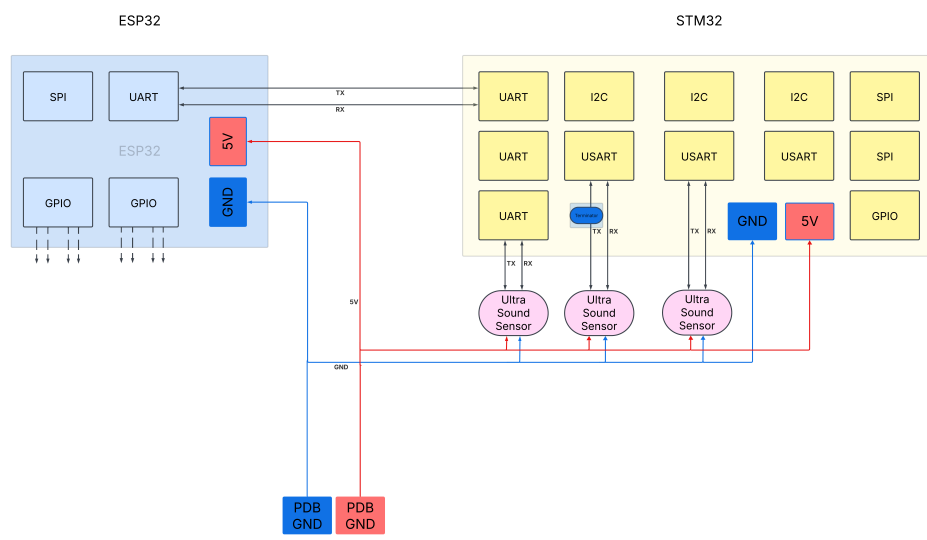


Figure 12: ESP32 and [STM32](#) configuration.

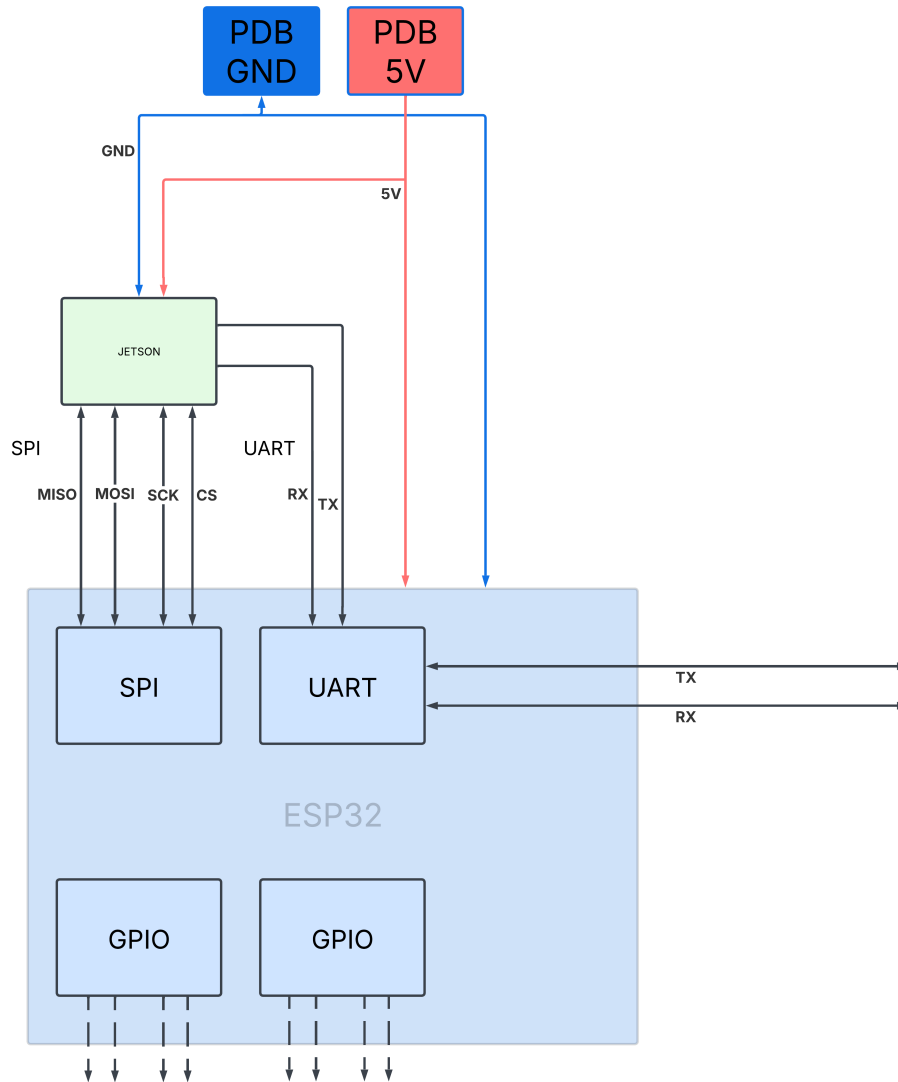


Figure 13: ESP32 configuration detail.

The [STM32](#) Carrier Board was developed to contain 7 [UART](#) connectors capable of connecting all the sensors needed in addition to maintaining communication with the Jetson Nano through an indirect connection with the ESP32 board. The physical SPI communication of the Jetson Nano and ESP32 required a 40-pin configuration in the carrier board, which required the [UART](#) communication protocol in the STM board to be attached to the ESP32 carrier board. The board is shown in [Figure 3.11.2](#).

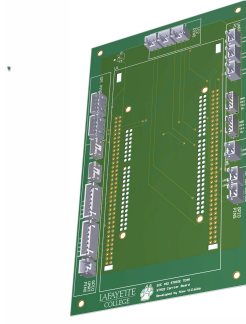


Figure 14: STM32 Carrier Board [PCB](#).

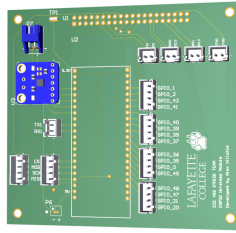


Figure 15: ESP32 Carrier Board [PCB](#).

The ESP32 Wireless Board contains the amplifier module connected to the main ESP32 dev kit with several communication ports such as [GPIO](#), [UART](#), and [SPI](#).

The decision on how the pin connections were set up was developed through the combination of a master pin list detailing the available communication protocol across the [MCUs](#), written by the sensor, microcontroller, and power teams. On top of this document, the team developed a connection design plan where the available pins from each board were connected and used as a reference in the development of the carrier boards. The carrier board is shown in [Figure 3.11.2](#).

As part of the thinking process, the following table displays the available communication protocols and how they connect to each other.

Table 11: Communication protocol connections.

Communication Protocol	ESP32	STM32	Jetson Nano
SPI	1	2	2
UART	2	7	1
I ² C	0	3	2
I ² S	1	0	1

Documentation of the master pin list is available in [Appendix 5](#).

4 Final Design Details

4.1 Frame

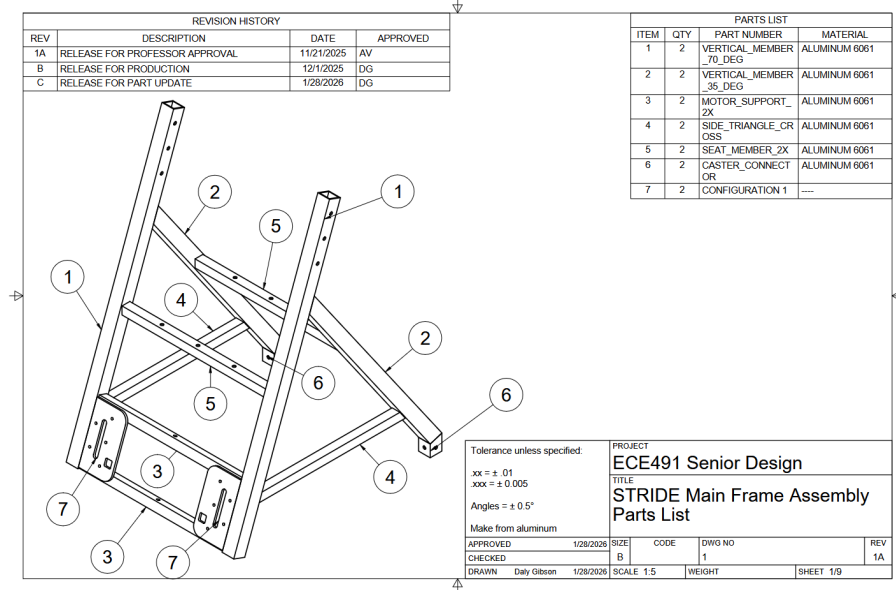


Figure 16: Final design of the structural frame, including structural beams and the finalized motor plates.

The structural frame, shown in Figure 16 is composed of square aluminum-6061 tubing cut and welded together. The design is symmetrical; there are mirrored triangular struts on either side, connected by four lengths of tubing. The triangular struts have attachment points at each vertex to accommodate two tri-wheels and a handle, respectively. This allows for an even distribution of weight across all four total tri-wheels and provides a steady base for level-ground traversal. The aluminum tubing is calculated to experience a maximum tensile stress of 3.8 MPa (full calculations in section 3.5), which is many margins below the maximum rated stress of 70 GPa. At the front of the rollator are two plates specifically designed for the motors to screw into; these are welded to two cross-bars and directly to the triangular struts. This provides a secure base for the motors and through them, the tri-wheels to be attached to the rollator. Two other cross-bars are aligned to provide a solid basis for the seat, as well as peripherals such as the depth camera.

Across the bottom of the horizontal beams is attached a bottom plate, cut from aluminum-6061. The main purpose of this plate is to provide a stable base to attach the battery on. At 16.5 lbs, the battery is the most massive object in the main chassis (excepting the motors, which are attached separately). The battery is ratcheted down using nylon straps strung through two sets of slots cut into the bottom plate. The plate also has a connection grid to allow for the installation of PCBs and MCUs to it through adapting brackets, as well as a grocery basket in the very back. The plate is bolted directly into the main frame.

A similar design is used for the seat plate, which is parallel to the bottom plate. This plate uses a smaller connection grid, as it is designed for smaller PCBs to be attached. On

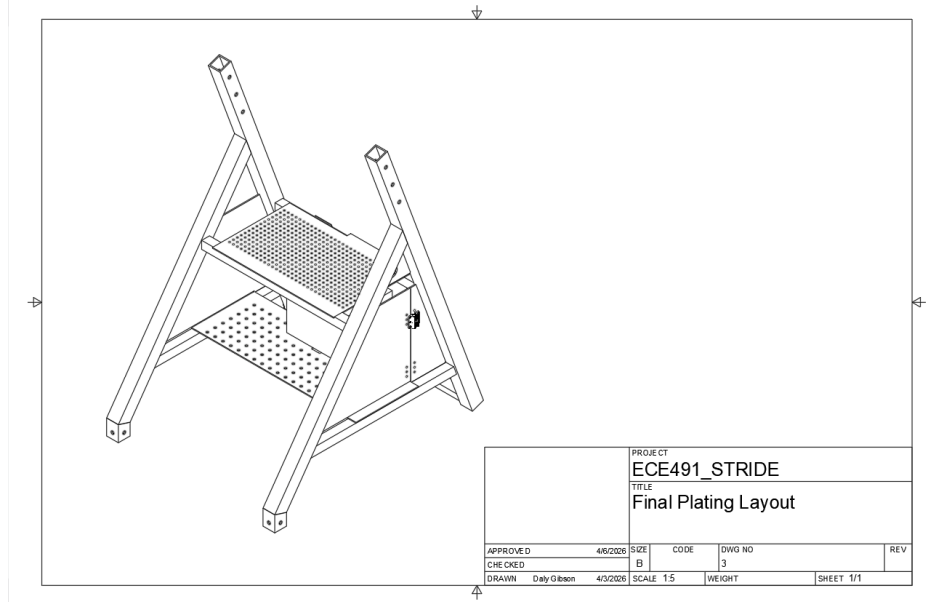


Figure 17: Final Plating Layout

top of this plate is attached a commercial rollator seat made of plastic. This provides a comfortable seat for the user when they wish to take a rest.

Finally, across the triangular struts and cross-beams are fastened laser-cut sheets of acrylic. These housings protect the sensitive electronics in the interior of the rollator. One side of the housing includes a hatch to allow the team access for any debugging or adjustment purposes. The acrylic sheets are attached with high-strength industrial foam tape; avoiding screws mitigates the risk of the acrylic cracking under tension, and allows for easy removal and replacement of the acrylic if necessary.

The handles are composed of steel brackets that can be bolted into any of the three handle attachment points. On these brackets are welded steel bars, which are covered on the end by high-friction rubber handles. The handles are also the mounting point for the brake system, which is the main method of dynamically controlling the rollator's speed. Further detailing of the brake system is provided in Section 4.8.1.

4.2 Wheels

This section presents the final design details of the front wheel, rear wheel, and motor shaft extension. A complete parts list and installation guide are provided in Appendix 6.

4.2.1 Front Wheel Assembly

The triangular side plates are the structural backbone of each tri-wheel assembly. Each assembly uses an outer plate and an inner plate. Both plates are machined from 10 in $\times$ 10 in $\times$ 0.75 in thick aluminum sheet and feature three vertex holes for the sub-wheel shafts and a central hole for the main drive shaft. The three corner mounting holes have a center-to-center distance of 8.4 in between adjacent vertex holes. Each corner of the triangular plate is blended

with an arc of radius 1.28 in. The central hole is placed at the geometric center of the plate. Its location is defined as the intersection of the three medians of the triangle formed by the three vertex hole centers. Figure 18 shows the plate drawings before and after bearing installation. Bearings are press-fit into the plate holes to support the rotating shafts. Unlike conventional wheel designs, the wheels do not contain internal bearings. Instead, the shafts rotate within the plate-mounted bearings, allowing the shafts, wheels, and sprockets to rotate together, while the plates remain stationary relative to each individual axle during normal operation.

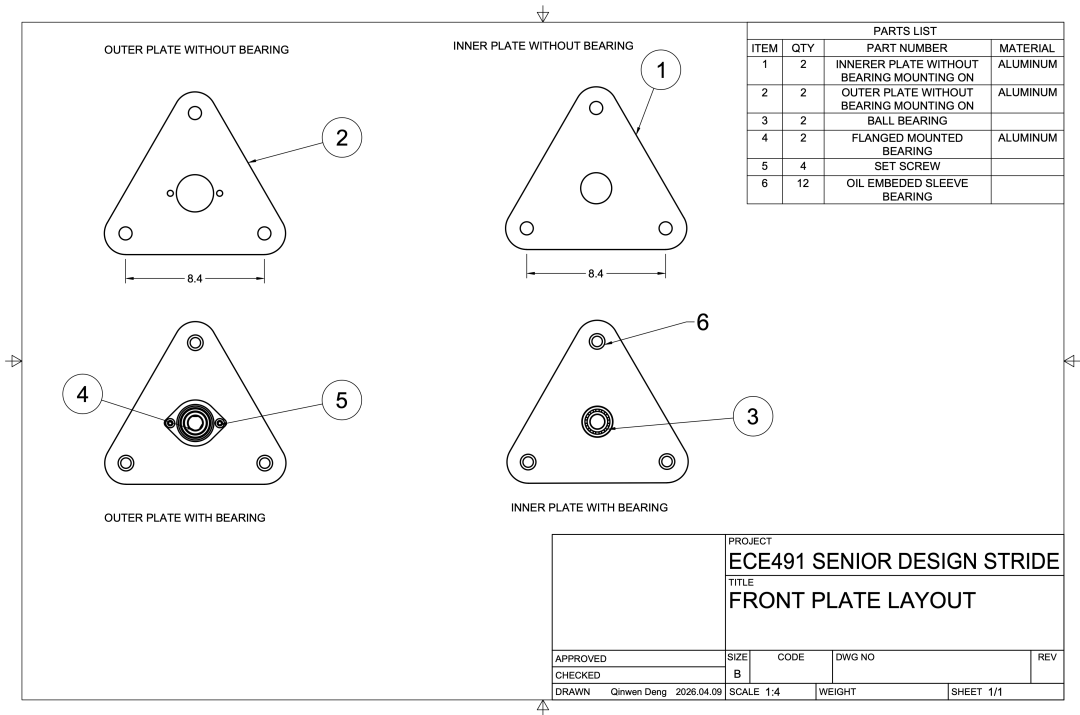


Figure 18: Front Tri Wheel Plate Layout

The outer plate uses a light-duty two-bolt flange-mounted ball bearing (inner diameter $\frac{7}{8}$ in) at the central drive shaft location. The flange bearing is bolted directly to the plate surface, providing a rigid connection that resists both radial loads and moment loads at the central shaft. The inner plate uses a standard press-fit ball bearing at the same location. While a flanged bearing would provide the same structural benefit on both sides, the overall assembly width is constrained to 5.5 in, measured from the outer face of the flange bearing on one side to the outer face of the opposite plate. A second flange bearing would protrude beyond this limit. The standard ball bearing sits flush within the plate thickness, reducing the assembly width to stay within this constraint. Each vertex hole has a press-fit oil-embedded sleeve bearing with an inner diameter of 0.32 in.

Figure 19 presents the final design layout of the tri-wheel assembly, shown through multiple orthographic and isometric views. At the core of the assembly are the inner and outer triangular plates, which provide structural support and alignment for the three-wheel arrangement. Each vertex of the triangular plate supports an 8-inch rubber wheel. Each

wheel has a diameter of 8 in and a width of 2.1in, and is mounted on a 0.32in (16mm) diameter, 5.16in long linear motion shaft that serves as the wheel axle. The spacing between adjacent wheel centers is 0.4in. The wheel shaft ends are machined with external threads (3/8-16 UNC-3A), so that, after the wheels, sprockets, and plates are assembled onto the shaft, locknuts can be installed to secure the components axially.

The drivetrain begins at the motor shaft extension, where a 15-tooth drive sprocket is mounted using a $\frac{3}{16}$ in $\times$ $\frac{3}{16}$ in the keyway and secured with a set screw. This sprocket drives a 30-tooth sprocket mounted on the master wheel axle, which is connected to the shaft via a 5 mm $\times$ 5 mm keyway and retained with a set screw. On the opposite side of the same master axle, a second 30-tooth sprocket is mounted with the same keyed, set-screw-secured interface. This sprocket is connected via #25 roller chain to the remaining orbital wheel sprockets, ensuring that all three wheel axles rotate synchronously. For the roller chain used in this design, the required length is approximately 55 outer links, with an additional offset half link used for final length adjustment. This orbital chain loop was assembled by breaking a 1-foot chain into 9 links and joining it to a 2-foot chain with 2 master links. The driven chain, which connected the 15T drive sprockets on the motor shaft extension to the 30T master wheel sprocket, was cut from a 1-foot chain to 11 links and closed with 2 master links. All chains were cut to length using a chain breaking tool.

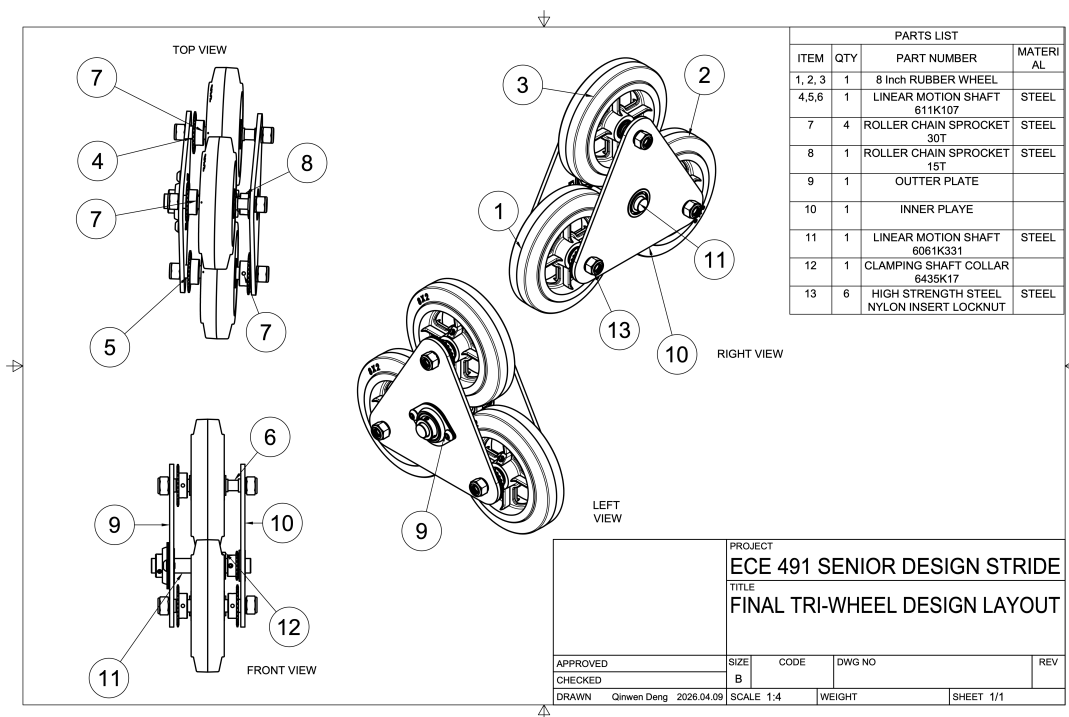


Figure 19: Front Tri-Wheel Design

4.2.2 Rear Tri-Wheel Assembly

The rear tri-wheel assemblies use the same triangular geometry and delta-side length (8.4in) as the front assembly, but differ in three ways: the sub-wheels are 8 in omnidirectional

wheels, instead of rubber wheels, the assemblies are unpowered, and the triangular side plates are replaced with spoke-shaped support plates to reduce weight.

Each rear assembly consists of three 8 in dual-coupled omni-wheels (2 in width) mounted between two spoke-shaped support plates cut from 0.12 in thick 10 in $\times$ 10 in steel plate. The omni wheels rotate freely on three hex head screw axles ($\frac{3}{4}$ in diameter, 3.7 in long, cut down from 4.25 in to satisfy the overall width constraint). Each wheel hub contains an oil-embedded bearing (0.75 in ID $\times$ 1 in OD $\times$ 2 in long) that rides on the smooth shaft of the hex head screw. The axles pass through both support plates and are secured on the far side with $\frac{3}{4}$ in-10 locknuts. Locknuts were used instead of standard hex nuts to prevent loosening under vibration and repeated loading during stair climbing. 1

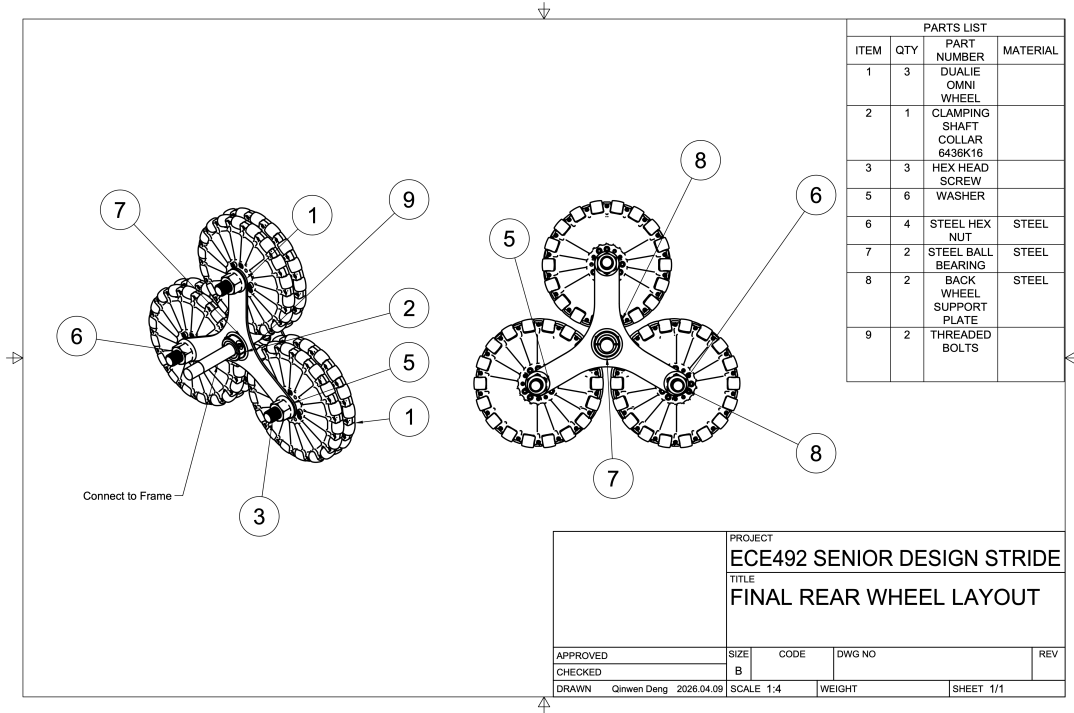


Figure 20: Final Rear Tri-Wheel Assembly Layout.

The assembly rotates about a central shaft (8 in long, $\frac{3}{4}$ in-10, with 6 in unthreaded and 2 in threaded). The unthreaded portion passes through ball bearings ($\frac{3}{4}$ in ID $\times$ $1\frac{3}{8}$ in OD $\times$ $\frac{1}{2}$ in width) press-fit into the center of each support plate. A clamping shaft collar is installed on the central shaft to prevent the ball bearings from sliding axially. The threaded portion extends through the rollator's aluminum frame tube (1.5 in long) and is bolted from the outside to fix the shaft to the frame. Because the central shaft is fixed to the frame, the ball bearings allow the entire tri-wheel assembly to rotate freely about the shaft during stair climbing. Figure 20 shows the final rear wheel layout.

4.2.3 Motor Shaft Extension

The original motor output shaft is a stepped cylindrical shaft consisting of a keyed section and a threaded end. The main shaft body has a diameter of 0.6677 in and extends approximately 1.84 in outward from the motor housing. A 0.2357 in wide, 1.634 in long, and 0.1387 in deep keyway is machined into this section. At the end of the shaft, the diameter is reduced to approximately 0.394 in, forming a threaded portion about 0.525 in long. The thread specification for this end section is $M10 \times 1.25$. [29]

Because the original motor output shaft was too short for the final layout, a custom motor shaft extension was designed to fit over the existing shaft and provide additional usable length. The extension was machined from a 1 in diameter 4140 alloy steel rod. One end of the extension was bored to create an internal cavity matching the diameter of the original motor shaft body, with the bore depth extended by 0.3 in beyond the original engagement length to provide assembly tolerance. A smaller coaxial internal section was then machined and threaded to match the motor shaft's threaded end ($M10 \times 1.25-6H$), allowing the extension to be screwed directly onto the original shaft. The detailed drawings of each part are provided in the Appendix 3.

To further secure the extension, the location corresponding to the original shaft keyway was marked, and two $M6 \times 1$ threaded holes were added to accept $M6 \times 10$ mm set screws. These set screws bear against the keyed section of the original shaft, helping prevent rotation or loosening under load. Figure 21 presents the final design layout of the motor shaft extension.

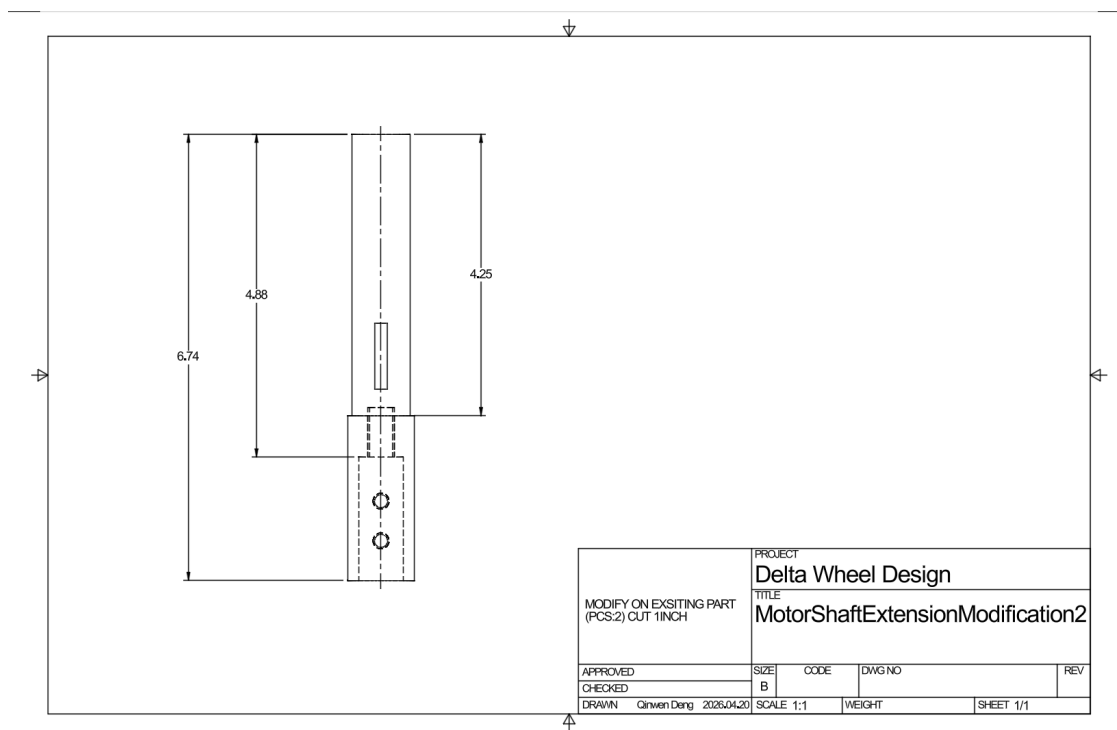


Figure 21: Motor Shaft Extension Design

While the base portion remains the original 1 in diameter, the remaining shaft section,

with a length of 4.25 in, was turned down to a diameter of $\frac{7}{8}$ in so that it could pass through the bearings used in the assembly. The larger 1-in diameter section acts as an integrated collar, which helps prevent the bearing from sliding axially and assists in locating the shaft within the mechanism. There is also a $\frac{3}{16}$ wide $\times$ $\frac{3}{32}$ deep $\times$ 1-in long keyway on this extension to secure the 15T driver sprocket.

4.3 Motor

4.3.1 Direct motor component.

The **STRIDE** system uses two Electrocraft **BLDC** motors, one on either side of the rollator driving the front tri-wheel systems. These motors are coupled with integrated right-angle gearboxes with a gear ratio of 21:1, delivering an output torque of 12.8 Nm continuously and 26.2 Nm at peak. They operate at 24 V DC and consume up to 12 A per motor under large load conditions. They are powered by the 24 V Bioenno **LiFePO₄** battery, which delivers up to 30 A, meeting the supply requirements for both motors. It is important to note that these motors are not powered directly from the power source but from the high-power 3-phase region of the **EZ DRIVE**. However, the electromagnetic brakes on these motors are powered directly from the power supply. The full connection diagram detailing this connection is shown in Figure 22.

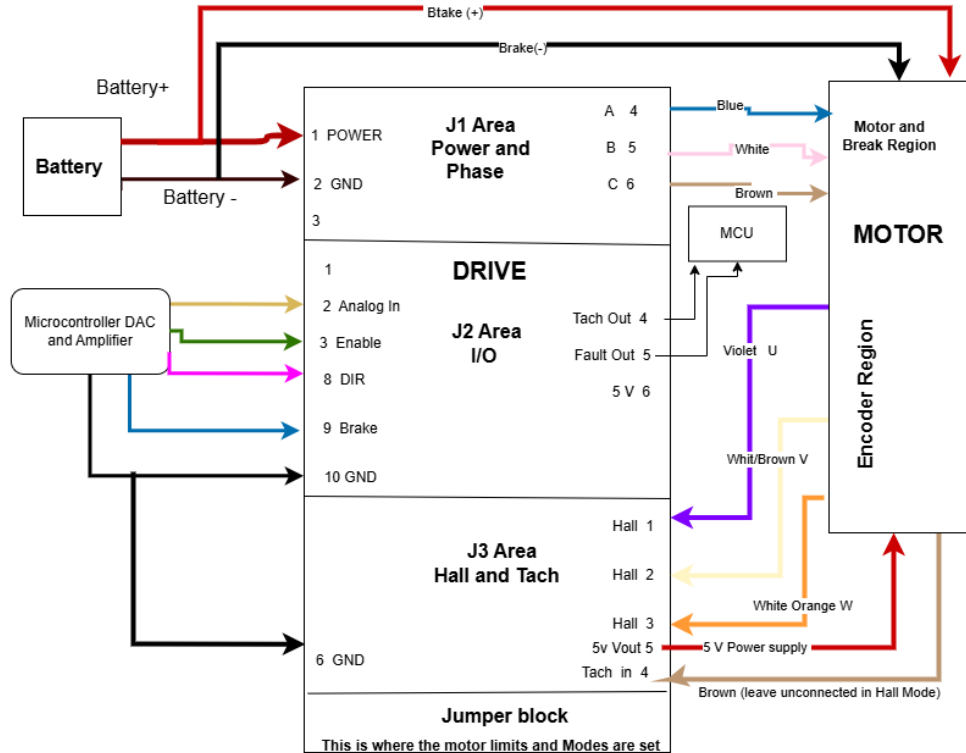


Figure 22: Motor Connection Diagram.

4.3.2 Motor control design.

At a high level, control begins at the Jetson NANO and trickles down the hierarchy to the motor [STM32](#), which generates control signals based on user input, system state, and sensor feedback. These signals are passed to a custom-designed motor controller interface board, before being sent to the ElectroCraft [EZ DRIVE](#). The [EZ DRIVE](#) then performs a power conversion and commutation required to drive the [BLDC](#) motors. This layered architecture separates low-power control logic from high-power motor actuation and was defined very early in the project design phase.

The [STM32](#) board generates both analog and digital signals required by the motor drive system. The voltage signal produced using the [MCU](#)'s internal [DAC](#), represents the desired motor speed. Its maximum value is 3.3 V but is amplified by the motor control board to a maximum of 10 V, representing the maximum 3 mph speed. The motor controller sets a maximum speed of 180 rpm; that maximum speed is by default mapped to a 10 V input signal. If the 3.3 V input signal from the [STM32](#) was used, the motor will not achieve the high rpm required to hit the 3 mph requirement. The digital outputs control functions such as enable, direction, and braking. These signals are derived from higher-level decision-making processes, including user input, obstacle detection, and battery status monitoring. The controller also checks for fault and speed inputs from the motor control board and sends the information to the Jetson Nano for processing.

The complete microcontroller to motor routing details are available in [Appendix 5](#).

4.4 Power

4.4.1 Battery

The [STRIDE](#) system uses a centralized 24 V power architecture from which all other electrical subsystems derive their required supplies. The main specifications used in battery selection were the required bus voltage, set primarily by the traction motors at 24 V, and the estimated peak system power, calculated at approximately 500 W under worst-case operating conditions. These calculations are provided in [Appendix 4](#).

The selected battery is a commercially manufactured Bioenno BLF-2430A 24 V [LiFePO₄](#) pack rated at 30 Ah. This choice is consistent with the [STRIDE](#) requirement that all electrical systems be powered by a single rechargeable battery and that the battery pack and battery-management hardware be commercially manufactured. The battery manual lists a nominal voltage of 24 V, a capacity of 30 Ah, a maximum continuous discharge current of 30 A, and internal BMS protection for overcharge, overdischarge, overcurrent, temperature, and cell balancing. At 24 V and 30 A, the battery can supply up to approximately 720 W continuously, which provides a margin above the estimated 500 W system requirement.

Normal operation was planned around the duty-cycle assumptions in the [STRIDE](#) requirements, namely, extended operation with intermittent active drive rather than continuous peak-load stair climbing. Before stair-climbing operation, the battery is intended to be charged to full capacity. This operating practice is also supported by the battery manual, which recommends full charging for normal use and notes that [LiFePO₄](#) balancing occurs at the top end of charge. The same manual also warns that terminal voltage should not be

treated as a reliable indicator of remaining charge over most of the discharge cycle, which motivated the inclusion of explicit battery monitoring rather than relying on voltage alone.

4.4.2 Final Power Distribution Board Design

In the final **PDB** architecture, the regulated rails remain 12 V, 5 V, and 3.3 V, but the board no longer relies on fully custom buck stages for each rail. Instead, the **PDB** uses integrated **DC-DC** power modules from the TDK-Lambda i6A4W family for the primary conversion stages and concentrates the custom design effort on the parts of the power system that remained unique to **STRIDE**: protected front-end power entry, input and output filtering, supervised rail control, trim adjustment, connectorization, Power Good reporting, and serviceable fault isolation [30]. This shift reduced implementation risk in the most layout-sensitive portions of the design while preserving the system-level behavior that the project still required.

That distinction is important. In the final board, the engineering effort is no longer centered on proving that a discrete buck regulator can be made to function. Instead, it is centered on making the low-voltage power system behave predictably and safely as part of the robot. The **PDB** is therefore not just a voltage-generation board. It is the supervised low-voltage distribution and monitoring node for the rest of the **STRIDE** electronics. The board receives raw battery power, conditions and distributes that power into the required rails, provides rail enable control, reports rail status through Power Good signals, and exposes the resulting power system in a way that is testable and maintainable at the subsystem level. Figure 23 summarizes the functional organization of the final **PDB**, including protected input handling, module-based rail generation, supervised enable and Power Good behavior, trim access, and fused low-voltage distribution.

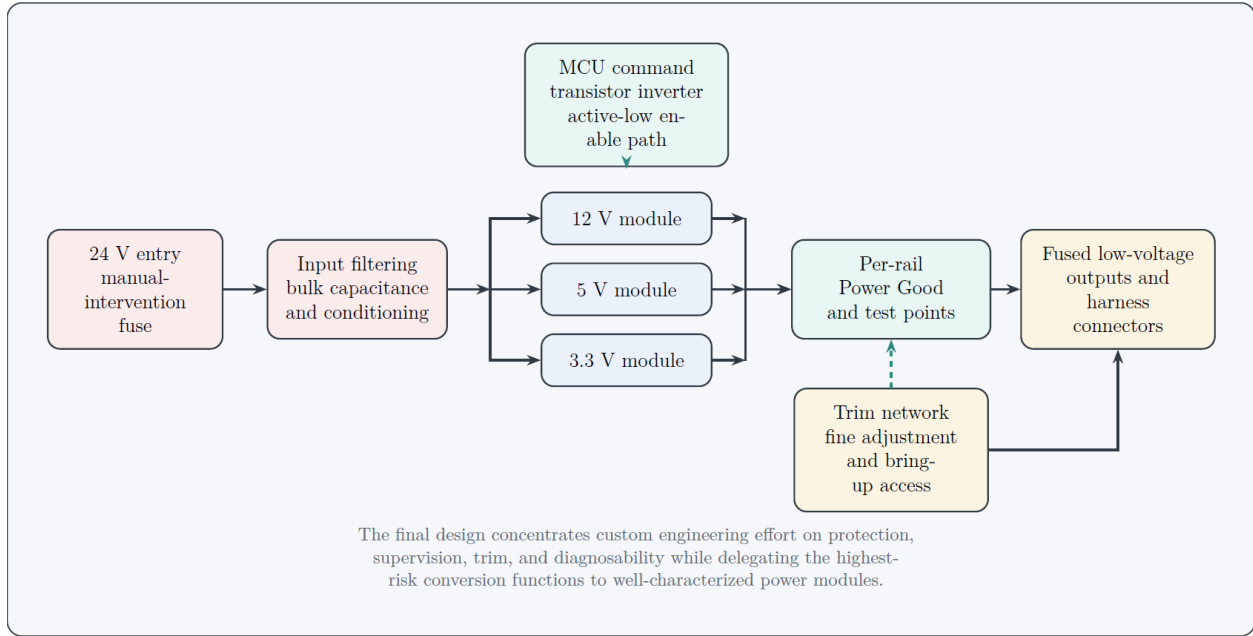


Figure 23: Functional organization of the final [PDB](#), showing protected 24 V entry, input conditioning, module-based generation of the 12 V, 5 V, and 3.3 V rails, transistor-inverted enable control, per-rail Power Good monitoring, trim and bring-up access, and fused low-voltage outputs.

The front end of the final design was chosen deliberately to favor protection and diagnosability over convenience. Input power from the battery passes first through dedicated protective and filtering elements at the [PDB](#) input, including bulk capacitance to buffer load transients on the 24 V line. A key part of this front-end strategy is the use of a fuse that requires deliberate human intervention if it opens. In other words, the board does not simply self-recover from every major upstream fault. If the input fuse blows, that event is intended to force inspection and corrective action before the board is returned to service. This is an important systems-level choice because it prevents repeated fault cycling and makes serious input-side problems more visible during debugging and maintenance. The power-entry stage therefore serves both an electrical role and a procedural one: it protects the board while also making severe faults explicit.

On the output side, the regulated rails are distributed through insulated harnesses and appropriately rated connectors sized to the expected load current and associated protection devices. Each branch is protected so that a sustained downstream fault can be isolated to the affected rail instead of collapsing the entire low-voltage system. Where appropriate, slow-blow behavior was favored so that normal startup surges and capacitor-charging transients would not produce nuisance trips. This was as much a serviceability decision as a protection decision. A manually replaceable fuse makes a fault visible to the operator and forces inspection before re-energization, which is preferable in this application to allowing an intermittent fault to cycle repeatedly without diagnosis. The final board also emphasizes connector commonality and field serviceability. Rather than scattering raw-bus wiring and conversion hardware throughout the robot, the [PDB](#) centralizes the low-voltage interfaces into a board that can

be inspected, tested, and replaced as a defined subsystem. This improves wiring clarity and helps keep the distinction between the battery domain, the supervised low-voltage rails, and downstream electronics unambiguous.

The supervised nature of the final design is one of its most important differences from the original concept. Each regulated rail is intended not only to exist, but to be controllable and observable. Rail enable control allows the system to deliberately bring rails up and down, isolate faults, and perform staged testing during bench bring-up. In the final implementation, this control is not handled as a direct logic pass-through. Instead, the enable structure uses a transistor-based inverter so that the controller can command rail behavior in the required logic sense while still interfacing cleanly with the module enable pins. That inversion stage is a small circuit detail, but it matters at the system level because it allows the board’s control philosophy to match the rest of the robot rather than forcing awkward firmware workarounds or unsafe assumptions about default states. It also means the enable path is part of the engineered supervisory structure of the board, not just a convenience feature appended to the module inputs.

Power Good serves the complementary role. A Power Good signal does not create power, but it gives the rest of the robot a trustworthy indication that a rail is alive and in regulation [31]. For a mixed system containing controllers, sensors, communication hardware, and user-interface electronics, that matters because downstream subsystems should not be treated as available simply because a battery is connected. They should be treated as available only after their supply rail is valid. In that sense, the final PDB behaves less like a passive supply board and more like a supervised low-voltage infrastructure board. The combination of per-rail enable control and per-rail Power Good monitoring gives the overall system a much more disciplined relationship to power-up, shutdown, and fault handling than a simple always-on regulator board would provide. The final board also incorporates trim and adjustment features so that each rail can be set and verified deliberately rather than treated as a fixed black-box output. This matters because the board was not intended only to power a single frozen configuration of electronics. It was intended to remain testable and adaptable as the larger STRIDE electrical system matured. In practice, that meant using calculated trim networks and limited post-assembly adjustment so that the 12 V, 5 V, and 3.3 V rails could be brought close to their intended targets without turning the board into an opaque module carrier. Together with test points and board-level access to the major nets, these adjustment and monitoring features make the board much more transparent during bring-up than a minimally exposed power-module carrier would be.

Since the full-load theoretical PDB output is 60.9 W for the original 3 A-per-rail design case, the corresponding ideal input current from a 24 V source would be

$$I_{\text{in,ideal}} = \frac{60.9}{24} \approx 2.54 \text{ A}$$

before accounting for conversion loss. Even after allowing for inefficiency and some transient headroom, this remains well below the battery’s 30 A continuous limit. That gives useful design margin while still justifying careful protection and filtering, especially because sensitive digital electronics can be disrupted by short disturbances even when average power consumption is modest.

A more realistic first-order efficiency estimate was taken from the datasheet efficiency

curves for the selected converter modules at $V_{\text{in}} = 24 \text{ V}$ [30]. Using

$$P_{\text{in}} = \frac{P_{\text{out}}}{\eta} \quad \text{and} \quad P_{\text{loss}} = P_{\text{in}} - P_{\text{out}},$$

the estimated per-rail input power and conversion losses can be calculated for both loading conditions.

For the 3 A case,

$$P_{\text{out}_{12}} = 12 \times 3 = 36 \text{ W}, \quad P_{\text{in}_{12}} = \frac{36}{0.955} \approx 37.70 \text{ W}, \quad P_{\text{loss}_{12}} \approx 1.70 \text{ W},$$

$$P_{\text{out}_5} = 5 \times 3 = 15 \text{ W}, \quad P_{\text{in}_5} = \frac{15}{0.905} \approx 16.57 \text{ W}, \quad P_{\text{loss}_5} \approx 1.57 \text{ W},$$

$$P_{\text{out}_{3.3}} = 3.3 \times 3 = 9.9 \text{ W}, \quad P_{\text{in}_{3.3}} = \frac{9.9}{0.875} \approx 11.31 \text{ W}, \quad P_{\text{loss}_{3.3}} \approx 1.41 \text{ W}.$$

Thus, for the 3 A case,

$$P_{\text{out}_{tot}} = P_{\text{out}_{12}} + P_{\text{out}_5} + P_{\text{out}_{3.3}} = 36 + 15 + 9.9 = 60.9 \text{ W},$$

$$P_{\text{in}_{tot}} = P_{\text{in}_{12}} + P_{\text{in}_5} + P_{\text{in}_{3.3}} \approx 65.59 \text{ W},$$

and

$$P_{\text{loss}_{tot}} = P_{\text{loss}_{12}} + P_{\text{loss}_5} + P_{\text{loss}_{3.3}} \approx 4.69 \text{ W}.$$

At a 24 V input bus, this corresponds to an estimated input current of

$$I_{\text{in}_{tot}} = \frac{P_{\text{in}_{tot}}}{V_{\text{in}}} = \frac{65.59}{24} \approx 2.73 \text{ A}.$$

For the 4 A case,

$$P_{\text{out}_{12}} = 12 \times 4 = 48 \text{ W}, \quad P_{\text{in}_{12}} = \frac{48}{0.96} = 50.00 \text{ W}, \quad P_{\text{loss}_{12}} = 2.00 \text{ W},$$

$$P_{\text{out}_5} = 5 \times 4 = 20 \text{ W}, \quad P_{\text{in}_5} = \frac{20}{0.925} \approx 21.62 \text{ W}, \quad P_{\text{loss}_5} \approx 1.62 \text{ W},$$

$$P_{\text{out}_{3.3}} = 3.3 \times 4 = 13.2 \text{ W}, \quad P_{\text{in}_{3.3}} = \frac{13.2}{0.875} \approx 15.09 \text{ W}, \quad P_{\text{loss}_{3.3}} \approx 1.89 \text{ W}.$$

Thus, for the 4 A case,

$$P_{\text{out}_{tot}} = P_{\text{out}_{12}} + P_{\text{out}_5} + P_{\text{out}_{3.3}} = 48 + 20 + 13.2 = 81.2 \text{ W},$$

$$P_{\text{in}_{tot}} = P_{\text{in}_{12}} + P_{\text{in}_5} + P_{\text{in}_{3.3}} \approx 86.71 \text{ W},$$

and

$$P_{\text{loss}_{tot}} = P_{\text{loss}_{12}} + P_{\text{loss}_5} + P_{\text{loss}_{3.3}} \approx 5.51 \text{ W}.$$

At a 24 V input bus, this corresponds to an estimated input current of

$$I_{\text{in}_{tot}} = \frac{P_{\text{in}_{tot}}}{V_{\text{in}}} = \frac{86.71}{24} \approx 3.61 \text{ A}.$$

Table 12 summarizes the estimated per-rail conversion results for both loading cases, while Table 13 summarizes the corresponding total PDB performance.

Table 12: Per-rail PDB conversion estimates at $V_{\text{in}} = 24 \text{ V}$

Rail	Current	η	P_{in}	P_{out}	P_{loss}
12 V rail	3 A	95.5%	37.70 W	36.0 W	1.70 W
5 V rail	3 A	90.5%	16.57 W	15.0 W	1.57 W
3.3 V rail	3 A	87.5%	11.31 W	9.9 W	1.41 W
12 V rail	4 A	96.0%	50.00 W	48.00 W	2.00 W
5 V rail	4 A	92.5%	21.62 W	20.00 W	1.62 W
3.3 V rail	4 A	87.5%	15.09 W	13.2 W	1.89 W

Table 13: Total PDB conversion estimates at $V_{\text{in}} = 24 \text{ V}$

Case	$\sum P_{\text{out},i}$	$\sum P_{\text{in},i}$	$\sum P_{\text{loss},i}$	$I_{\text{in,tot}}$	η_{eff}
3 A per rail	60.9 W	65.59 W	4.69 W	2.73 A	92.9%
4 A per rail	81.2 W	86.71 W	5.51 W	3.61 A	93.6%

These results show that even under the heavier 4 A loading condition, the estimated low-voltage conversion burden remains modest relative to the battery’s 30 A continuous current capability. They also show that the expected converter losses are only on the order of a few watts total, which is useful both for power budgeting and for estimating the thermal load that must be managed on the PDB. More importantly, they show that the final board’s main challenge is not the raw power level itself, but the requirement to generate, supervise, distribute, and validate multiple low-voltage rails in a way that is electrically robust and operationally transparent to the rest of the robot.

4.4.3 Battery Monitoring Board

The STRIDE system is powered by a 24 V LiFePO₄ battery pack, which requires continuous monitoring to prevent unsafe operating conditions and to provide accurate information about remaining energy capacity. Since the BMS of this battery pack is internal and does not allow for any external read, a battery status monitoring system is included in STRIDE to enable real-time measurement of voltage, current, and state of charge. Electrically, the system measures currents up to no more than 30 A at 24 V while maintaining a measurement error below 10%. It also operates with 3.3 V logic to remain compatible with the STM32 MCU platform. Functionally, the circuit provides real-time current sensing, voltage monitoring, and state-of-charge estimation, while communicating efficiently with the MCU through I²C protocol.

The circuit consists of a $1\text{ m}\Omega$ 100W shunt resistor placed in series with the battery and load so that a current flowing through the system develops a small voltage drop across the resistor's terminals. This drop is directly proportional to the voltage according to Ohm's law. An LTC2944 battery gas gauge chip is placed across the resistor to measure this differential voltage and current. Figure 4.4.3 shows the charge monitoring circuitry with the battery gas gauge chip (LTC2944).

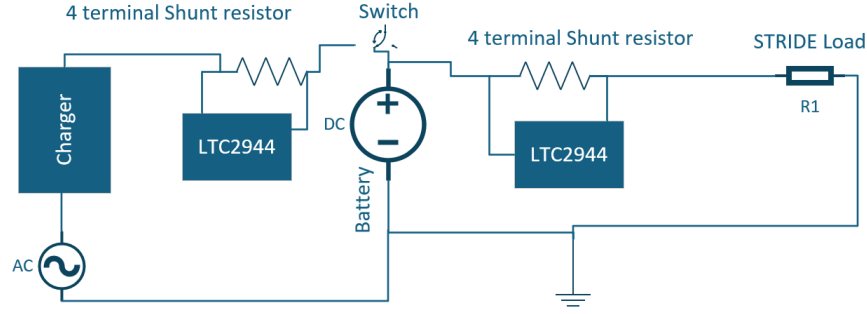


Figure 24: Battery Monitor Circuit Diagram.

The LTC2944 chip measures the 24 V supply voltage and the current traveling in the system. It has a precision analog Coulomb counter that integrates the current through the sense resistor between the battery's positive terminal and the load or charger. Battery voltage, battery current, and silicon temperature are measured with an internal ADC and these values are written to registers on the chip. The system MCU then directly reads these registers over a two-wire interface via the I²C protocol and sends this information up the hierarchy to the Jetson or uses it locally in the backup logic scenario.

Given that the Bioenno BLF-2430A 24 V LiFePO₄ has separate physical connections for charge and discharge paths, the STRIDE system uses two different battery status monitoring boards for each path. The discharging board measures the accumulated charge over the discharge period while the charging board measures the accumulated charge over the charging period when connected to a charger. The difference between these two is added to the previously available charge on the battery and that becomes the new available capacity. The detailed design principle and history is found in Appendix 18.

The board provides a 3.3 V regulated output for the I²C pull up resistors with a 24 V input. The board turns on with a bright red glowing LED with a separate bright red LED to alert the user of overvoltage and overcurrent threshold violations. Though the board is expected to dissipate a maximum power of approximately 1 W, the on-board shunt resistor is heat-sunk to improve the thermal dissipation. The system MCU communicates with this board with a 2-pin 2.54 mm JST-XH connection housing with Serial Data (SDA) and Serial Clock (SCL) lines. The detailed design of the board can be found in Appendix 5. The library to read and write to the board was developed by the team and is located in Appendix 7.

4.5 Processors

4.5.1 Jetson Orin NANO

The Jetson Orin Nano is a compact embedded platform featuring an NVIDIA Ampere GPU with [CUDA](#) support, enabling efficient parallel processing required by the depth camera. It runs a Linux-based operating system, which simplifies environment setup, software integration, and debugging. [CUDA](#) compatibility is particularly important, as it allows direct support for the depth camera’s [SDK](#) and accelerates vision-related processing.

In this system, the Jetson Nano acts as the main “watchdog” and decision-making unit. It collects and processes all incoming data, including motor speed, battery status, signals from the control cluster, braking inputs, server communication, and depth camera outputs. Based on this data, it generates speed commands and sends them to the Motor [STM32](#). (Detailed decision logic and implementation are described in the Software Section.) The Jetson Nano communicates with the Motor [STM32](#) via [UART](#), the ESP32 via [Serial Peripheral Interface \(SPI\)](#), and the depth camera via [USB](#), forming the central hub of the system.

4.5.2 STM32

The [STRIDE](#) system uses the STM32H7A3ZI-Q, which features a single-core ARM Cortex-M7 processor operating at up to 280 MHz, with approximately 2 MB of flash memory. It includes high-resolution [ADCs](#) (up to 16-bit) and two 12-bit [DAC](#) peripherals for analog interfacing. The [MCU](#) provides multiple communication interfaces, including several [UART](#), [I²C](#), and [SPI](#) peripherals, as well as over 100 [GPIO](#) pins. It also integrates advanced timer peripherals for precise control applications and operates within a voltage range of approximately 1.7 V to 3.6 V.

The [STRIDE](#) system has 2 sensor [MCU](#) boards and a system [MCU](#) board. The sensors [MCU](#) boards read and write data and instructions to and from the ultrasonic sensors. The system [MCU](#) board performs all the tasks related to system level-low write functions. These functions include writing motor controller signals, reading battery measurements, getting sensor values from the sensor board, writing to the control cluster, reading user brakes, and receiving and sending information to the Jetson Nano board. Since this [MCU](#) handles most of the peripheral devices, it is referred to as the System [MCU](#) throughout the scope of this document and all calls to that name should point to this specific device. The Jetson Nano is essentially the brain of the system and the system [MCU](#) is the backbone, in that the Jetson Nano makes the decisions and passes those decisions down to the [MCU](#).

4.5.3 ESP32

The ESP32 is a self-contained System-on-Chip that includes built-in Wi-Fi and dual-mode Bluetooth. It can enable any [MCU](#) to connect to a Wi-Fi network, such as the case of the Jetson Nano. This is used because it can run applications independently while maintaining a reliable connection. The 8MB flash drive memory allowed the ESP32 to contain sound instructions that were deployed through a digital-to-analog [I²C](#) protocol meant to connect a mono audio amplifier (MAX98357A) to an off-the-shelf 4-ohm speaker.

The ESP32 serves as a communication interface between the user and the Jetson Nano. The main communication protocol between the wireless communications between the rollator and mobile phone is through a Wi-Fi access point. On top of that, this module contains several extra communication protocols as alternatives to communication with the Jetson Nano, such as [SPI](#), [UART](#), and standard [GPIO](#) that could be used if needed.

The Wi-Fi functionality was used on the ESP32 to facilitate remote commands. Commands are sent from a smartphone over a local Wi-Fi network and are received by the ESP32. These commands are then passed to the Jetson Nano, which interprets and sends them to the motor driver circuit through a [UART](#) communication protocol embedded in the ESP32 carrier board.

4.6 Sensors

4.6.1 Depth Camera

In the final design, the depth camera is used as the primary sensing modality. The system leverages the ZED depth camera to generate a dense 3D point cloud, which provides per-pixel spatial information (X , Y , Z in millimeters). This allows the rollator to move beyond simple distance sensing and instead interpret the geometric structure of its environment. Rather than relying on sparse measurements, the system evaluates thousands of sampled 3D points per frame, enabling robust detection of ground, walls, and obstacles simultaneously. The codes are stored in GitHub repository [7](#).

A key design decision in the implementation is the use of explicit detection constraints derived from earlier system-level analysis. For example, the obstacle detection range is limited to 4 m, which directly corresponds to the previously calculated emergency braking distance. Similarly, a larger bound (8 m) is used to filter out irrelevant distant points for ground and environmental understanding, as well as relieving the pressure of computational resources. The system also applies a horizontal Region of Interest constraint using `ROI_HALF_WIDTH_MM = 406.4f`, effectively restricting obstacle and ground detection to a 32-inch-wide corridor in front of the rollator (aligned with ADA doorway requirements). These constraints ensure that computation is focused only on actionable regions, improving both efficiency and reliability. At the core of the algorithm is the computation of Euclidean distance for each 3D point:

$$\text{dist} = \sqrt{X^2 + Y^2 + Z^2}$$

This distance metric is used to determine whether a point lies within the critical safety zone. Points that fall within the obstacle range (i.e. `dist < 4000mm`) and are not classified as ground or walls are labeled as obstacles. This simple yet effective formulation allows the system to directly map raw depth data into safety-critical decisions. Additionally, invalid or noisy measurements (e.g., non-finite depth values) are filtered out early in the pipeline to maintain robustness.

To improve classification accuracy, the system incorporates plane detection for both the floor and surrounding walls. The floor is detected using the built-in `findFloorPlane` function, and points close to this plane (within a 3cm threshold) are classified as ground and excluded from obstacle consideration. This prevents false positives caused by uneven surfaces or depth noise. Similarly, left and right walls are detected using sampled regions and



Figure 25: Depth Camera Color Coded Obstacle Detection

plane fitting `detectWallPlane`. By calculating the perpendicular distance from each point to the detected wall planes, the system can identify and ignore structural boundaries such as hallway walls. This is particularly important for integrating with ultrasonic sensors, as it prevents redundant or misleading detections in narrow environments.

Another important aspect of the implementation is computational optimization. Instead of processing every pixel, the algorithm uses a downsampling strategy `SAMPLE_STEP = 2`, reducing the number of evaluated points while maintaining sufficient spatial resolution. This significantly improves real-time performance without sacrificing detection accuracy. Combined with `ROI` filtering and early rejection of irrelevant points, the system achieves efficient processing suitable for deployment on an embedded platform such as the Jetson Nano. Runtime measurements (FPS calculation in the loop) further support performance validation and highlight potential limitations under dynamic conditions.

4.6.2 Ultrasonic Sensors

The A02YYUW ultrasonic sensor was selected based on the requirement of obstacle detection. With a 60-degree beam width, a single ultrasonic sensor is used to detect a wide range at the front of the rollator. Two ultrasonic sensors are placed at the back, which serve as the backward obstacle detection sensors when the rollator mode is set to remote control mode. The sensor outputs 1 byte, `Header`, `Data_H`, `Data_L`, and a checksum byte at a baud rate of 9600 `BPS` with a 100 ms response time without filtering. The distance formula is:

$$\text{Distance} = \text{Data_H} \times 256 + \text{Data_L} \quad (1)$$

The checksum is used to verify whether the received frame is valid before the distance value is accepted by the `MCU`. Specifically, the `MCU` adds the first three bytes of the frame

— namely the header, `Data_H`, and `Data_L` — and then keeps only the lowest 8 bits of the result:

$$\text{Sum} = (\text{Header} + \text{Data_H} + \text{Data_L}) \& 0\text{x}00\text{FF} \quad (2)$$

This calculated checksum is then compared with the checksum byte transmitted by the sensor. If the two values match, the frame is considered valid, and the distance measurement is accepted. If they do not match, the frame is treated as corrupted and discarded. In this way, the checksum provides a simple error-detection method for identifying corrupted serial data caused by communication noise or transmission faults.

During operation, however, ultrasonic measurements can still be affected by environmental echoes, multipath reflections, and mechanical vibrations transmitted through the rollator frame. These effects reduce the reliability of the raw distance readings. To mitigate the inherent noise characteristics of ultrasonic distance sensors, a motion-aware asymmetric nonlinear lowpass filter was implemented to condition raw sensor readings before they are passed to the decision logic. The filter applies direction-dependent exponential moving average smoothing, tracking distance decreases to preserve the latency sensitivity required for real-time obstacle detection, while attenuating distance increases through a slower smoothing coefficient combined with a spike confirmation mechanism that requires large positive jumps to persist across a configurable number of consecutive samples before being accepted as valid. Sensor read failures are handled by treating erroneous samples as no-ops, ensuring the last valid filtered output is always retained. This design balances responsiveness to genuine environmental changes with robustness against transient measurement artifacts that are common in ultrasonic sensing under dynamic conditions. See figure 26 for a visual representation of this process.

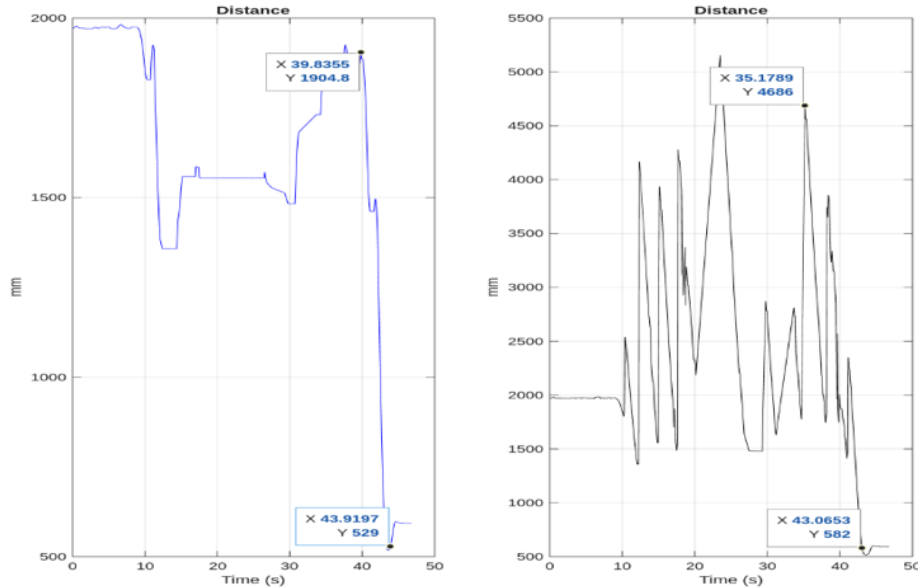


Figure 26: Ultrasonic Sensor Output Waveform Filtering

4.7 Software

The Jetson Orin Nano runs a customized Linux-based operating system, providing a flexible and developer-friendly environment for this application. The software is implemented in C++ and built using CMake, allowing modular development and efficient compilation. Version control and code management are handled through GitHub, enabling consistent updates and collaboration during development. However, in the final deployed system, the Jetson Nano is not exposed to the public internet and only communicates with the designated server on the ESP32.

To improve usability, the system is configured to automatically execute the main program on boot. This ensures that once the user powers on the device, all required software components start running without any manual intervention, resulting in a seamless and user-friendly operation.

STRIDE's main embedded control is carried out by the **STM32**. The system software is written in C/C++, mainly because the team is familiar with these languages and they provide a suitable level of control to work directly with hardware like **ADCs**, **DACs**, timers, and communication interfaces. Development is done using STM32CubeIDE version 1.19, which integrates well with ESP32 devices and simplifies peripheral setup through CubeMX. The project is built using the GNU Arm Embedded Toolchain (arm-none-eabi-gcc), using C11 and C++17, which offer a good balance between modern features and reliability. CubeMX is used to generate the initial project structure, including startup files and drivers, giving the team a solid starting point for development. These tools are all available from the official ST Microelectronics webpage [32].

The software itself is organized into modular pieces so that different parts of the system can be developed and tested independently. The team wrote custom libraries for important subsystems like motor control and battery monitoring to ensure the behavior matches the specific needs of **STRIDE**. In other cases, existing **STM32** HAL libraries are used to save time and improve reliability. This setup allows the system to handle multiple responsibilities at once, such as controlling the motors, reading battery data over **I²C**, managing communication, and handling safety features, all while keeping the code structured and manageable.

To ensure reliable and timely execution, the system **MCU** uses FreeRTOS with the CMSIS-RTOS v2 interface, allowing the system to be divided into separate tasks such as motor control, sensor reading, and communication, each with its own priority. This makes the system more organized and ensures that critical functions receive consistent attention. However, the design remains flexible, and the system can easily revert to a main super-loop architecture if needed. This provides a fallback option in case of RTOS-related complexity, ensuring that the system can still operate reliably using a simpler execution model if required. The RTOS operates entirely within the **MCU**, managing task scheduling and execution, while all external communication remains handled through physical peripheral interfaces. As a result, the RTOS does not affect device compatibility and instead serves only to organize and control the internal operation of the **STM32** system **MCU**.

The ESP32 was programmed through the Arduino IDE v2.3.4. The motivations behind choosing this IDE were access to Espressif board management software built for Arduino IDE and connection to a **USB** 2.0 cable (Standard-A to Micro-B) in the ESP32, allowing instruction upload and debugging of the board with the IDE. To handle the complexity of

wireless networking, the project leverages several specialized libraries:

- WiFi.h & WebServer.h: Core ESP32 libraries. The first one manages the soft access point through the antenna available in the device. The second library manages HTTP requests to handle different functions such as /play or /stop.
- LittleFS.h: This library provides a lightweight flash file system.
- ESP8266Audio Suite (AudioGeneratorMP3, AudioFileSourceLittleFS, AudioOutputI2S):
 - AudioFileSourceLittleFS: Reads MP3 file from flash memory.
 - AudioGeneratorMP3: Using the compressed MP3 data leverages a software decoder to generate audio.
 - AudioOutputI2S: Connects to the digital-to-Analog Converter (DAC) using the I2S (Inter-IC Sound) protocol.

The ESP32 connects using its local Wi-Fi hub to an external website, which can be accessed from a laptop or mobile device. This website can be used to view information about and send commands to the rollator.

The ESP32 behaves as an interface between the caregiver and Jetson Nano. The website offers access to different information such as rollator speed, battery level and current mode. An emergency option is available at all times for the caregiver to stop the rollator in case of emergency.

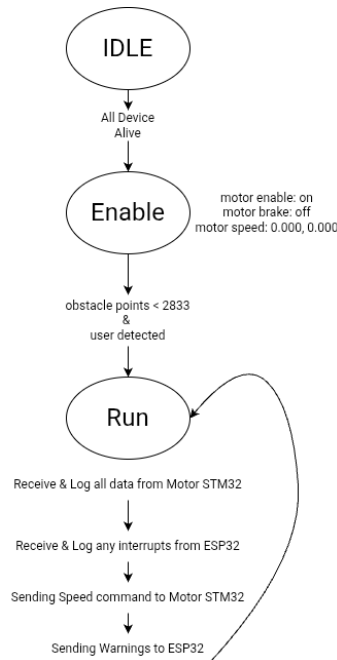


Figure 27: Software FSM on Jetson Orin Nano.

4.8 User Control & User Interface

4.8.1 Physical Control Design

The two major components to the physical controls are the control cluster and the braking system.

The control cluster is driven by a custom PCB, as detailed in Section 4.8.4. The physical control cluster encases this board in a 3D-printed housing, which is attached above the handles for easy access to the system. The three standard control buttons are installed flush into the housing and are labeled through attached labeling.

The buttons carry out the following functions:

- Mode Control: This button cycles the rollator between the three main modes of Standard, Caregiver, and Stair Climbing.
- Stair Climbing Confirm: This button is pressed to begin the stair climbing process.
- Reset: This button cancels any running processes (e.g. stair climbing) and resets the system to Standard mode, stopped.

The control cluster also contains the E-stop button, in order to provide easy access for the user or a bystander to see and access the E-stop.

The braking action of the rollator is controlled by the brake levers installed on the handles. These brakes use a Hall-effect sensor to detect when the brake is pressed, sending digital signals to the motor STM32 to indicate a requested braking action. Each side's brake lever affects the speed of the corresponding motor. Pressing one brake lever slows the corresponding motor, causing the system to turn to that side; pressing both brake levers slows and stops the system. Because the brake levers send digital signals, the full braking action (e.g. speed attenuation) is controlled by software.

On the handles are also mounted two touch sensors. These touch sensors need to be pressed to confirm the user is in position, holding the handles, and actively prepared for the rollator to begin moving.

4.8.2 User Interface

The device gives a range of feedback to the user in various situations. Whenever the device is powered on, the safety lights on the side activate, notifying the user and bystanders that the device is active even if it is not moving. When moving, the system plays a musical tone to alert bystanders of the device's motion.

The device provides audio and visual feedback during normal operation through the speaker and control cluster. The Mode Control button is illuminated continually during operation according to the diagram in Figure 4.8.2. When the mode is switched, the speaker plays a corresponding audio cue to indicate the updated mode setting.

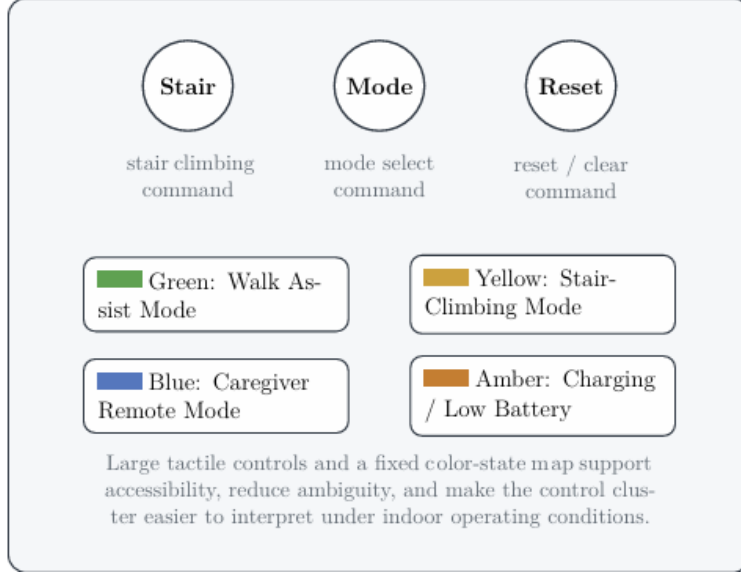


Figure 28: User-facing control cluster.

If an obstacle is detected within the warning zone of 4 m, a button on the control cluster flashes, and the device plays a message: “Warning: obstacle detected ahead.” If the user does not slow the device down within the requisite reaction time, the rollator automatically brakes, which slows the rollator down to a stop if necessary.

When running stair traversal, the device provides feedback to the user on its starting position. If the starting position is insufficient, the device repeats an error message through flashing and audio signals. If the starting position is sufficient, the stair climbing button flashes green, and an audio confirmation plays before starting stair traversal. During stair traversal, the musical tone increases in intensity to signal the heightened need to allow space for the device. After the stair traversal has ended and the device has stopped, another audio confirmation plays.

4.8.3 Remote Control

The user can access the ESP32 to send commands to the device through a device-hosted Wi-Fi hub. An HTML-based website is deployed with multiple options from the ESP32 with different modes (climbing and driving), audio, device parameters such as speed and battery, and an emergency stop. The user can access through any device with an HTML-compatible navigation, such as a smartphone or laptop.

4.8.4 Final User Interface Board Design

The final UI board is a two-rail, I²C-controlled control-cluster interface built around the [Next eXperience Semiconductor \(NXP\) PCA9685PW](#) [27]. In the final architecture, the logic domain is supplied from a nominal 3.3 V rail, while the user-facing illuminated hardware is supplied from a separate 5 V LED rail. This separation allows the board to interface cleanly with controller-side digital logic while still driving the illuminated control elements at a useful

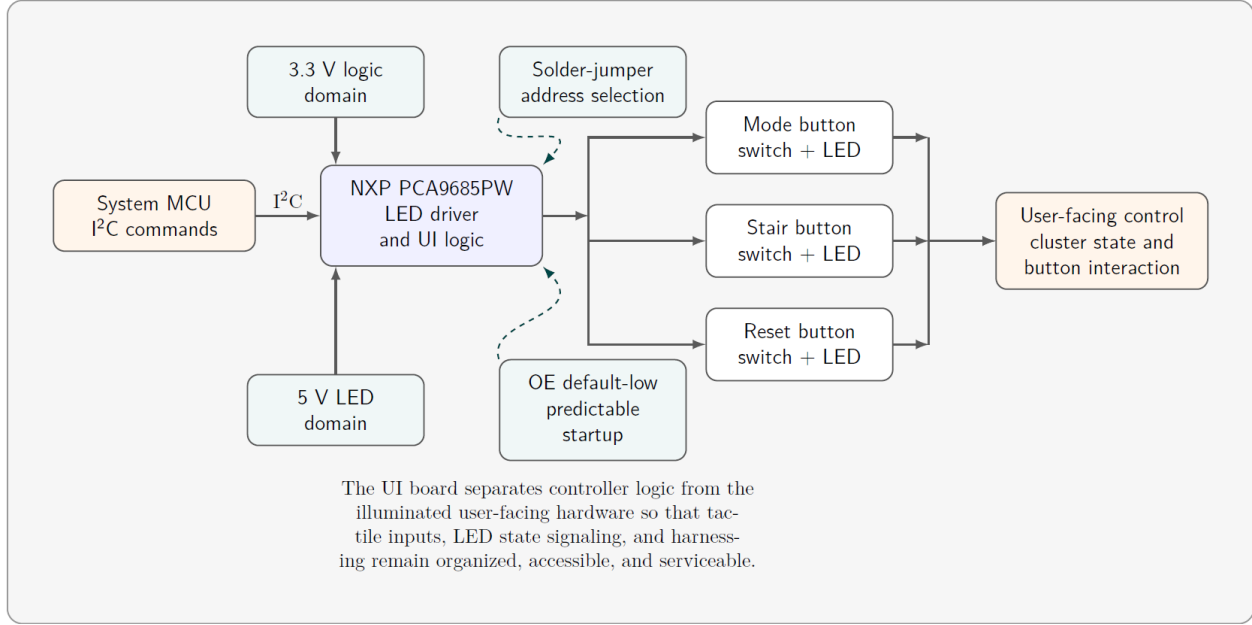


Figure 29: Final UI board functional architecture, showing the controller-side I²C interface, PCA9685-based LED control, separated 3.3 V and 5 V domains, and the three illuminated button paths.

indoor brightness. The design therefore treats the board not as a generic LED breakout, but as the electrical backplane for the **STRIDE** control cluster.

At the center of the board is the PCA9685, which provides a structured multi-channel LED-control architecture over I²C [27]. This was especially well-suited to the final control-cluster design because the LEDs were not arbitrary decorative elements. They were assigned specific operational meanings and therefore needed to be controlled in a repeatable and organized way. By offloading that function to a dedicated LED driver, the final design reduced controller-side pin consumption, improved architectural cleanliness, and created a clearer separation between system logic and user-visible behavior [27]. The functional architecture of the board is shown in Figure 29.

The final board supports three principal user buttons, Mode, Stair, and Reset. Each of these has both a switch function and an illumination function, and the board treats those as distinct electrical paths. The switch lines are grouped into the controller-side interface, while the illuminated button assemblies are routed through dedicated output-side connectors. This organization helps preserve clarity at both the schematic and harness levels. It also improves maintainability, because switch-path and LED-path faults can be isolated more cleanly than they could in a direct-wire implementation. That distinction is particularly valuable in a user-facing subsystem, where a board may need to be debugged or replaced without reinterpreting a one-off wiring scheme.

The final implementation also reflects the intended operating environment of the control cluster. **STRIDE** is an indoor-use device, and the board was designed to provide a clear visible indication under indoor lighting without adding unnecessary output-stage complexity. As a result, the PCA9685 outputs are used directly rather than being followed by additional

buffering stages. That decision was appropriate for the current control-cluster load while preserving a simpler, more transparent board architecture. In other words, the board was sized for the real user-interface problem it had to solve, not for a hypothetical worst-case lighting system that the project did not require.

The final connector strategy reflects the same system-level discipline. Across the [STRIDE](#) electrical system, output-side interconnects were standardized around JST connectors, while board power input and output connections were standardized around the Molex Micro Lock - Plus family [28]. The [UI](#) board follows that convention so that it remains electrically and mechanically consistent with the rest of the project. This connector commonality improves cable-management clarity, reduces harness ambiguity, and simplifies maintenance and spares. In practical terms, it helps the board function as part of a coordinated electrical architecture rather than as an isolated custom design.

The final design also reflects several choices made specifically for robustness and serviceability. The PCA9685 address configuration is handled with solder jumpers rather than removable shunt headers, which makes the addressing arrangement much less susceptible to vibration or accidental disturbance. Likewise, the output-enable line is pulled low by default so that the LED outputs assume a predictable enabled state unless deliberately overridden. The board also uses clear silkscreen identifiers and a cleaner harnessing arrangement than a direct-wire implementation would have allowed.

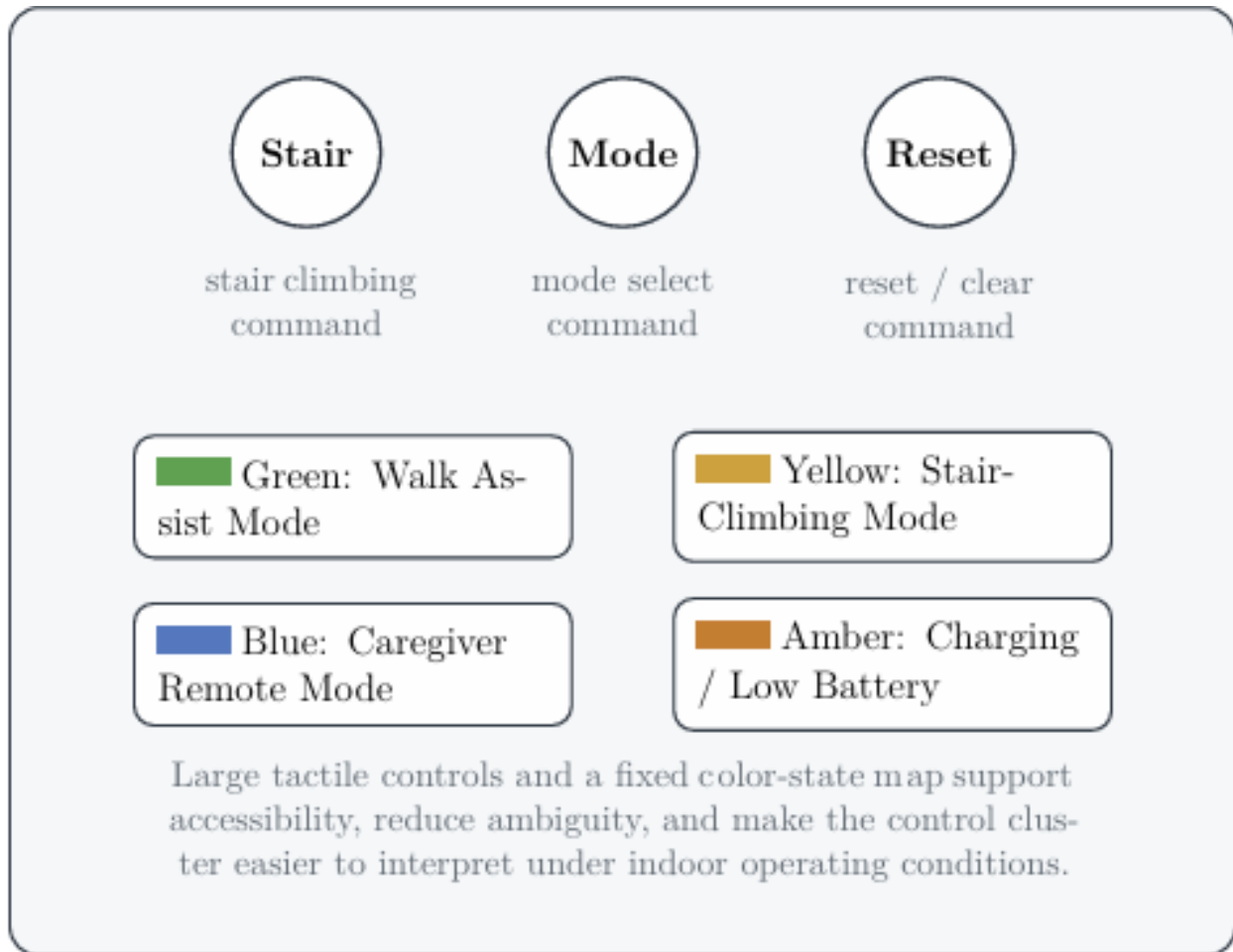


Figure 30: User-facing STRIDE control cluster, showing the physical Mode, Stair, and Reset controls and the associated color-state mapping used for user interpretation

The board’s role in accessibility is also more explicit in the final design than in the early concept. The combination of large illuminated buttons, defined color-state mapping, and centralized control logic makes the control cluster easier to understand and validate as a user-facing interface. The final hardware therefore supports not only electrical correctness, but also the larger project expectation that the interface be usable and interpretable in a consistent way [25], [26]. At the same time, the board reflects the kind of disciplined PCB engineering associated with IPC-2221C and IPC-2152. For a low-power interface board like this one, those standards are less about heroic current-carrying calculations and more about framing the board as a properly designed printed-circuit assembly with deliberate routing, appropriate conductor sizing, maintainable connectorization, and reliable manufacturability [24]. In that sense, the final UI board is not simply a low-power support board. It is the electrical realization of the STRIDE control cluster as an accessible, maintainable, and professionally integrated subsystem.

5 Testing & Evaluation

All testing and evaluation procedures followed the [STRIDE](#) Safety Plan and the [STRIDE](#) Testing and Integration plan, which lists best practices for conducting tests. The full Safety Plan and Testing & Integration Plan are available as Appendix [16](#).

5.1 Microcontrollers

5.1.1 STM32

The [STM32 MCU](#) was initially validated using a sample user application generated by the development environment at start-up, which successfully controlled onboard LEDs, confirming correct board setup and basic functionality.

The [DAC](#) channels were then configured to produce analog outputs within the 0–3.3 V range, and measured values using a digital multimeter confirmed accurate and consistent voltage generation.

Digital I/O functionality was verified by interfacing [GPIO](#) pins with LEDs and pull up resistors on a breadboard, confirming correct active-low operation.

Finally, peripheral functionality was tested by configuring the [UART](#) interface using a dedicated connector. Input was provided via a keyboard, and output was successfully observed on a terminal, confirming reliable serial communication.

These tests collectively verified that the [STM32 MCU](#) operates correctly and is suitable for the intended control applications.

5.1.2 Jetson Orin Nano

The Jetson Orin Nano was first validated through a boot test using the customized Ubuntu operating system image. The system successfully powered on, completed the boot process, and displayed a functional desktop environment, confirming that the operating system image, storage device, display output, and basic board configuration were operating correctly.

After boot validation, the Stereolabs [SDK](#) was installed to provide software support for the ZED 2i depth camera. The depth camera was then connected to the Jetson through the [USB](#) interface, and the Stereolabs Depth Viewer application was used to verify camera operation. The application successfully detected the connected camera and displayed the expected image and depth output, confirming that the Jetson could interface with the depth camera and support the required perception hardware.

These tests verified the basic functionality of the Jetson Nano as an embedded processing unit and confirmed that it was properly configured for later integration with the depth-sensing system. [1](#)

5.1.3 ESP32

The ESP32

The I/O pins were verified by interfacing [GPIO](#) pins with a multimeter and toggling High and Low states in software. Wi-Fi connectivity was tested by hosting a webpage using the default `WiFi.h` and `WebServer.h` libraries and connecting to the ESP32 from a smartphone

via its default IP address. The main communication protocol ([SPI](#)) was tested by exchanging a string message at a common baud rate and reviewing the output using PuTTY as a serial monitor. For the audio interface, the internal flash memory was loaded with a series of MP3 audio files using the LittleFS library and processed with the AudioGeneratorMP3 library for [Inter-Integrated Sound \(I²S\)](#) output, connected to a coupled amplifier that converts the digital signal into an analog output for the 4 Ω speaker.

These tests collectively verify that the ESP32 is functional.

5.1.4 Microcontroller Integration

The SPI communication protocol was used across the Jetson Orin Nano and ESP32. For the ESP32, the software was written using the Arduino IDE V2.3.4. Testing specified connecting the ESP32 [SPI](#) pins (MOSI, MISO, SCLK, and CS) through the JST connector directly into the Jetson Nano JST connector on the same ESP32 Carrier Board. The following list specifies the technical communication points done:

- Both configurations are set up for full-duplex master.
- Data size is set to 8 bits and first bit to MSB First.
- Baud rate of 115200 [BPS](#) per second.
- PuTTY, configured with the same baud rate, was used on the laptop as a monitor terminal to evaluate communication.

[UART](#) communication between the Jetson Nano and the [STM32](#) was tested and confirmed that the data could be transmitted in both directions and parsed correctly. The C++ software was written in VS Code. The Jetson to [STM32 UART](#) link successfully transmitted and received test values at the expected 10 Hz rate. Test speed command "1.234, 3.000" were displayed correctly on both devices. The sensor decision logic will be tested once the control logic is implemented.

The main program on the Jetson Orin Nano was also tested as the central communication process for collecting data and sending control commands. During the test, the program received and parsed incoming data from the STM32 through [UART](#), received additional data from the ESP32 communication link, and then transmitted the normal default speed command back to the motor controller. The terminal output was used as a runtime log to verify that each stage of the communication sequence occurred in the expected order. [1](#)

The log structure stored on Jetson followed three main stages: first, the received data were printed to confirm that incoming test messages were successfully captured and parsed; second, the depth camera callback status was displayed to show whether the perception module had updated the brake-related signal; finally, the command being sent was printed to confirm the outgoing speed command. This log sequence verified that the main program could coordinate communication inputs, include the depth camera callback result, and transmit the appropriate command during each control cycle. [1](#) A full list of all microcontroller testbenches, including test-by-test records, is available in [Appendix 9](#).

5.2 Sensors

5.2.1 Ultrasonic Sensors

The ultrasonic sensor was evaluated through static accuracy, repeatability, and dynamic tracking tests. In the static tests, the sensor successfully detected stationary targets within the tested 10 mm to 2000 mm range, with distance error remaining below 3%. This confirmed that the sensor can provide reliable short- to mid-range measurements under a static conditions.

To improve measurement stability during operation, the raw ultrasonic readings were processed using a lightweight two-stage filter. A three-sample median filter was first applied to reject isolated single-sample spike readings, followed by a light moving average that weighted the previous output more heavily than the newest filtered value. This approach reduced short-term jitter while preserving the main distance trend.

The filter was intentionally kept lightweight because the ultrasonic sensor is used mainly to trigger warning states, not to directly control the motor speed. A stricter maximum-jump threshold would remove more spikes, but it could also reject real jump-in obstacles that suddenly enter the sensing range. Similarly, increasing the median-filter window would remove more multi-sample disturbances, but it would also increase response delay. Therefore, isolated one-point spikes are rejected, while longer disturbances are smoothed rather than completely removed. This tradeoff is shown in Figure 31, where the filtered output removes the isolated spike behavior while still preserving the sensor response needed for warning threshold comparison.

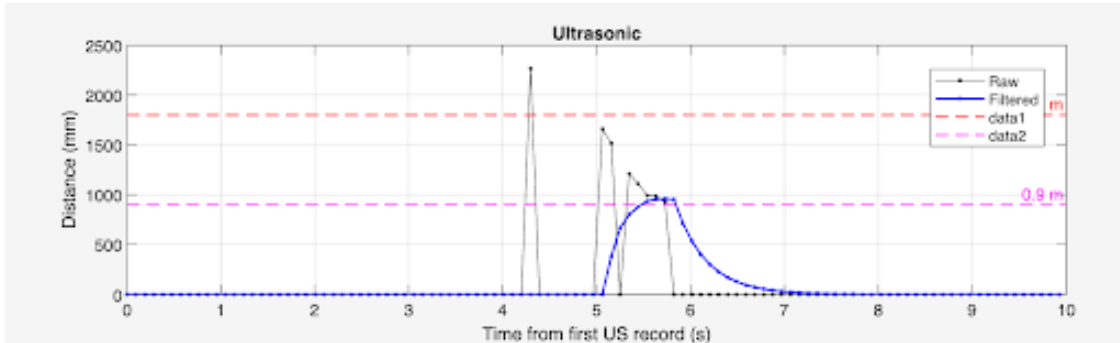


Figure 31: Ultrasonic hallway filter test. The filtered signal rejects isolated single-point spikes and smooths longer disturbances, while still allowing comparison against the 1.8 m and 0.9 m warning thresholds.

The dynamic hallway test produced less consistent raw readings than the static test because the hallway environment introduced reflections, multipath echoes, and off-axis detections. These effects explain why not every spike was removed by the final filter. So when the ZED 2i detects two walls on side, the sensor will be bypassed.

The hallway filter result in Figure 31 shows the intended use of the filtered ultrasonic output for warning decisions. When a usable filtered distance falls below the 1.8 m threshold, the system can generate a warning; when it falls below the 0.9 m threshold, the system can generate a stronger warning. The separate glass wall detection test also showed that the ultrasonic sensor can detect transparent surfaces that are difficult for the depth camera, but

the detailed glass-test records are kept in Appendix 10. Overall, the ultrasonic sensor is suitable as a supplemental warning sensor, especially for transparent or reflective surfaces, but its readings should be filtered and combined with other sensing information before being used in the final obstacle-detection system.

5.2.2 Depth Camera

The ZED 2i Depth Camera was evaluated to validate its functionality, accuracy, and reliability under various conditions. Initial power-up tests confirmed proper device enumeration and stable startup behavior.

Depth performance was verified across multiple distances (1–10 m) and surface types, demonstrating accuracy within the expected $\pm 2\%$ range for opaque objects. Transparent surfaces (e.g., glass) failed to produce reliable data, which was anticipated by the team and the reason for incorporating ultrasonic sensors. Later tests confirmed consistent multi-depth detection, robustness under varying lighting conditions, and strong repeatability with minimal variance across extended trials. A result from these tests is shown in Figure 32.



Figure 32: Depth camera detection test. Obstacles are shown in red; the ground plane and non-interfering objects are shown in blue.

Dynamic testing, including both moving objects and camera motion, showed stable depth tracking with errors remaining within acceptable bounds ($\hat{2}\%$), indicating reliable real-time performance.

5.2.3 Sensors Integration

The sensor integration test evaluated whether the depth camera and ultrasonic sensors could operate at the same time without interfering with each other. The depth camera output

was checked as a point-based obstacle signal, where detected objects produced obstacle points in the processed point cloud. The ultrasonic sensor output was checked as a filtered distance measurement in millimeters.

During the test, both sensors produced clear and readable outputs simultaneously. No interference was observed between the two sensing methods. The depth camera continued to report point-cloud based obstacle information, while the ultrasonic sensor continued to report a stable filtered distance value. A full list of sensor testbenches, including detailed test-by-test records and result figures, is available in Appendix 10.

5.3 Motor Testing

5.3.1 Motor

The motor test verified that the ElectroCraft [EZ DRIVE](#) and MP24 [BLDC](#) motor operated correctly before connecting the [STM32 MCU](#). In this phase, the [MCU](#) was replaced with manual switches, a breadboard, a lab power supply, a multimeter, and an oscilloscope so that the drive inputs could be tested directly. The test confirmed that the drive powered on correctly, the green power LED turned on, the red fault LED remained off, and the internally generated 5 V supply was within the expected range. The enable input also worked as expected when pulled low, confirming that the active-low control logic was functioning properly.

The first motion test confirmed that the motor shaft rotated smoothly when controlled using the [EZ DRIVE](#) internal potentiometer. As the potentiometer input increased, the motor speed also increased, showing that the drive was correctly translating the speed command into motor motion. The electronic brake test also passed, with the motor decelerating quickly without reverse spin or overshoot. The external analog input test confirmed that the motor speed increased proportionally when the reference voltage was raised, and the direction reversal test confirmed that the motor could spin in the opposite direction when the direction input was changed.

The speed linearity test showed that the motor speed had a clear linear relationship with the reference voltage. With the drive configured for a 4000 rpm maximum speed setting and a 12 A current setting, the no-load motor speed increased predictably as the input voltage increased. This confirms that the motor and drive can be controlled reliably using a voltage command. The only issue observed during the motor test was the tach-mode test, where no valid tach signal was detected. Because Hall-mode operation worked successfully, the system can continue using Hall feedback while the tach signal issue is investigated. A full list of all motor testbenches, including test-by-test records, is available in Appendix 11.

5.3.2 Motor Controller

The motor control board test verified that the board can correctly interface the [STM32 MCU](#) signal levels with the [EZ DRIVE](#) analog speed input. Since the [STM32 DAC](#) can only output approximately 0–3.3 V, the motor control board uses a non-inverting op-amp amplifier to scale this signal to the 0–10 V range required by the [EZ DRIVE](#) analog input.

The amplifier output test showed a strong linear relationship between input voltage and output voltage. The measured trendline was approximately

$$y = 3.0174x - 0.0307 \quad (3)$$

with an R-squared value of 0.9999. This confirms that the board provides a gain of about 3, which is the intended scaling factor. At full [MCU DAC](#) output, the board can therefore produce nearly 10 V for the motor drive speed reference.

Overall, the motor control board performed as expected. The output voltage followed the input voltage very closely, with minimal offset and excellent linearity. This confirms that the board is ready to be used as the interface between the STM32 [DAC](#) and the [EZ DRIVE](#) during the next integration phase.

A full list of all motor testbenches, including test-by-test records, is available in [Appendix 11](#).

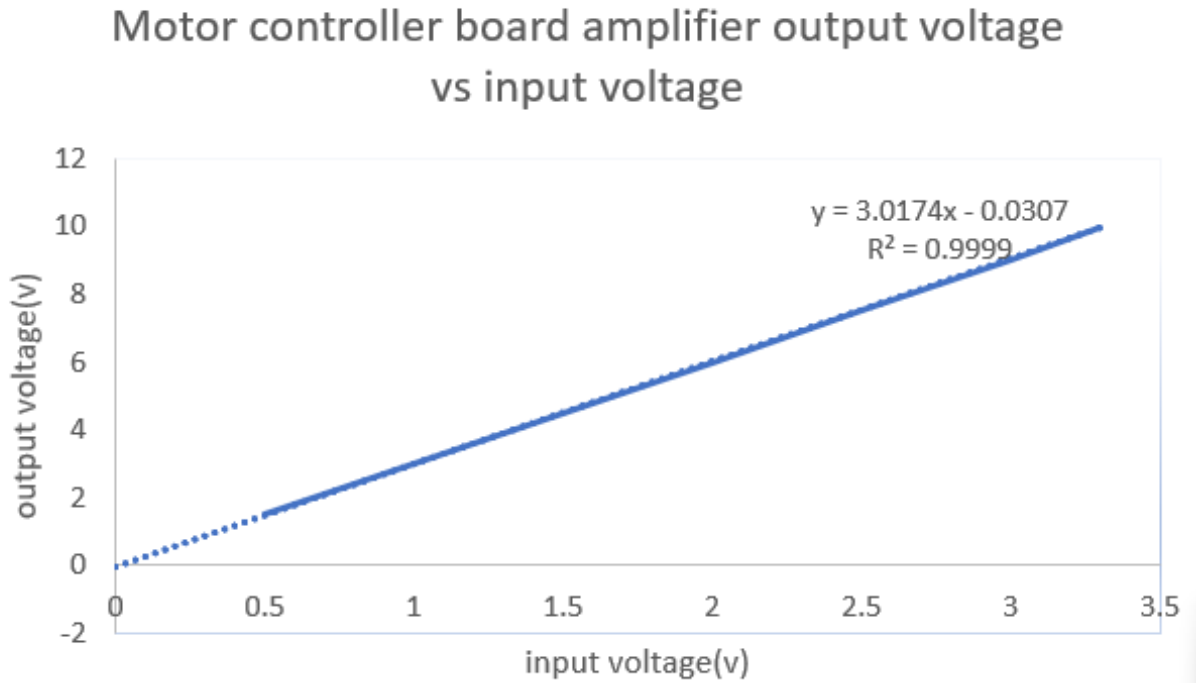


Figure 33: Motor controller board amplifier output voltage versus input voltage. The amplifier output increases linearly with the input voltage, showing a measured gain of approximately 3.02. This confirms that the motor controller board successfully scales the [MCU DAC](#) input range of 0–3.3 V to nearly 0–10 V for the [EZ DRIVE](#) speed command input.

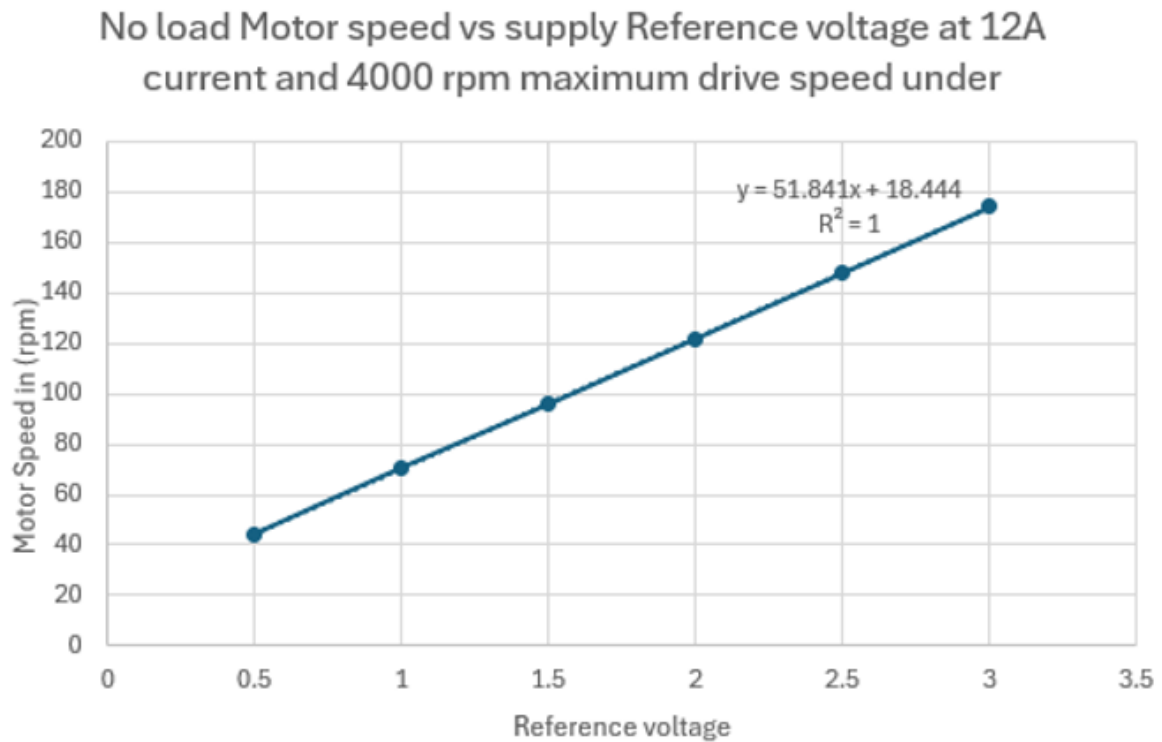


Figure 34: No-load motor speed versus reference voltage at 12 A current limit and 4000 rpm maximum drive setting. The motor speed increases linearly as the reference voltage is raised from 0.5 V to 3.0 V. The trend line shows a strong proportional relationship between voltage command and motor speed, confirming predictable speed control under no-load conditions.

5.4 Frame & Wheel Testing

5.4.1 Frame Acceptance

The frame was tested for sufficient strength. All welds were inspected and manipulated to ensure proper compliance. A weight of 120 lbs was applied to each square tubing member to verify strength.

5.4.2 Wheel Acceptance

The front and rear tri-wheel assemblies were tested individually before integration. After fully assembling the front tri-wheel assembly, the motor shaft was rotated manually to verify that all sub-wheels rotated together without the chain skipping and that all fasteners remain secure. Each rear tri-wheel assembly was similarly assembled and manually rotated to confirm free rotation about the central shaft, smooth spinning of all omni-wheels, and also lateral rolling capability. Any errors at this stage were treated as design errors, and resulted in revisions made to the tri-wheel design.

After individual verification, the front tri-wheels were connected to the motor shafts. The back tri-wheels were installed at the back of the system. Verification was performed at this

stage to ensure the tri-wheels were properly attached; that no fasteners were loose, that the back tri-wheels could still rotate freely; that the front tri-wheels were locked in place by their connection with the static motor shaft.

Further tri-wheel testing falls under integration testing and is detailed in Section 5.8.

A full list of all frame and wheel testbenches, including test-by-test records, is available in Appendix 12.

5.5 Power Distribution Board (PDB)

The PDB was evaluated using a staged bench bring-up and acceptance approach rather than a single power-on check. This was done to reduce risk to the board and to produce documentation that would remain useful to future team members or reuse engineers. The released bench [Acceptance Test Plan \(ATP\)](#) for the module-based PDB defines a sequence beginning with pre-power visual inspection, unpowered resistance and continuity checks, trim-network verification, first power-up from a current-limited 24.0 V bench supply, pre-trim voltage sanity checks, controlled trim adjustment, ON/OFF verification, Power Good verification, rail-by-rail load testing, combined-load validation, startup and ripple inspection, thermal checks, battery-source confirmation, and a two-hour board-level soak test. The same document also defines explicit pass/fail criteria and the measurements to be recorded at each step, so the test process is repeatable and not dependent on undocumented tribal knowledge.

The test methodology was also chosen to align with the broader [STRIDE](#) project requirements and Lafayette senior design expectations. The [STRIDE](#) requirements state that technical requirements are to be validated through analysis, demonstration, or inspection, and that the results are to be documented in formal test reports. The General Project Requirements likewise require complete documentation and formal test reports, and they specifically require a written test plan, a risk assessment, and faculty approval for prototype testing.

Safety during PDB testing was treated as part of the engineering process rather than as a separate afterthought. The initial energization was performed from a current-limited bench supply with downstream mission loads disconnected, safety glasses worn, and an immediate power cutoff available. This approach is consistent with the Lafayette Electrical Safety Program, which states that students are considered unqualified persons for exposed energized electrical equipment of 50 V or more and that energized parts should be de-energized, locked out, tagged out, and verified as de-energized before work proceeds. Even though the PDB operates well below that threshold, the same safety mindset was retained during bring-up, especially for first power application, trim work, and any later battery-fed validation.

Initial bench results were encouraging. The board reflowed successfully and completed first power-up without an immediate catastrophic fault. All three output rails came up and responded to trim adjustment, which confirmed correct basic module population, functional trim paths, and successful front-end power entry. During early bench characterization, the 12 V rail was observed over an adjustable range of approximately 12.065 V to 12.650 V, the 5 V rail over approximately 5.0350 V to 5.0650 V, and the 3.3 V rail was brought close to its nominal target. At this stage, the most important result was not merely that voltages were present, but that the board could be powered safely, the outputs were controllable, and no gross assembly fault prevented basic operation.

The formal trim targets used for bring-up were $12.00\text{ V} \pm 0.05\text{ V}$, $5.00\text{ V} \pm 0.03\text{ V}$, and $3.30\text{ V} \pm 0.02\text{ V}$. After trimming, the operational acceptance bands under load were defined as 11.80 V to 12.20 V for the 12 V rail, 4.90 V to 5.10 V for the 5 V rail, and 3.23 V to 3.37 V for the 3.3 V rail. These limits were chosen so that the board would be judged not only on nominal no-load voltages, but on whether it could remain in a useful and safe regulation window once connected to realistic downstream loads.

Beyond simple voltage verification, the [PDB](#) test plan intentionally evaluates the supervision features that distinguish this board from a generic power module carrier. Separate tests are defined for rail ON/OFF behavior and for Power Good signaling, so that the system can confirm not only that a rail exists, but that it can be deliberately enabled, disabled, and reported to the rest of the robot in a predictable way. This is particularly important for [STRIDE](#), where the low-voltage power system must behave as a supervised subsystem and not merely as a passive collection of always-on rails.

At the time of writing, the board has passed the earliest and most important bench milestones: successful assembly, safe first energization, rail presence, and trim responsiveness. The remaining formal validation follows the released [ATP](#) and includes rail-by-rail load testing, representative and worst-case combined-load testing, oscilloscope-based startup and ripple inspection, thermal screening, confirmation using the actual 24 V [LiFePO₄](#) battery through the intended protection path, and a two-hour representative-load soak test. Those later tests are important because they move the board from “it powers up” to “it remains stable, safe, and integration-ready under realistic operating conditions.”

Battery-source confirmation is especially important because [STRIDE](#) is required to operate from a single rechargeable battery, and the selected Bioenno BLF-2430A battery is rated for nominal 24 V operation with a maximum continuous discharge current of 30 A . The [ATP](#) therefore calls for repeating the core no-load and representative-load checks using the actual battery through the intended protection path after bench-supply validation is complete. This provides a much more realistic confirmation that connector losses, fuse behavior, and source impedance do not introduce surprises that were hidden during laboratory supply testing. The testing procedure and other documentation can be found in [Appendix 14](#).

5.6 Power Distribution Board (PDB) Results

The module-based [PDB](#) was assembled, powered, trimmed, loaded, and validated before subsystem integration. The complete bring-up procedure, acceptance test record, detailed [General Purpose Interface Bus \(GPIB\)](#) plots, raw measurement summaries, and final board disposition are included in [Appendix 14](#). This section summarizes the board-level results needed for the final [STRIDE](#) report.

The [PDB](#) passed the initial electrical bring-up. Unpowered resistance checks found no hard shorts between the input bus, regulated rails, or ground. During first power-up from a current-limited 24.0 V bench supply, the board powered without an immediate fault while downstream loads were disconnected. The filtered input node measured approximately 23.97 V , and the regulated outputs measured 12.004 V , 5.01 V , and 3.308 V . The trim networks were functional, the ON/OFF controls toggled the intended rails, and the Power Good signals tracked the rail states. These checks confirmed correct assembly, converter operation, trim response, enable control, and supervisory signal behavior before load testing.

Rail-load testing was performed using fixed power-resistor loads and an HP/Agilent 34401A DMM. For the official GPIB logs, the meter was connected at the PDB output test point for the rail under test. This measurement point was used so that the logged voltage represented the board output rail rather than the voltage at the far end of temporary load wiring. Startup and shutdown samples, when present, were retained for event analysis but excluded from the steady state regulation statistics. Table 14 summarizes the principal rail and combined-load results.

Table 14: PDB validation results summary.

Test condition	Load condition	Duration	Key result	Result
12 V rail soak	3.3 Ω , 50 W	61.70 min	Mean = 12.000053 V, min/max = 11.990338/12.006340 V, 0 failed samples	Pass
5 V rail soak	3.3 Ω $\parallel$ 5 Ω = 1.988 Ω	61.97 min	Mean = 4.962613 V, min/max = 4.955741/4.971738 V, 0 failed samples	Pass
3.3 V rail soak	1.0 Ω	60.56 min	Mean = 3.23706 V, min/max = 3.22479/3.24631 V, 19 marginal low-limit samples	Pass
Combined-load stress test	3.3 Ω , 3.2 Ω , 1.0 Ω	$\approx$ 57.3 min	Estimated regu- lated output power $\approx$ 61.69 W, all rails passed steady state analysis	Pass

The 12 V rail produced the strongest regulation result. With a 3.3 Ω , 50 W resistor load, the rail averaged 12.000053 V, with a minimum of 11.990338 V, a maximum of 12.006340 V, and a peak-to-peak spread of 16.002 mV. The bench supply reported approximately 1.97 A at the 24 V input bus, corresponding to about 47.3 W input power. Using a representative 12 V conversion efficiency of approximately 96%, this corresponds to about 45.4 W delivered to the rail. The resistor estimate, 12 V/3.3 $\Omega \approx$ 3.64 A, places the test in the same high-current operating region. The 12 V rail remained within the selected 11.94 V to 12.06 V band for all valid samples.

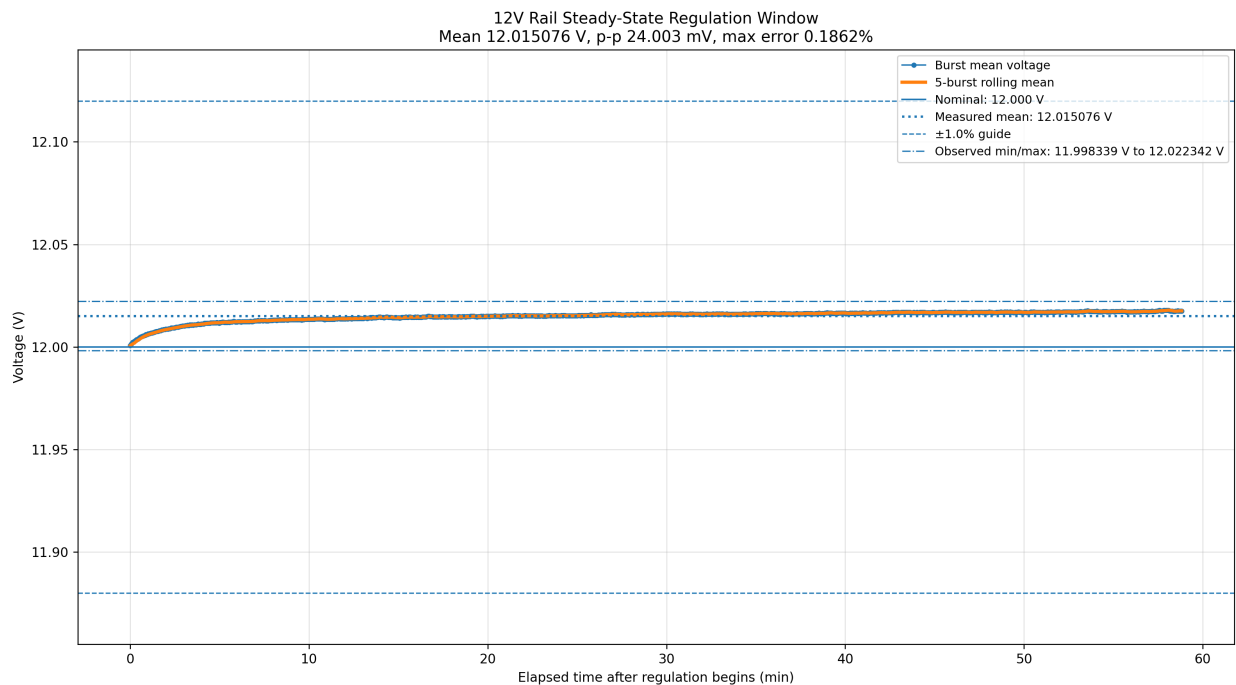


Figure 35: 12 V rail steady state regulation window during the high-current GPIB soak test.

The 5 V rail also passed its corrected high-current test. With the load return connected to PDB ground and the DMM measuring at the PDB output, the rail averaged 4.962613 V over 61.97 min. The minimum was 4.955741 V, which is approximately 0.89% below nominal, and the maximum was 4.971738 V. The result stayed well inside the selected 4.90 V to 5.10 V acceptance band. The $3.3\ \Omega \parallel 5\ \Omega$ load gives an estimated 2.50 A from V/R , while the input-power estimate places the operating point near 3.0 A. The difference is expected because the estimates depend on resistor value, supply readback, efficiency, and wiring loss. The important result is that the 5 V rail remained stable and within approximately a 1% regulation envelope under a real high-current load.

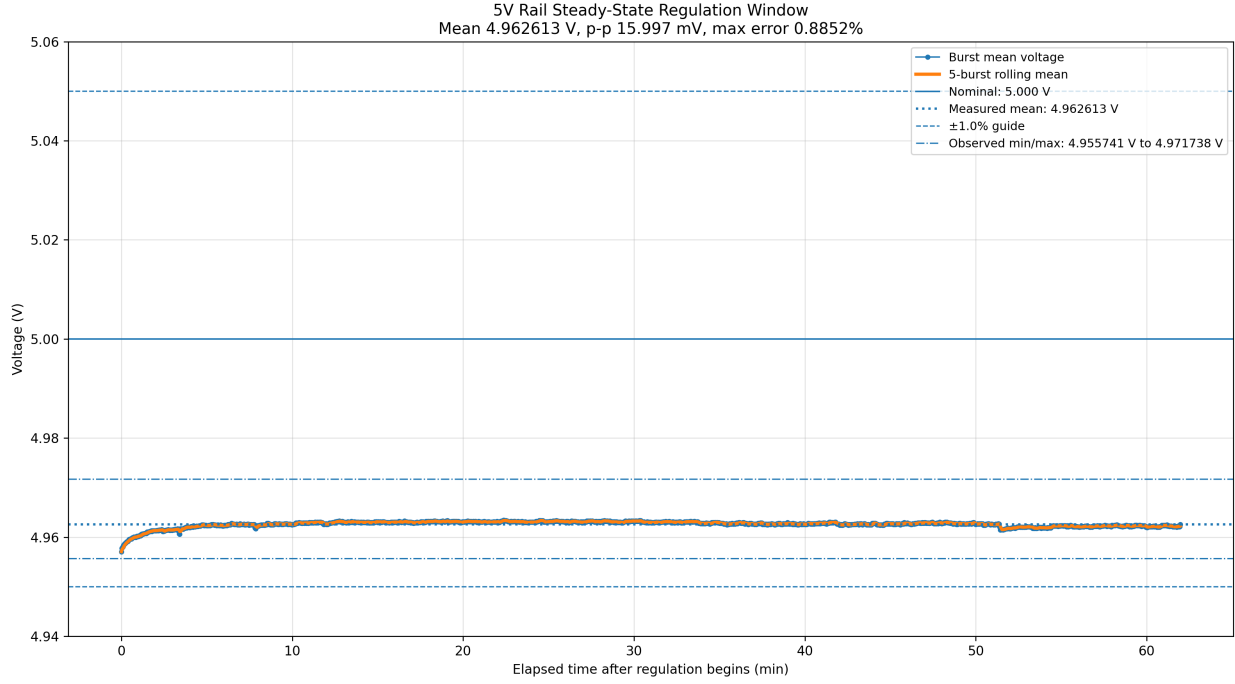


Figure 36: 5 V rail steady state regulation window during the corrected high-current GPIB soak test.

The 3.3 V rail was tested with a $1.0\ \Omega$ resistor, which produced an average load current of approximately

$$I_{\text{load},3.3} = \frac{3.23706\ \text{V}}{1.0\ \Omega} \approx 3.24\ \text{A}.$$

This was a conservative stress condition for a low-voltage logic rail. The rail averaged 3.23706 V over 60.56 min, with a minimum of 3.22479 V and a maximum of 3.24631 V. The analysis recorded 19 marginal low-limit samples, equal to 0.00356% of the valid samples. The rail did not shut down, restart, oscillate, smoke, smell, or show unsafe thermal behavior. Since the 3.3 V rail is expected to support lower-power logic, sensing, and control circuitry in the integrated system, this result is best interpreted as a conservative stress test rather than a normal continuous operating condition.

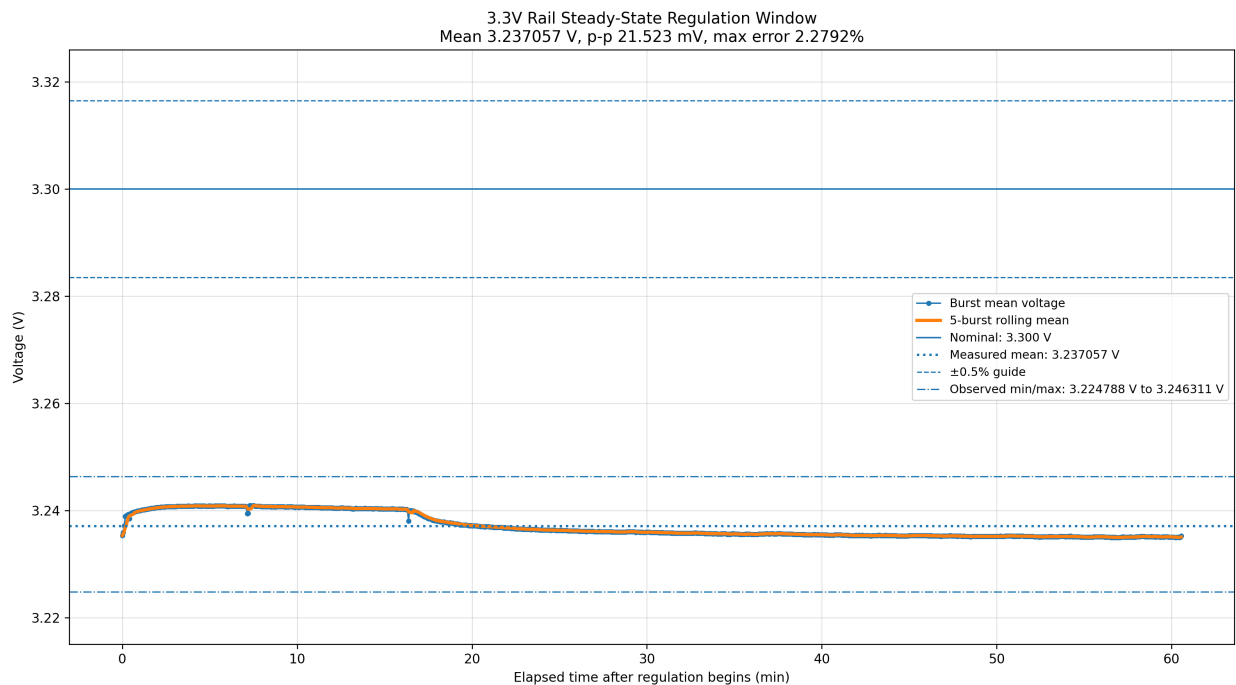


Figure 37: 3.3 V rail steady state regulation window during the high-current GPIB soak test.

The combined-load test applied simultaneous fixed-resistor loads to all three regulated outputs: $3.3\ \Omega$ on the 12 V rail, approximately $3.2\ \Omega$ on the 5 V rail, and $1.0\ \Omega$ on the 3.3 V rail. Using the measured steady state means, the estimated rail currents were approximately 3.63 A, 1.56 A, and 3.23 A, respectively. The corresponding regulated output power was approximately 61.69 W. During the same run, the bench supply reported approximately 24.0 V and 2.68 A, or 64.32 W input power. This is an engineering estimate rather than a calibrated efficiency measurement, but it confirms that the board remained stable during a demanding simultaneous rail-load condition.

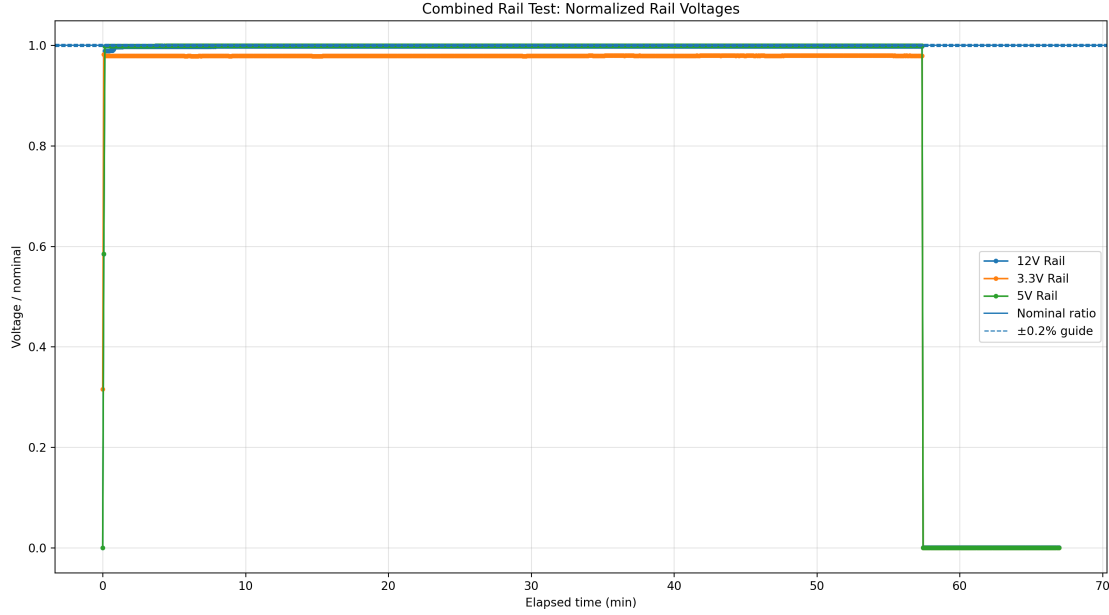


Figure 38: Combined-load normalized voltage comparison for the 12 V, 5 V, and 3.3 V rails. The complete combined-load plot set and steady state summary table are provided in Appendix 14.

The combined-load data did not show cross-regulation failure, input-bus instability, nuisance shutdown, or rail collapse during the processed steady state interval. The 12 V rail carried the largest share of regulated output power, the 5 V rail remained close to nominal, and the 3.3 V rail remained stable under a heavier-than-expected electronics-rail load. No abnormal heating, smell, discoloration, shutdown, or thermal runaway was observed during the rail-level or combined-load tests. Battery-source confirmation also passed for the completed checks, showing that the [PDB](#) remained stable when supplied from the intended nominal 24 V [LiFePO₄](#) battery path.

The final [PDB](#) bench disposition was **Pass**. Based on the completed bring-up checks, trim verification, ON/OFF and Power Good checks, rail-level [GPIB](#) soaks, combined-load stress test, qualitative thermal screen, startup and ripple checks, and battery-source confirmation, the module-based [PDB](#) is ready for subsystem integration. The supporting evidence is documented in Appendix 14, including the bench [ATP](#), detailed test tables, plots, schematics, board plots, tester and witness sign-off, and final disposition. The remaining reliability work belongs at the full-system level, where the assembled [STRIDE](#) rollator should be evaluated over the required continuous-operation interval.

5.6.0.1 Automated data collection and graph generation.

To make the [PDB](#) rail tests repeatable, a Python-based logging and analysis script was developed for the [STRIDE](#) power tests. The script supports [GPIB](#) preflight checks, single-rail acquisition from an HP/Agilent 34401A [Digital Multimeter \(DMM\)](#), analysis of previously collected [.csv](#) files, analysis of legacy Tera Term logs, and combined acquisition from up to three [GPIB](#) meters. During a rail test, the script records timestamped voltage bursts, sample indices, rail name, load notes, input-voltage/current metadata, and analysis settings. It then

exports the raw `.csv` data, cleaned steady-state samples, burst statistics, summary statistics, percentile data, startup/shutdown event estimates when enabled, and presentation-ready voltage regulation plots.

This software was used for long-duration [PDB](#) rail regulation and soak testing, where numerical stability, drift, and pass/fail margins were more important than sub-millisecond transient timing. The generated plots summarized rail behavior over time, including burst-mean voltage, all-sample voltage, detrended residuals, within-burst variation, rolling mean voltage, and voltage percentiles. Oscilloscope captures were still used separately for startup, shutdown, overshoot, and other fast transient behavior because the [GPIB DMM](#) logger records burst-level measurements rather than oscilloscope-speed waveforms.

5.7 Battery Monitor

The [STRIDE](#) Battery Status Monitoring Board was tested to verify that it could safely power on, communicate over [I²C](#), and accurately measure voltage, current, power, and remaining battery charge before being integrated into the full [STRIDE](#) system. The board uses the LTC2944 battery gas gauge with a high-side current-sensing setup and a 1 m Ω shunt resistor. During bench testing, the board first passed basic bring-up checks, including no shorts on the power rails or [I²C](#) lines, a stable regulated 3.3 V rail, correct power LED operation, and successful serial monitor output from the STM32 host. The test plan results showed that the input voltage and regulated rail measured about 23.9 V and 3.28 V, and the current-sense test produced about 0.423 A for an expected 0.4 A reading, confirming that the board was functioning correctly at the hardware and communication level.

For the main performance test, the board was connected to a variable [DC](#) supply and a fixed 12 Ω resistive load. As the supply voltage was gradually increased, the LTC2944 logged voltage, current, power, and charge data. The measured current followed the expected linear relationship from Ohm's law, while the measured power followed the expected curved trend because power increases with the square of current or voltage. The measured current had a mean absolute error of about 5.58%, giving an average accuracy of about 94.42%. Power measurements also closely matched the expected theoretical behavior when comparing theoretical vs. experimental I-V curves, showing that the board was able to track changing load conditions realistically. The results of this test are shown in [Figure 39](#).

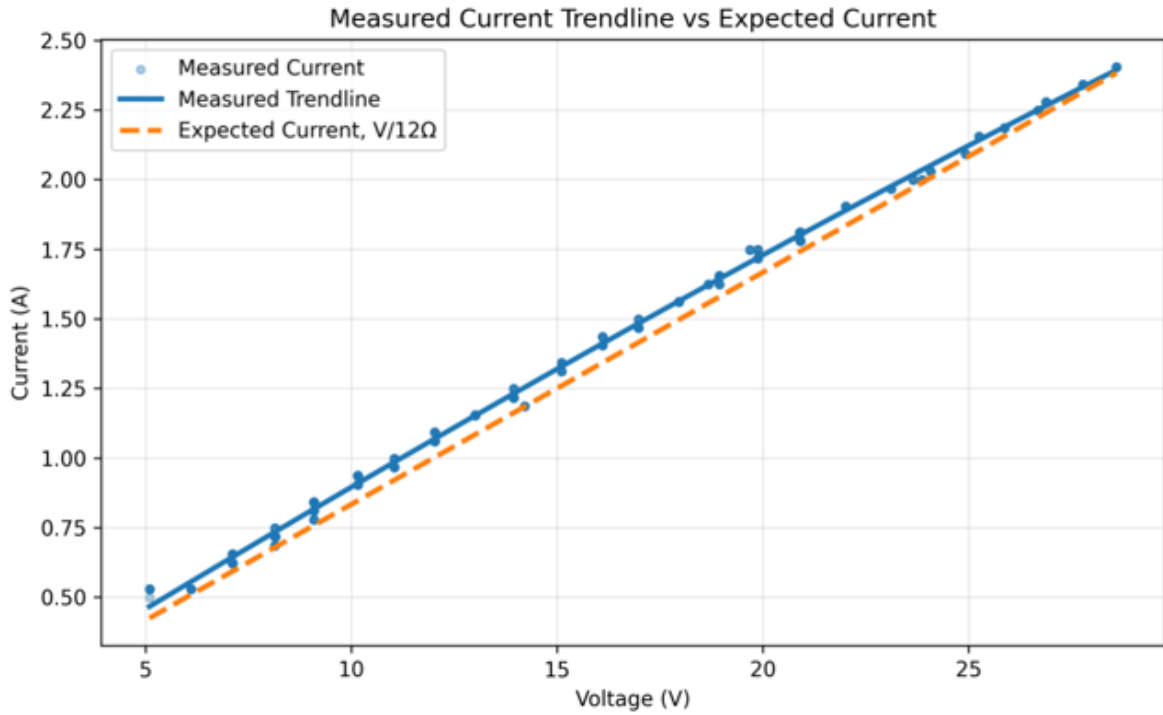


Figure 39: Remaining battery charge over time: measured versus expected. The measured remaining charge closely follows the expected discharge curve over the test duration. The small difference between the two curves indicates that the battery monitor’s coulomb-counting performance is consistent with the theoretical charge depletion based on the applied load.

The remaining-charge test showed one of the strongest results. Instead of assuming constant current, the expected charge consumption was calculated dynamically by integrating the changing current over time. The measured remaining charge curve closely overlapped with the expected curve throughout the roughly 800-second test window. This confirmed that the LTC2944 was not just estimating battery state from voltage, but was properly tracking discharge behavior through coulomb counting. The results of this test are shown in Figure 40.

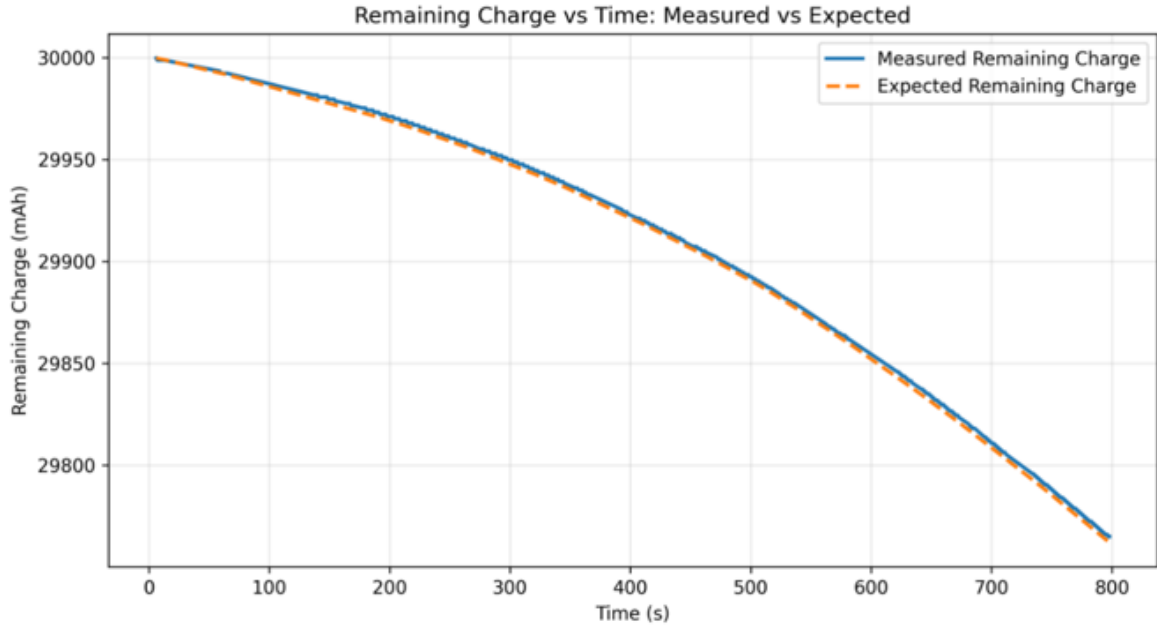


Figure 40: Measured current trendline versus expected current. The measured current increases approximately linearly with voltage and closely matches the expected current calculated using the $12\ \Omega$ load resistance. This validates the battery monitor’s current measurement accuracy across the tested voltage range.

Overall, the test results showed that the Battery Status Monitoring Board met its intended requirements and is suitable for integration into the [STRIDE](#) system for real-time battery monitoring and power management. [1](#)

The full testing plan, as well as results for the battery status monitoring board, are available in [Appendix 14](#).

5.8 Full-System Integration

Integration testing was performed at several points as different subsystems were verified in order to maintain momentum and catch errors as soon as possible.

5.8.1 Sensor Decision Making

Prerequisites: Sensors, Microcontrollers

The sensor decision-making integration test verified that the sensing subsystem could transmit usable obstacle information to the microcontroller-level control system and that the control system responded correctly to that information. This test was performed after the depth camera, ultrasonic sensors, Jetson Nano, and [STM32](#) had each been validated independently.

The depth camera was used as the main control-decision sensor. The Jetson Nano processed the ZED 2i point-cloud data and converted it into obstacle, ground, and wall

classifications. Obstacles were placed inside and outside the configured 4 m detection range, and the Jetson output was monitored to confirm that the obstacle state changed only when an object entered the safety region. When the number of obstacle points exceeded the configured threshold, the depth camera decision logic generated a stop condition and commanded the rollator speed to zero.

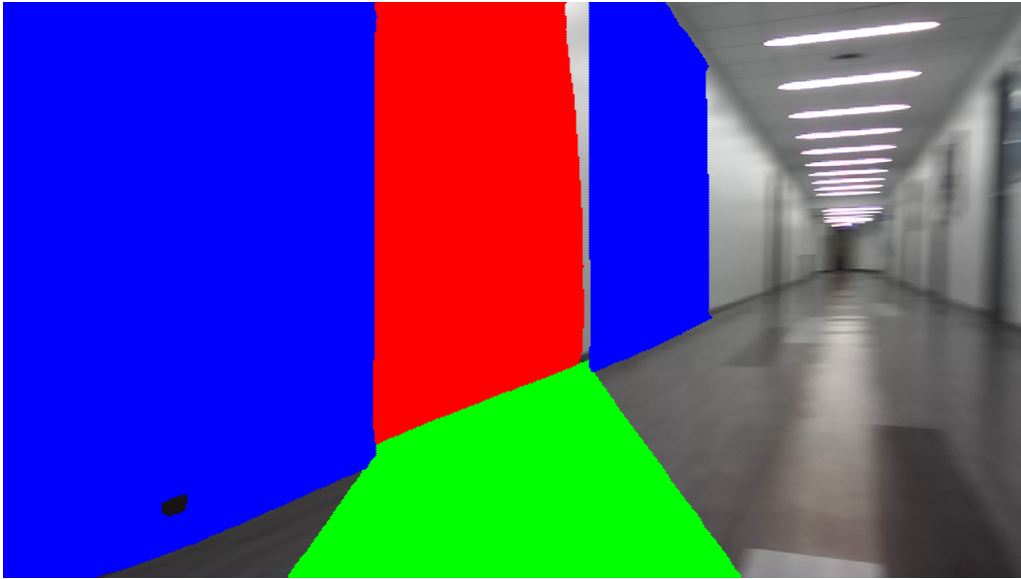


Figure 41: Depth camera decision output showing the point-cloud based classification used for control decisions. The detected ground, walls, and obstacle regions allow the Jetson Nano to determine whether the rollator should continue, warn, or stop.

The ultrasonic sensor was used as a warning sensor rather than the primary control-decision sensor. Each ultrasonic sensor was connected to the [STM32](#) through [UART](#), and the [STM32](#) verified each received frame using the header, high and low distance bytes, and checksum byte. After filtering, the ultrasonic output was interpreted as a distance in millimeters. If the filtered distance was below 1.8 m, the system generated a warning. If the filtered distance was below 0.9 m, the system generated a stronger warning to indicate a closer obstacle. These warning thresholds allowed the ultrasonic sensor to support the user without directly replacing the depth camera's point-cloud based control decision.

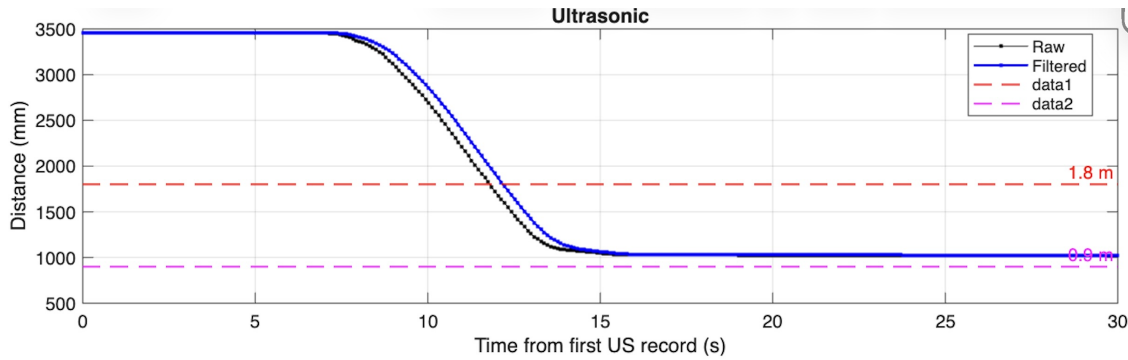


Figure 42: Ultrasonic warning output showing the raw and filtered distance measurements. The filtered millimeter reading is compared against the 1.8 m warning threshold and the 0.9 m stronger-warning threshold.

All sensor information was ultimately transmitted to and processed by the Jetson Nano. The depth camera sent processed point-cloud information directly to the Jetson, while the ultrasonic sensor distances were first read and filtered by the sensor [STM32](#) before being forwarded to the Jetson. The Jetson then combined these inputs with the current operating state and generated the resulting motion command. When no obstacle condition was present, the Jetson allowed the commanded driving speed to continue. When the depth camera detected an obstacle above the point threshold, the Jetson sent a zero-speed command to the motor [STM32](#). When the ultrasonic reading crossed the 1.8 m or 0.9 m warning thresholds, the Jetson generated the corresponding warning state while still leaving the primary control decision to the depth-camera logic.

5.8.2 Driving

Prerequisites: Motor, Power, & Wheels

The rollator was placed on flat ground to test forward driving, steering, and stopping. Forward driving was evaluated through consistency of experienced speed and average deviation from a forward path. Turning was evaluated through the delay between a turn request and a turn beginning, and the turn radius. Qualitative metrics were also evaluated regarding the smoothness of the turn and the smoothness of acceleration/deceleration. Stopping was evaluated through the delay between a stop and the rollator coming to a full stop. These measurements were repeated at several base speeds.

The rollator was next evaluated on sloped ground. The slope used for this was the ADA-compliant ramp outside of [Acopian Engineering Center \(AEC\)](#). The same forward driving and stopping tests were performed to confirm that the rollator could, without sensor and [MCU](#) interference, still safely traverse a ramp in both directions without running away or stopping.

5.8.3 Stair Traversal

Prerequisites: All subsystems.

The unloaded rollator was then brought to the [AEC](#) front stairs to test stair traversal. At this point, stair traversal was evaluated for possibility, rather than full-fledged sensor-controlled traversal. To facilitate this, the rollator was placed directly at the point relative to the staircase where stair traversal procedures would begin, properly oriented.

First, stair traversal was performed ascending. The team verified that the front tri-wheels tumbled over the 6.75-inch step, the rear tri-wheels passively followed, and all components remain intact after climbing. Once climbing was confirmed, the load was incrementally added to determine the maximum climbing capacity.

Next, stair traversal was performed descending. The team verified that the rollator descended in a controlled manner.

5.8.4 Full System Test

Prerequisites: All subsystems.

The full integration testing plan and results are available in [Appendix 16](#).

6 Conclusions & Future Work

The [STRIDE](#) device combines three components—a powered mobility assist system, a stair-climbing mechanism, and a sensor detection array—into a single cohesive device that addresses needs and challenges faced by seniors and individuals who have lost mobility due to injury or illness. On flat ground, the motorized wheels and simple user-interaction system allows for easy navigation. The control of wheel speed allows for the safe ascent and descent of slopes.

[STRIDE](#)'s autonomous navigation of stairs removes the need to lift the device up and down stairs. Additionally, the device can lift up other small loads, such as groceries, further reducing the burden on the user.

The sensor array allows for obstacle detection and avoidance, promoting the safety of the user at all times. The two control options of analog, on-device selections and a digital remote terminal allow for the user to control the device in whatever way best serves their specific situation. The on-device controls are simple, allowing hassle-free operation of the device, while the remote terminal allows for more settings to be viewed, accessed, and changed if desired.

Through addressing these needs, [STRIDE](#) provides users with a way to gain more independence in their daily lives.

The team has identified several items that could be implemented in the future to increase the usefulness of the rollator:

- Traveling backwards.

The current design does not allow the rollator to travel backwards. This constraint was chosen as adding the ability to move backwards presents a much bigger control problem in terms of controlling wheel speed, and opens up more potential risks for user safety (increased risk of falling while walking backwards, etc). In most scenarios, the user will have more than enough clearance to perform a 180° turn. There are still theoretically situations where having the ability to back up could be useful, so while not within the scope of the system, it is a possible design expansion for any future work.

- Expanding navigable surfaces.

For the purpose of testing and focusing on the feasibility of the climbing mechanism, the surfaces and staircases the system is expected to clear have been limited. Testing was focused on the indoor surfaces and staircases within [AEC](#), and may reasonably expand within the course of the project to the paved outdoor surfaces directly outside. However, the system is not rated to traverse rough terrain, such as gravel or cobblestones, or more varied stair rises and runs. If a second mark of the system was produced, this would be one of the primary objectives in order to allow users to travel where they please; however, this is less important for the purposes of creating a working prototype.

- Expanding object detection, tracking, and avoidance.

The current design only measures the distance of any detected obstacle from the rollator; for the initial design, this is sufficient to operate the device safely. To further improve navigation

safety and environmental awareness, object detection and body tracking from the Stereo Lab [SDK](#) with YOLO (You Only Look Once) could be incorporated. Beyond simply detecting obstacles, the enhanced module will identify object types and track their motion over time. This contextual understanding would allow the system to anticipate potential collisions and adapt its behavior dynamically based on the environment. This advanced function would significantly increase the robustness of the system.

7 Team Management

7.1 Team Roles

7.1.1 Motor Team

This team led specification of the motors; unit testing of the motors; developing the motor controller software; designing [PCBs](#) related to the motors; integrating the motors into the system.

Lead: Maurice Otieno

Member: Steven Pang, Sam Satalof

7.1.2 Frame Team

This team led design of the frame; design of the tri-wheel; unit testing of the tri-wheel; assembly of the frame and tri-wheel; placement of components on the rollator.

Lead: Daly Gibson, Alex Villalba Gonzalez

Member: Qinwen Deng

7.1.3 Power Team

This team led design of the power system; design of the [PDB](#); wiring design.

Lead: Sam Satalof

Member: Alex Villalba Gonzalez, Maurice Otieno

7.1.4 Sensors Team

This team led specification of the sensors; unit testing of the sensors; development of sensor software procedures; integration of the sensors.

Lead: Frank Jiang

Member: Qinwen Deng, Steven Pang

7.1.5 Software Team

This team led design of the user interface; development of the remote control terminal; design and testing of user interaction.

Lead: Alex Villalba Gonzalez

Member: Qinwen Deng, Sam Satalof

7.2 Expenses

This section outlines the projected and actual expenses incurred during the design and fabrication of [STRIDE](#). All components were sourced with consideration for cost effectiveness, availability, and compatibility with the design requirements. The project expenses total \$9120.86. The majority of the cost was attributed to frame & wheel components and the motor & power system.

Table 15: Expense breakdown.

Category	Expected	Actual	% of total spent
Motor & Power	\$3105	\$2905.07	31.9%
Frame Wheels	\$1135	\$2781.98	30.5%
Sensors	\$860	\$672.59	7.4%
PCB Components	-	\$1582.26	17.3%
Microcontrollers	\$600	\$397.01	4.4%
Total Cost	\$6600	\$9120.86	

(Expected expenses include 15% addition for expected overrun, tariffs, etc. [PCB](#) components were initially specced under all categories and later reclassified.)

The Frame & Wheel subsystem accounted for the largest budget variance between the expected and actual budget. This overrun was driven by two primary factors. First, the commercially available wheels that met the project’s size and load requirements were metric-dimensioned, and the wheel bore diameter propagated through much of the drivetrain design, dictating shaft sizes, bearing selections, and associated hardware. Since the college machine shop was equipped only with imperial tooling, approximately \$400 in metric broaches, reamers, and related cutting tools had to be purchased to accommodate these dimensions. Second, a mid-project design revision from a caster-based rear wheel to a free-rotating tri-wheel configuration introduced additional material and fabrication costs. The custom omni-wheel assemblies alone accounted for over \$750 in material and production costs.

Full documentation of project expenses, including line-by-line items, is available as [Appendix 17](#).

7.3 Courses

ECE 414 - Embedded Systems: This course teaches how to program microcontrollers using the C programming language. The course also explains how to handle data from various sensors using common communication protocols and utilize them as inputs to software structures such as state machines.

ECE 322/323 - Solid State Electronics I and II (PCB Design): These courses focus on hardware design principles with practical experience in voltage regulation and PCB development. They build a strong foundation in circuit theory, SPICE simulation, and hardware implementation.

ECE 331/332 - Signals & Systems, Communication Systems: These courses develop an understanding of time- and frequency-domain signal analysis and their applications in system design. Includes exposure to core communication concepts, including modulation techniques, error detection and correction, and fundamental communication protocols.

ECE 341 - Electromagnetics: This course provides an extensive overview of the application of electrical and magnetic properties in electrical systems. This includes telecommunications, azimuth antennas, transformers, magnetic properties of materials, and transmission lines.

CS 205 - Software Engineering: This course provides foundational knowledge on web

development tools, including the use of cloud services, error analysis, database management, and handling of software systems, both back-end and front-end.

ES 226 - Statics: This course provides foundational guidelines for the analysis of mechanical forces on beams and members of a system.

ECE 433 - Industrial Electronics and Control Systems: This course provides foundational knowledge on designing motor controllers, including feedback loop concepts, controller characterization, controller structures and usages, and simulation techniques in MATLAB.

ECE 427 - Sensors & Electronic Systems: This course provides foundational knowledge on basic sensor operations, including principles of ultrasonic sensors, DAC and Time-of-Flight sensors. It also introduces some examples for noise analysis technique and resolution analysis for DAC/ADC.

7.3.1 Concepts Learned

- How to specify a motor for a desired system.
- How to specify a battery for a desired system.
- Analysis of capacitor dielectric behavior, including C0G/NP0 stability, X7R/X5R DC-bias derating, aging, voltage rating, and temperature coefficient.
- Evaluation of passive-component parasitics, including equivalent series resistance, equivalent series inductance, capacitor anti-resonance, and layout-dependent filtering behavior.
- Review of high-current test-path losses, including resistor self-heating, wire resistance, connector contact resistance, fuse voltage drop, and measurement-location effects.
- Analysis of trim and regulation sensitivity, including resistor tolerance, potentiometer range, local versus remote sense behavior, and load-dependent voltage droop.
- Evaluation of power-resistor load-bank behavior, including hot resistance drift, continuous versus pulse power rating, heat spreading, and derating during long soak tests.
- Review of capacitor placement and package limits for switching-regulator input/output filtering, including ripple current, loop area, and effective capacitance under bias.
- Analysis of protection-component behavior, including fuse cold resistance, hold current margin, voltage drop, thermal rise, and coordination with expected rail current.
- Review of connector and wiring choices for serviceability, current capacity, voltage drop, strain relief, and repeatable test measurements.
- PCB Design
 - 2-layer PCB design and fabrication
 - Development of carrier boards based on manufacturer documentation
 - 4-layer PCB design & fabrication.

- PCB norms and nuances.
 - RF/EMI minimization & size optimization.
- Design & fabrication through the machine shop.
 - Proper CAD protocols for machine shop use.
- Collating research.
- How to write and format a scientific paper.

8 Appendixes

I. Design documents provided to the team:

Lafayette College Department of Electrical Engineering. [ECE 491-492 General Project Requirements](#). This document provides the general project requirements for the Senior Design courses at Lafayette College, including expectations, assignments, and general project guidelines and safety standards.

Lafayette College Department of Electrical Engineering. [STRIDE Specific Project Requirements](#). This document provides all of the specific project requirements for the [STRIDE](#) senior design project, including success metrics and constraints.

Lafayette College. [Electrical Safety Program](#). All general rules and regulations that must be abided by when developing, testing, and operating an electrical project on Lafayette College's campus.

II. Final mechanical design drawings for the frame.

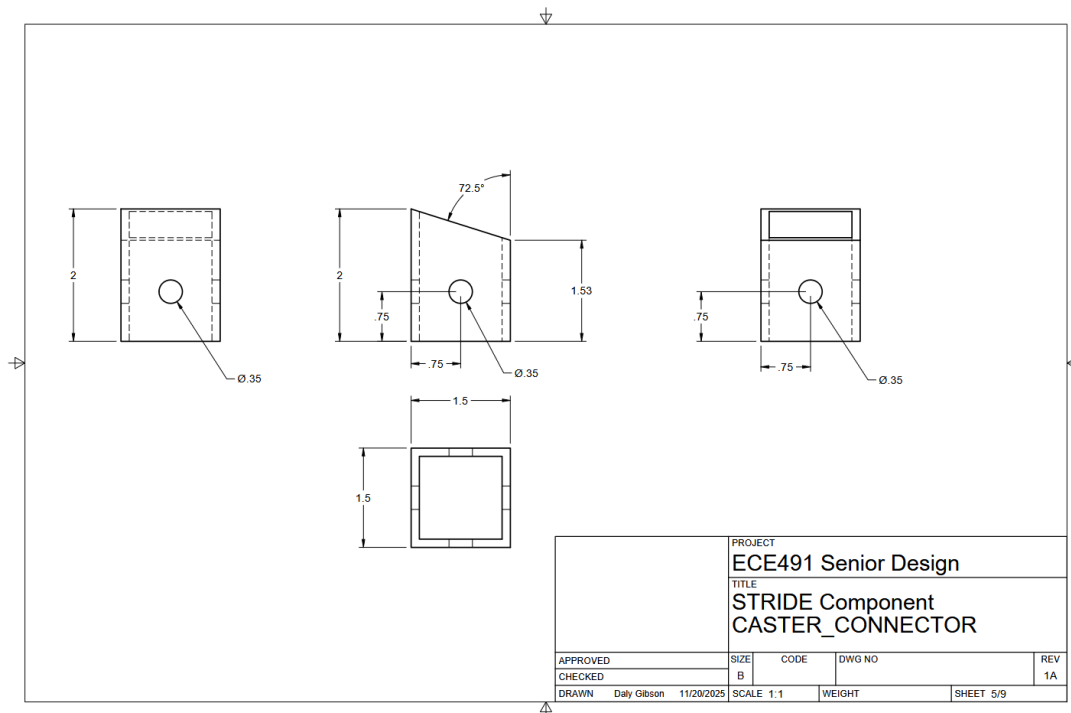


Figure 43: Caster Connection.

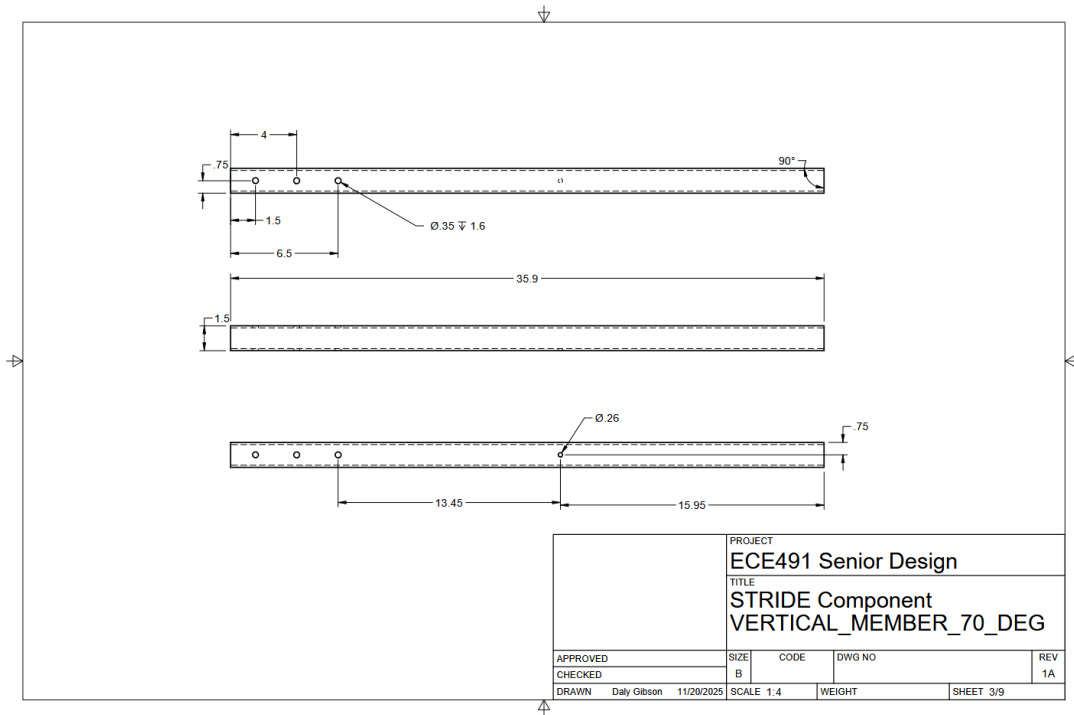


Figure 44: Vertical member: 70 degrees.

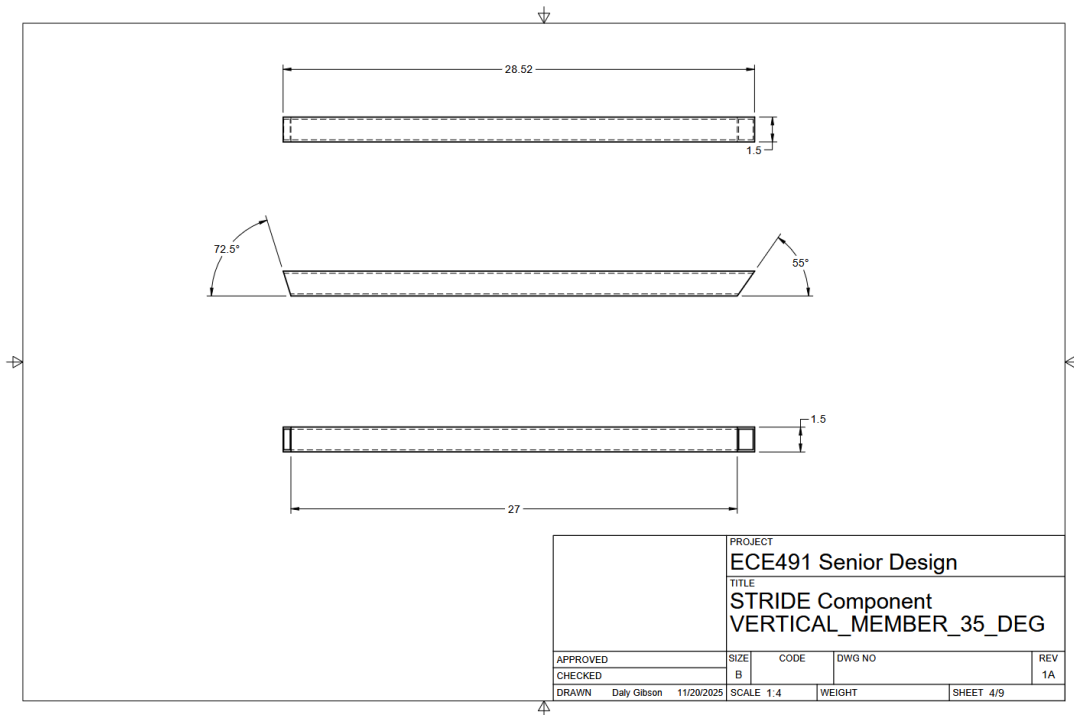


Figure 45: Vertical member: 35 degrees.

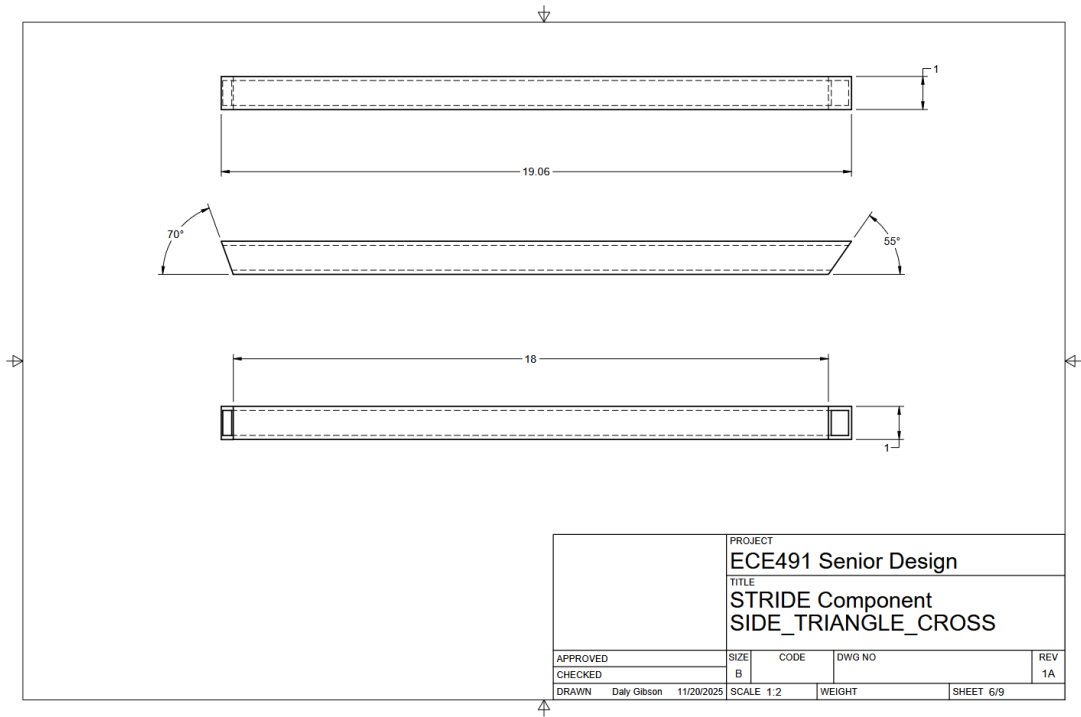


Figure 46: Side support member.

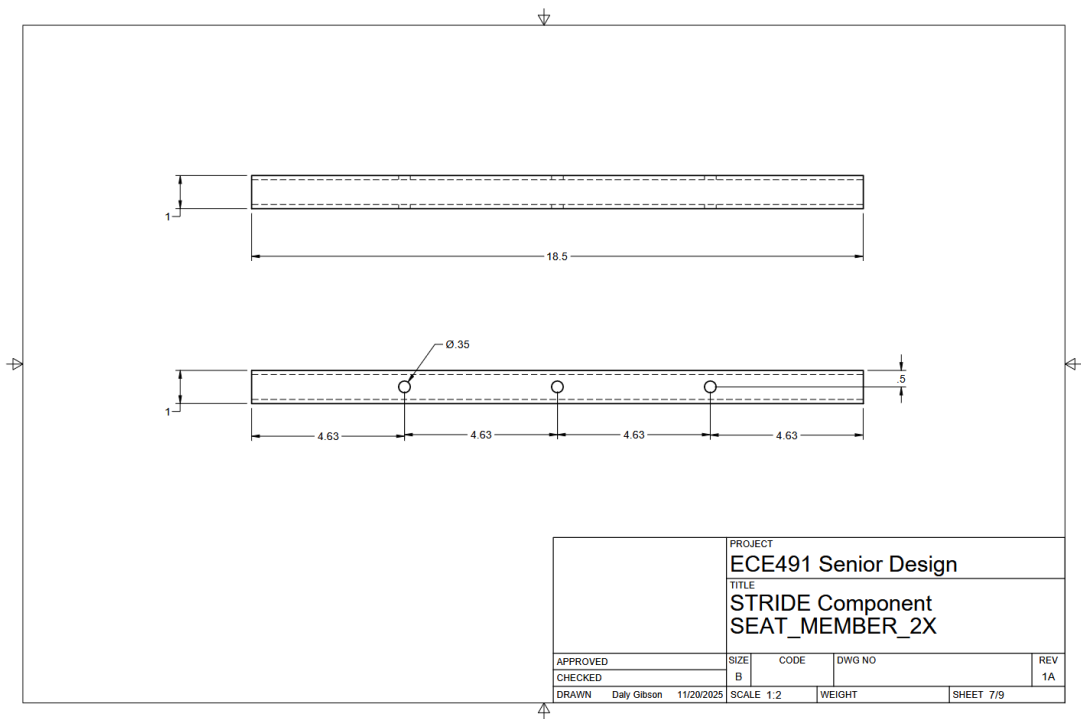


Figure 47: Seat support member.

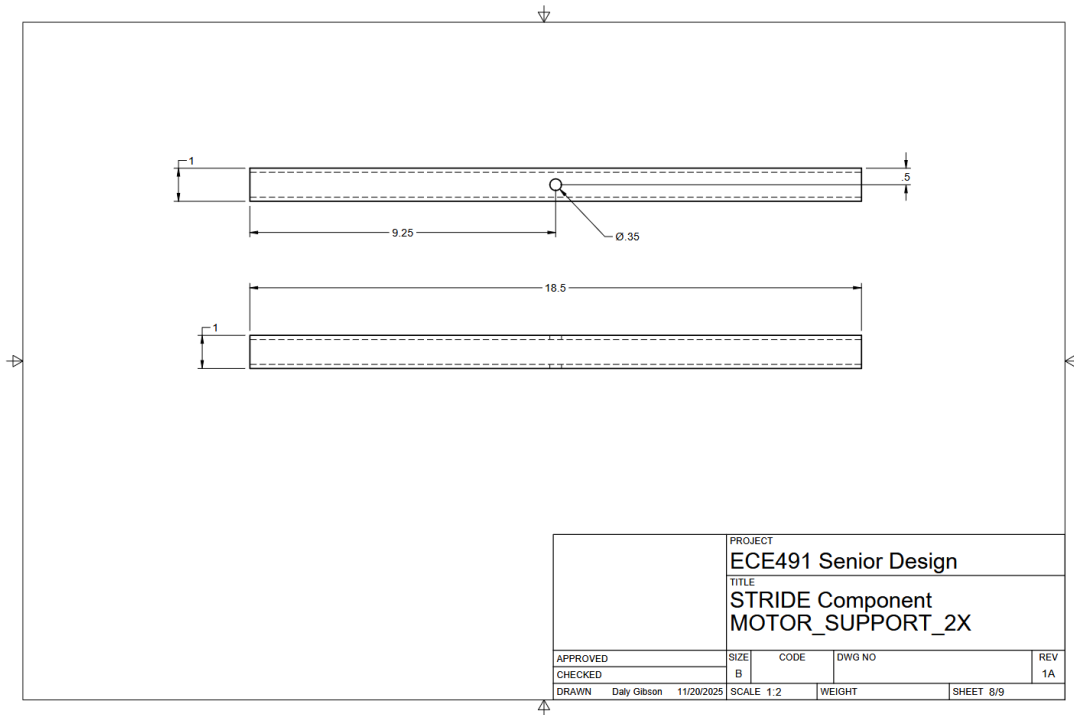


Figure 48: Motor support member.

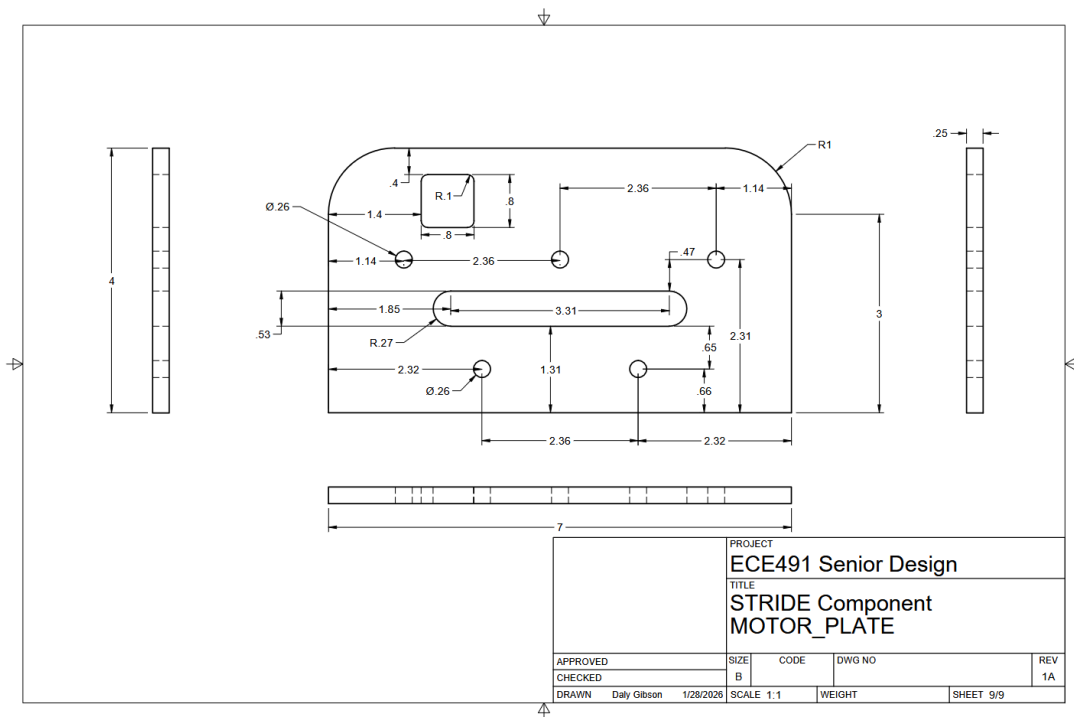


Figure 49: Motor plate.

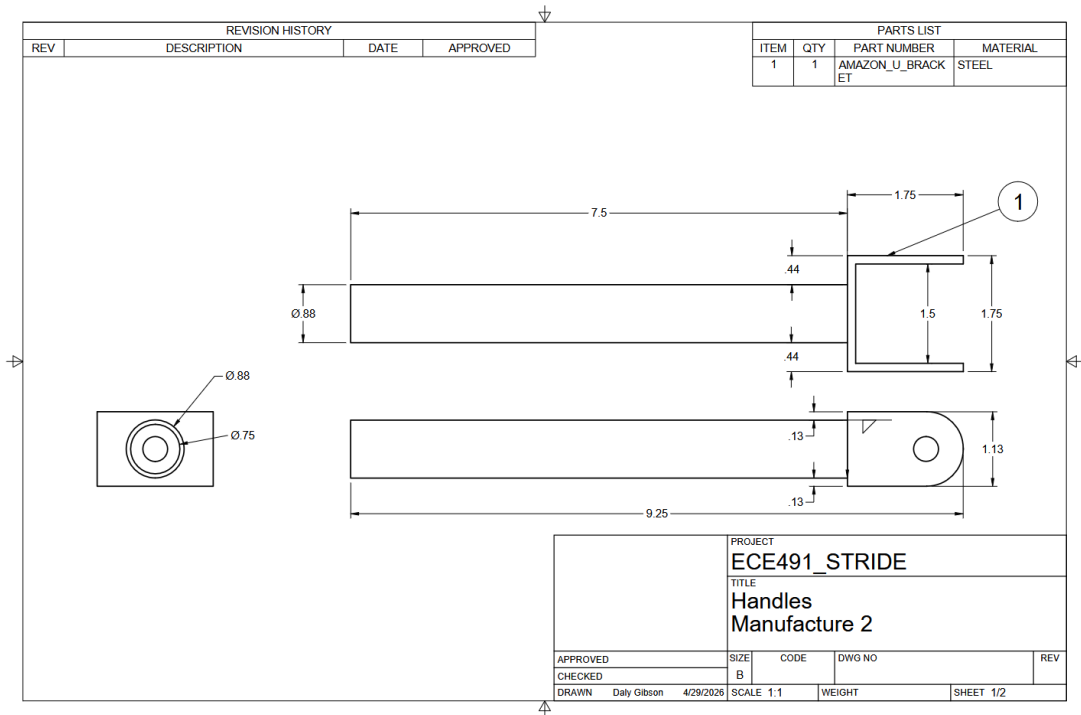


Figure 52: Handles.

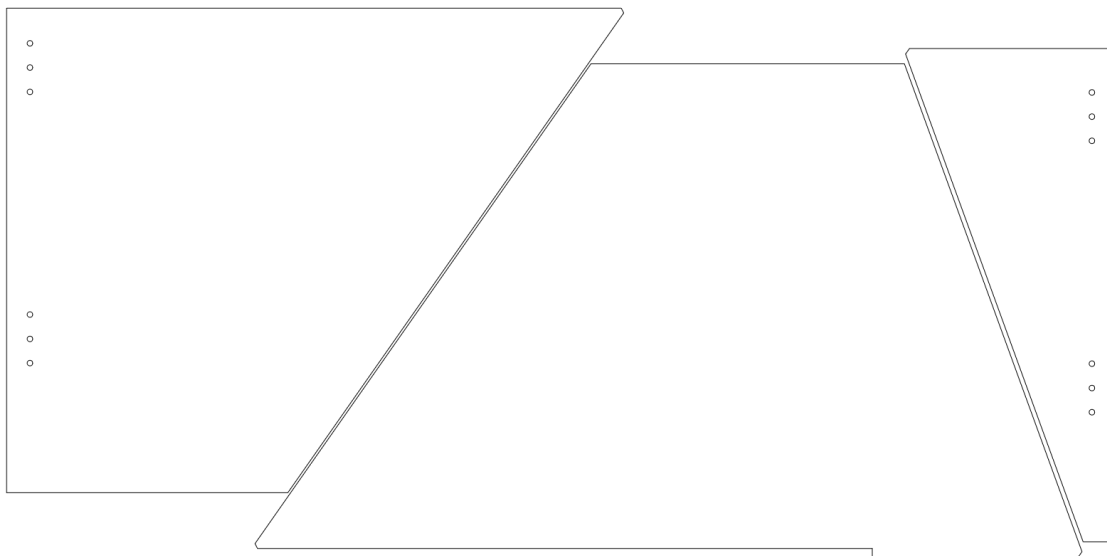


Figure 53: Layout for lasercut side panels.

III. Final mechanical design drawings for the triwheel.

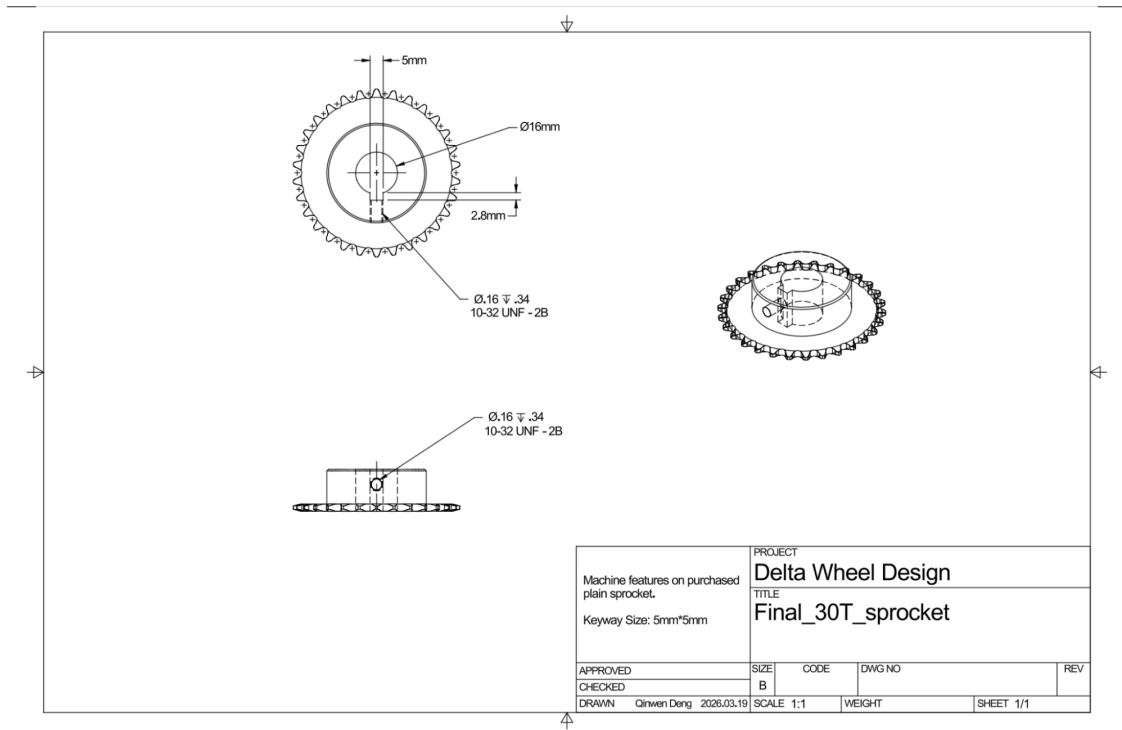


Figure 54: Final 30-tooth sprocket.

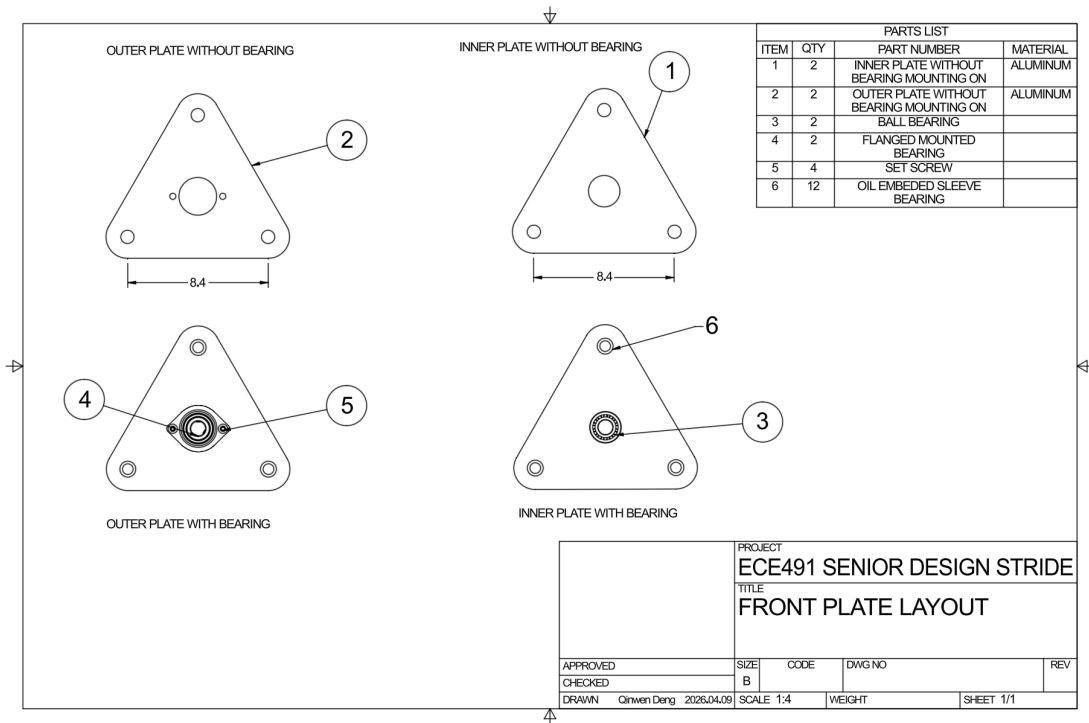


Figure 55: Front plate layout.

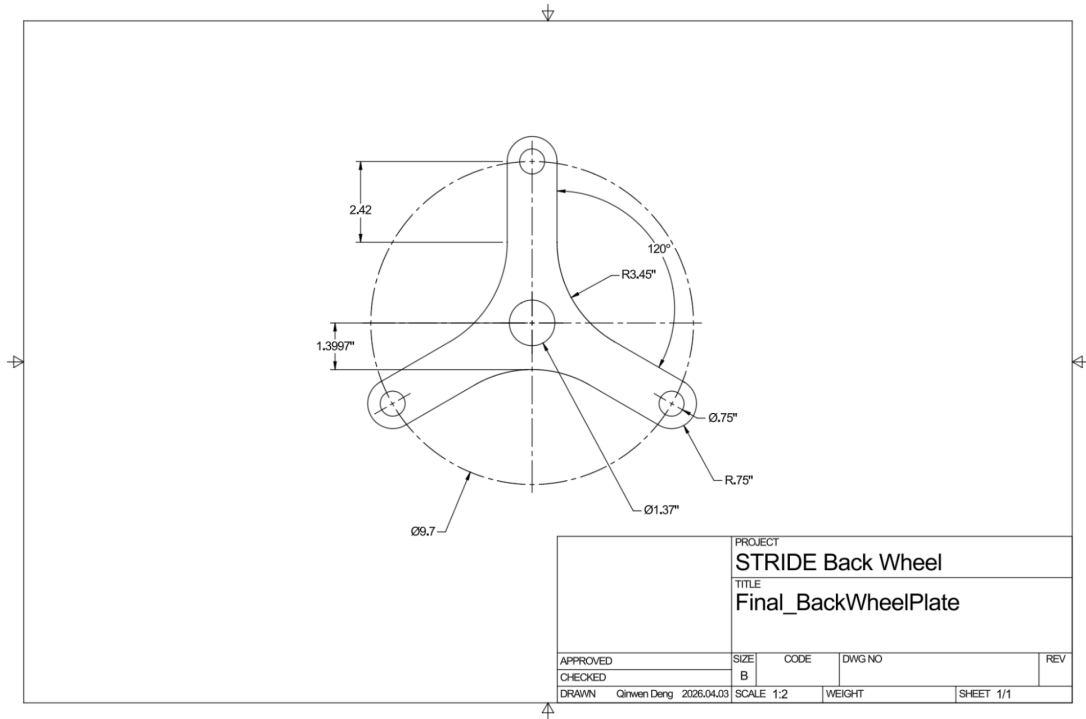


Figure 56: Final back wheel plate.

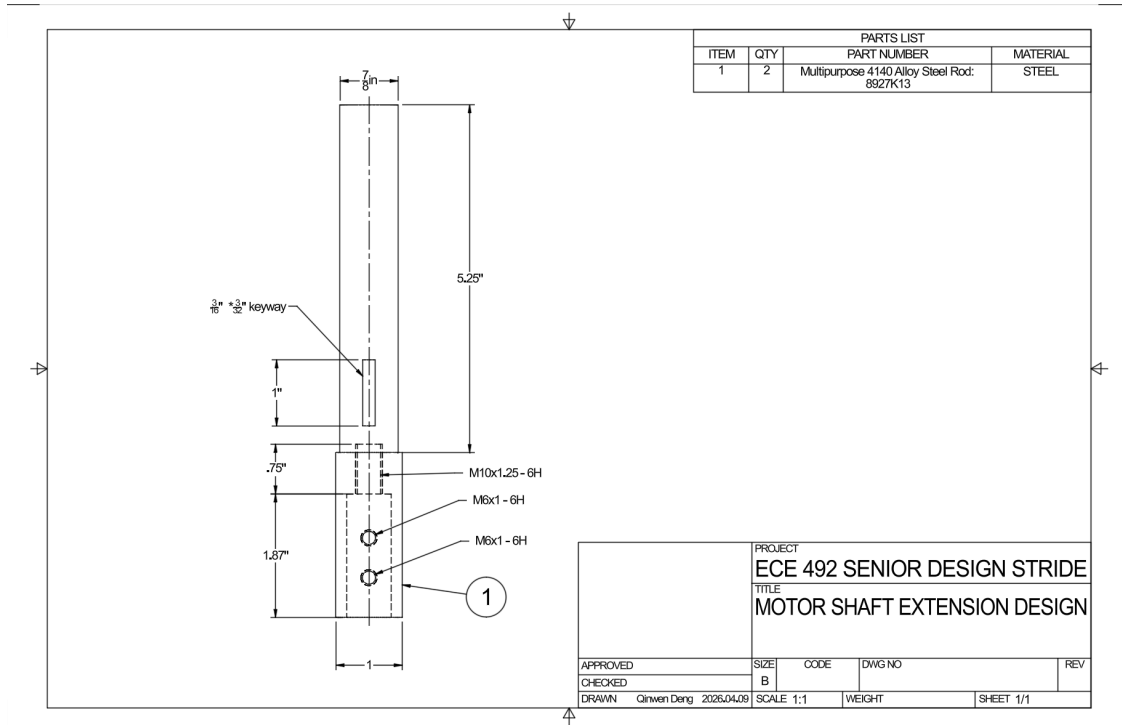


Figure 57: Motor shaft extension.

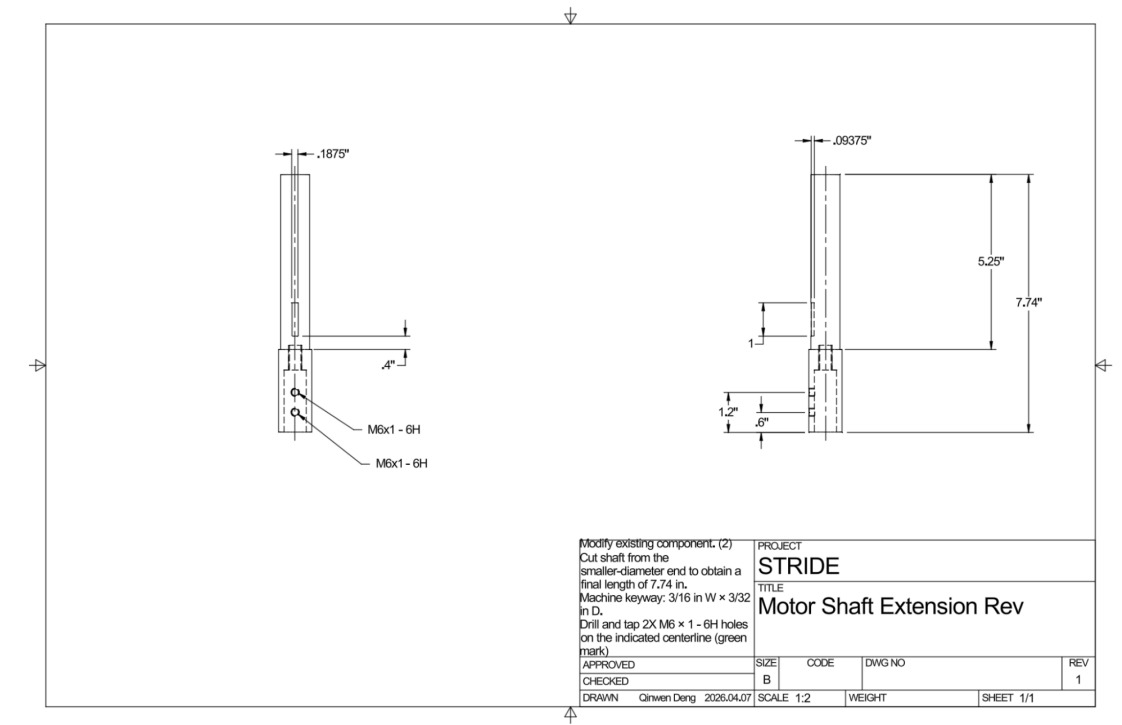


Figure 58: Motor shaft extension revision.

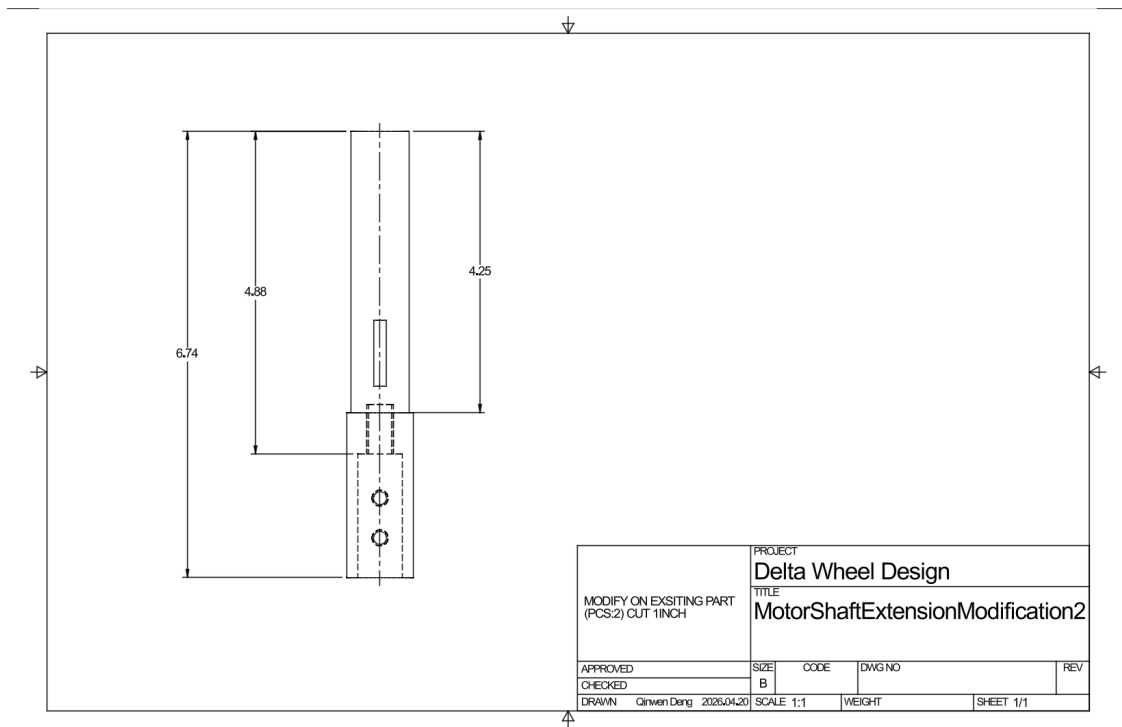


Figure 59: Motor shaft extension revision 2

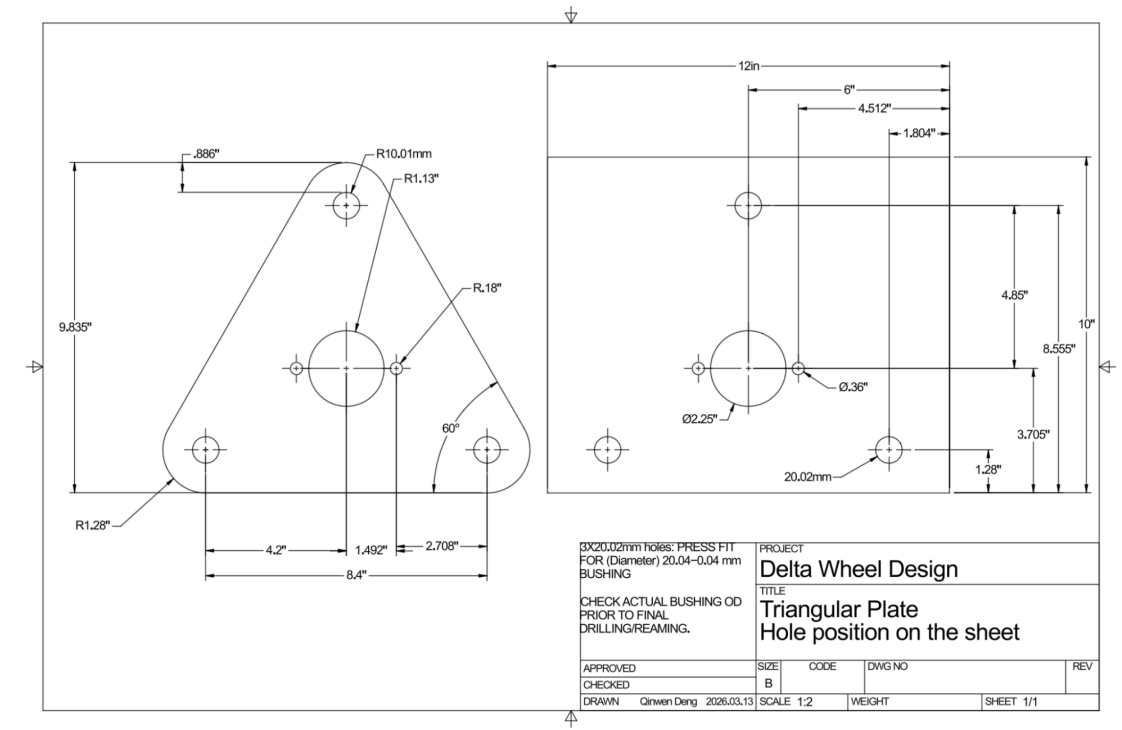


Figure 60: Triangular plate hole position.

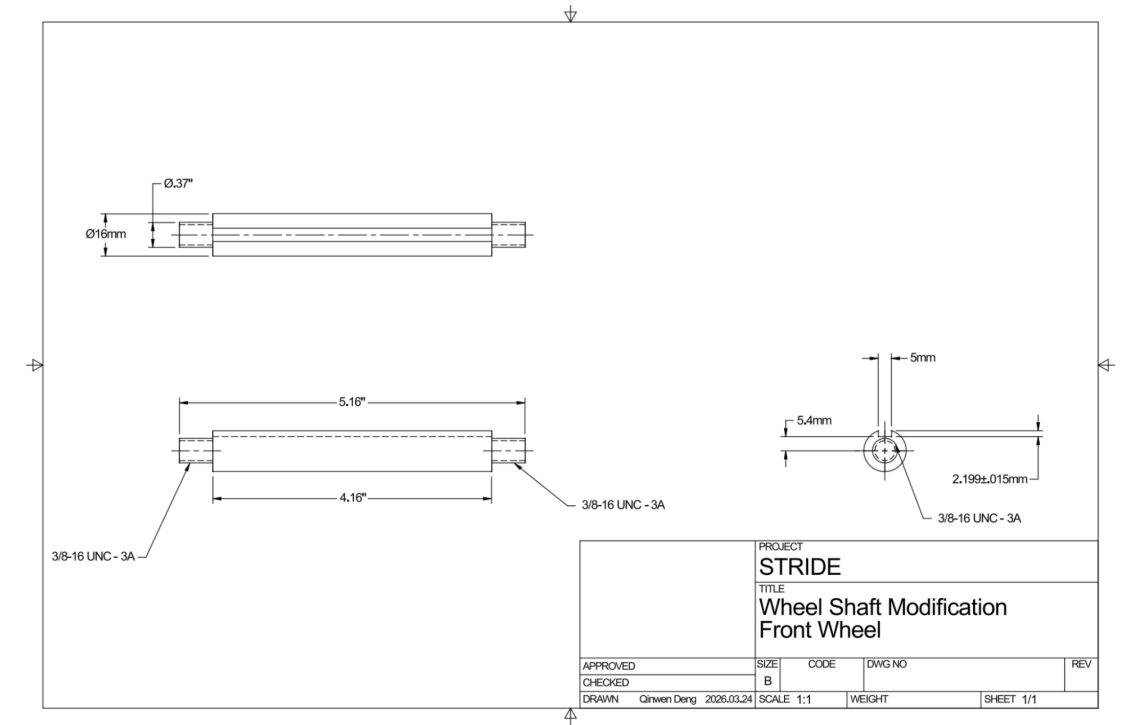


Figure 61: Wheel shaft modification.

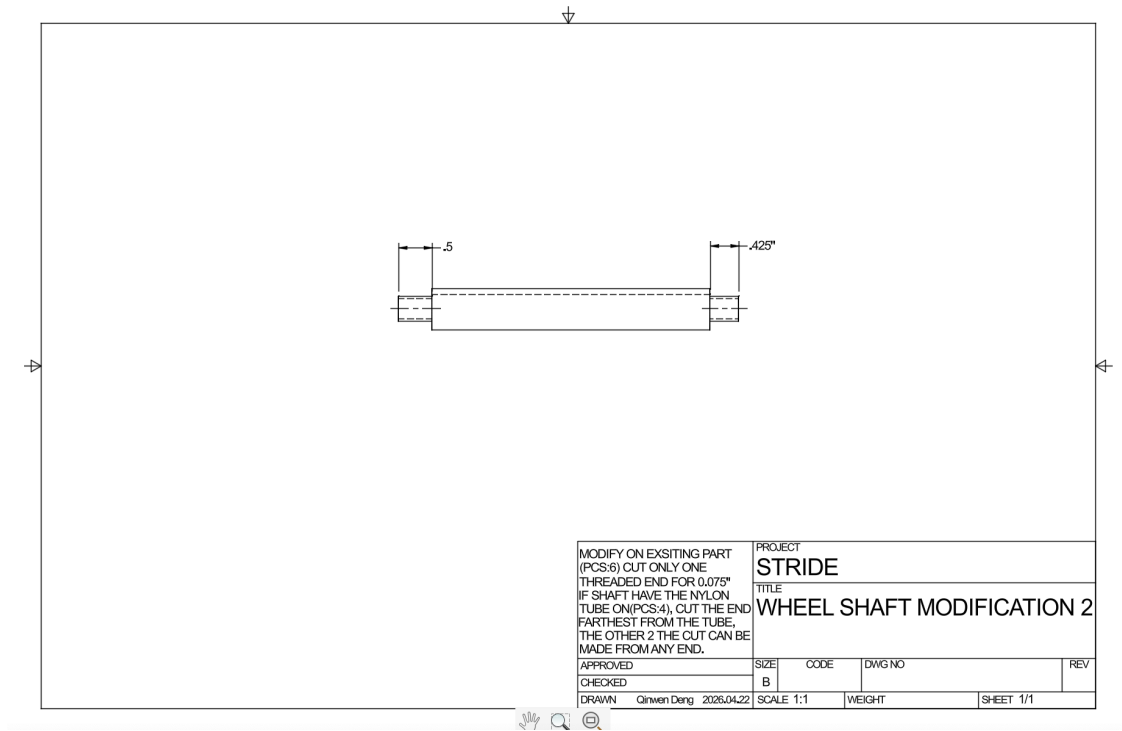


Figure 62: Wheel shaft modification 2.

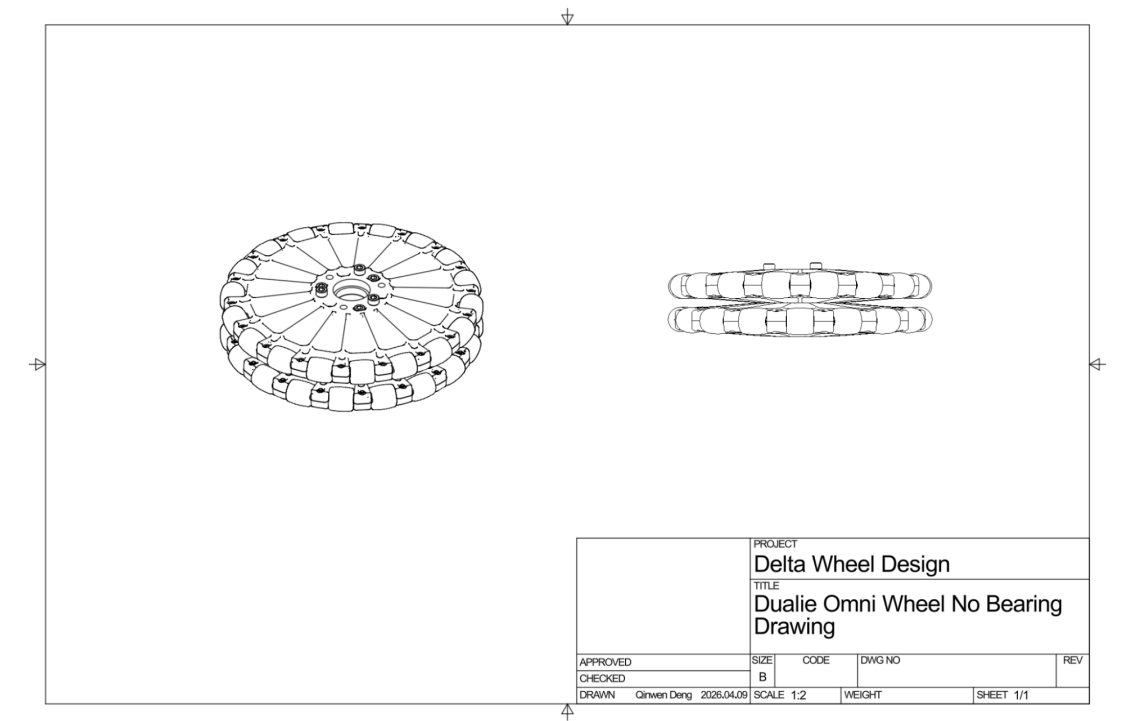


Figure 63: Omni-wheel.

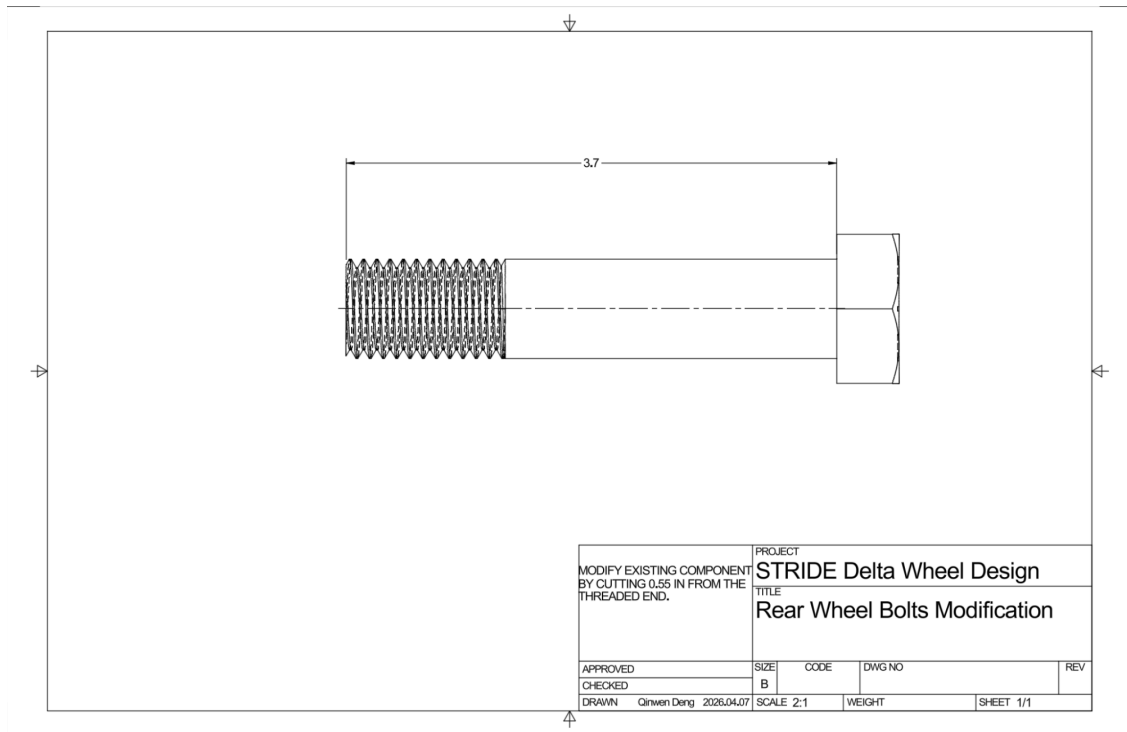


Figure 64: Rear wheel bolts (wheel shaft).

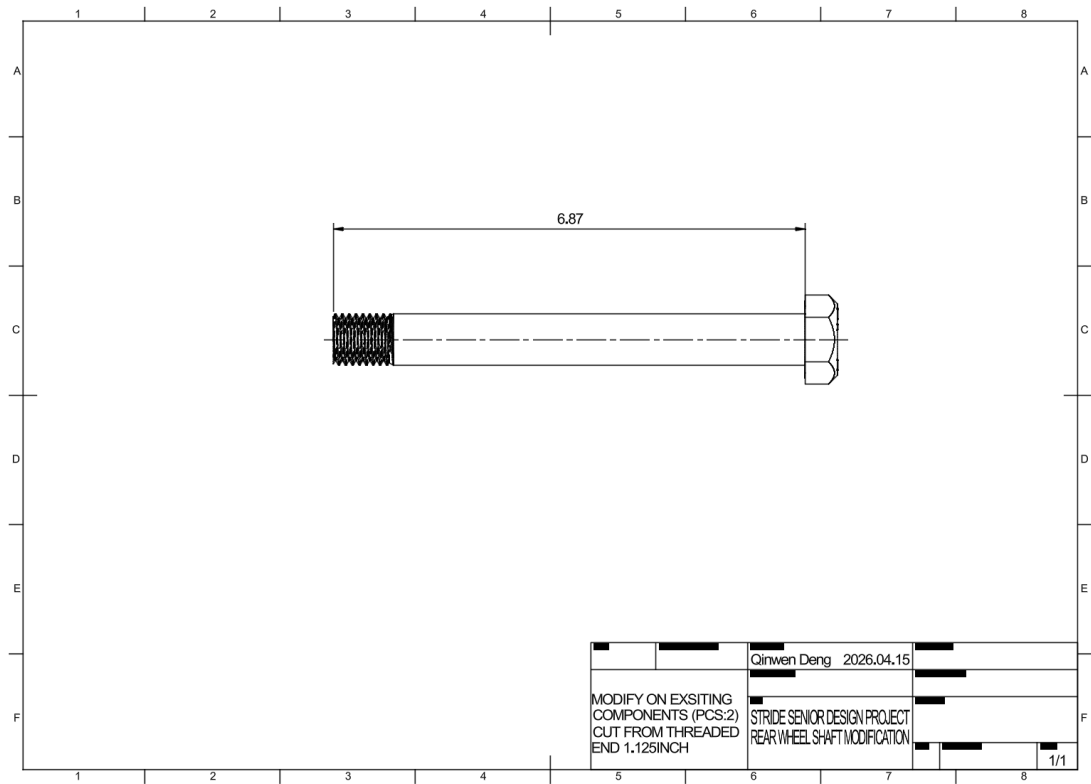


Figure 65: Rear wheel central bolts (central shaft).

IV. Calculation documentation:

STRIDE Senior Design Team. [Material Parameter Calculations](#). Stress-strain calculations for aluminum tubing specification.

STRIDE Senior Design Team. [Wheel Design Note](#). Calculations, analysis, and comparison of tri-wheel design.

STRIDE Senior Design Team. [Company Compare/Contrast: Power Budget](#). Calculations of estimated max wattage requirements of all major systems, used to determine battery parameters. Also includes comparison matrices for different vendors supplying motors and batteries.

STRIDE Senior Design Team. [Torque Calculation](#). Calculations of required torque for the stair-climbing process.

V. Full diagrams:

STRIDE Senior Design Team. [Master Pin List](#). Full documentation of all used pins, their voltages, and their communication protocols (if applicable).

STRIDE Senior Design Team. [Motor Wiring and Wiring Procedures](#). Full diagram of the motor wiring system and procedures for properly wiring the motor and motor controller.

VI. Assembly Guides:

STRIDE Senior Design Team. [General Frame Assembly](#). General guidelines for attaching peripheral components to the frame.

STRIDE Senior Design Team. [Wheel Parts List & Assembly Instructions](#). Details complete part lists and assembly guide.

VII. Software:

STRIDE Senior Design Team. [GitHub: Battery Monitor](#). Code to control the battery monitor board.

STRIDE Senior Design Team. [GitHub: Depth Sensing](#). Code to perform depth sensing using ZED 2i Depth Camera and its [SDK](#).

VIII. Safety and testing general documents:

STRIDE Senior Design Team. [Safety Plan and Review](#). Details safety procedures specific to the fabrication of the [STRIDE](#) senior design project.

STRIDE Senior Design Team. [Testing & Integration checklist](#). A checklist for general tests, including unit and integration tests.

IX. Microcontroller Testing Documents:

STRIDE Senior Design Team. [Microprocessor tests](#). Unit tests for the microprocessors.

STRIDE Senior Design Team. [Microprocessor tests](#). Unit tests for the MCU carrier boards and communication between MCUs.

STRIDE Senior Design Team. [Microprocessor tests](#). Unit tests for STM32-Jetson UART communication.

X. Sensor Testing Documents:

STRIDE Senior Design Team. [Sensors unit tests](#). A document detailing the testing procedure, test matrices, and error tolerance for the various sensors.

STRIDE Senior Design Team. [OPT 3301 \(TOF\) Performance Test](#). A performance test of the ToF sensors, which concluded that the sensors were unsuitable for this system.

STRIDE Senior Design Team. [Ultrasonic Sensors unit tests](#). A document detailing the testing procedure, test matrices, and error tolerance for the ultrasonic sensors.

XI. Motor Testing Documents:

STRIDE Senior Design Team. [Motor unit test](#). A document detailing the testing procedure, test matrices, and error tolerance for the motors.

XII. Frame Testing Documents:

STRIDE Senior Design Team. [Frame unit test](#). A document detailing the testing procedure, test matrices, and error tolerance for the frame.

STRIDE Senior Design Team. [Wheel unit test](#). A document detailing the testing procedure, test matrices, and error tolerance for the Wheel.

XIII. PCB Testing Documents:

STRIDE Senior Design Team. [PCB unit test](#). A document detailing the testing procedure, test matrices, and error tolerance for the various PCBs.

XIV. Power Testing Documents:

STRIDE Senior Design Team. [Power unit test](#). A document detailing the testing procedure, test matrices, and error tolerance for the power systems, including the battery.

STRIDE Senior Design Team. [STRIDE Module-Based PDB Bring-Up Procedure and Bench ATP PDB Bring Up Procedure & Acceptance Test Plan](#)

STRIDE Senior Design Team. [STRIDE PDB Test Report PDB Test Report](#)

XV. UI Testing Documents

STRIDE Senior Design Team. [UI unit tests](#). A document detailing the testing procedure, test matrices, and error handling for the UI.

STRIDE Senior Design Team. [STRIDE Module-Based UI Bring-Up Procedure and Bench ATP UI Bring Up Procedure & Acceptance Test Plan](#)

STRIDE Senior Design Team. [Battery Status Monitoring Board Test Plan](#). Testing procedure for the battery status monitoring board.

STRIDE Senior Design Team. [STRIDE Battery Status Monitoring Board Test Result](#). Results for the battery status monitoring board tests.

XVI. Integration Test:

STRIDE Senior Design Team. [STRIDE Acceptance Testing](#). A document detailing the final acceptance testing procedure and results for the full system.

XVII. Budget documentation:

STRIDE Senior Design Team. [STRIDE Budget Tracker](#). An updating, itemized list of every purchase for the purposes of the STRIDE senior design project.

XVIII. Miscellaneous design documents:

STRIDE Senior Design Team. [Problems Sensors Could Solve](#). A doc listing out potential challenges the system could face, then grouping the problems which could be solved by a particular sensor.

STRIDE Senior Design Team. [Depth Camera Comparisons](#). A doc listing and comparing the different depth cameras the team could purchase.

STRIDE Senior Design Team. [Battery Status Monitor Design Notes](#). Full notes of the design process behind the battery monitor circuit.

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Artificial Intelligence (AI) References:

1. AI has been used on this subsection of text for formatting and grammatical check purposes. The intent of the text has not been changed.
2. AI was used on the contents of this table to properly format it into L^AT_EX. The content of the table was not changed.